Network Theory

Detailed Solutions • 1995–2026

GATE INSTRUMENTATION ENGINEERING

(IN)

Himanshu Agarwal & Anish Saha

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Where Concept Meets Clarity!

Preface

Welcome to PrepFusion™!

This book works, in full, every question GATE has actually asked in Network Theory (IN), gathered from the original papers and arranged unit by unit and year by year. Nothing is paraphrased or reconstructed: every question appears as it was set, and every solution is taken step by step rather than asserted.

Each solution carries the same identifier as the question it answers in the companion questions volume, so you can move between practice and explanation without a lookup table. Where the examining authority revised an answer or awarded marks to all (MTA), that is noted with the question it affects.

Used end to end, the book shows you what the paper rewards — which topics repeat, how the framing has shifted over the years, and where your preparation still has gaps.

Meet your Authors

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> *"We built this book the way every aspirant deserves one to be built — organized strictly by unit and by year, so that every hour of practice maps directly onto how GATE actually tests you."*
>
> — Himanshu Agarwal & Anish Saha

How to Read the Question Numbers

Every question in this book carries a three-part identifier instead of a plain serial number:

Q3.24.7

3	Unit (chapter) number — Unit III of this book
24	GATE exam year — 2024 (two digits; 98 = 1998)
7	Question number within that unit and year

So **Q3.24.7** is the 7th question that GATE 2024 contributed to Unit III. Numbers are therefore not sequential through the book — they tell you the unit and the exam year at a glance. The same identifier is used in the questions book, the solutions book and the answer keys, so you can jump between them without a lookup table. In the PDF bookmarks panel, expand a unit to see every question ID and click one to go straight to it.

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4. Energy Sources in Electrical Circuits
5. Series-Parallel Connections
6. Methods of Analyzing Electrical Circuits — Nodal and Mesh Analysis

Q1.26.1 MCQ 2026 2M Ans: B ● ○ ○

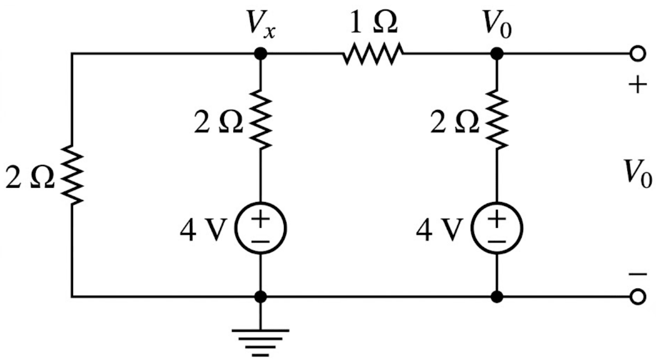


Fig. Solution Q1.26.1

At node V_x,

$$\frac{V_x}{2} + \frac{V_x - 4}{2} + (V_x - V_0) = 0 \Rightarrow 2V_x - V_0 = 2 \quad \text{(i)}$$

At node V_0,

$$(V_0 - V_x) + \frac{V_0 - 4}{2} = 0 \Rightarrow -2V_x + 3V_0 = 4 \quad \text{(ii)}$$

From (i) and (ii),

$$2V_0 = 6 \Rightarrow V_0 = 3\text{ V}$$

$\boxed{\text{B) 3}}$

Q1.25.1 MCQ 2025 2M Ans: C ● ● ○

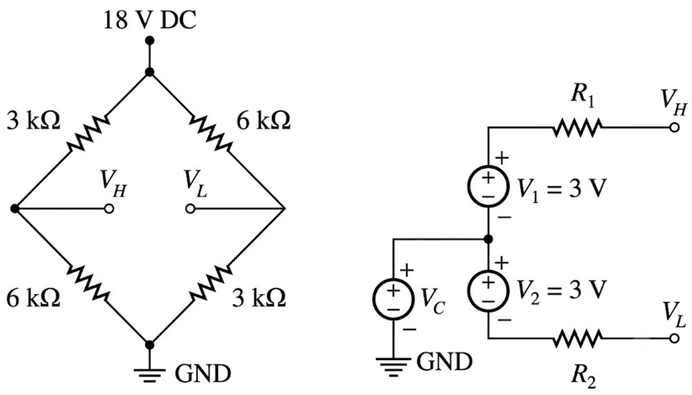


Fig. Solution Q1.25.1

From Figure (a),

$$V_H = 18\left(\frac{6}{6+3}\right) = 12\text{ V}, \qquad V_L = 18\left(\frac{3}{3+6}\right) = 6\text{ V}$$

From Figure (b),

$$V_H = V_C + 3, \qquad V_L = V_C - 3$$

Hence,

$$V_C = 9\,\text{V}$$

Also,

$$R_1 = 6 \parallel 3 = 2\,\text{k}\Omega, \qquad R_2 = 6 \parallel 3 = 2\,\text{k}\Omega$$

$$\boxed{\text{C) } R_1 = 2\,\text{k}\Omega,\ R_2 = 2\,\text{k}\Omega \text{ and } V_C = 9\,\text{V}}$$

Q1.23.1 NAT 2023 1M Ans: 1

Given circuit is as shown below.

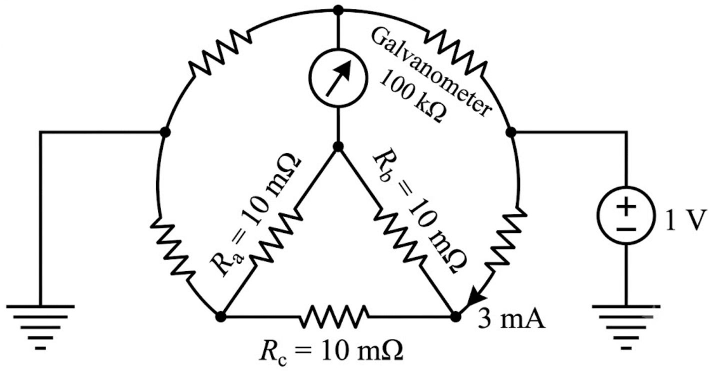


Fig. Solution Q1.23.1

As the bridge is balanced, the reading of galvanometer will be zero.

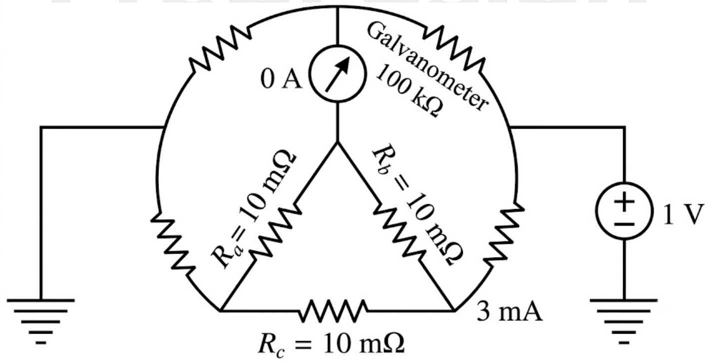


Fig. Solution Q1.23.1

Using current division rule,

$$I_{R_a} = 3 \times \frac{10}{30} = 1\,\text{mA}$$

$$\boxed{1}$$

Q1.20.1 MCQ 2020 1M Ans: A ● ○ ○

Maximum current for 100 Ω, 1 W:

$$I = \sqrt{\frac{P}{R}} = \sqrt{\frac{1}{100}} = 0.1\ \text{A}$$

Maximum current for 2 Ω, 0.5 W:

$$I = \sqrt{\frac{0.5}{2}} = 0.5\ \text{A}$$

Maximum current for 1 Ω, 0.25 W:

$$I = \sqrt{\frac{0.25}{1}} = 0.5\ \text{A}$$

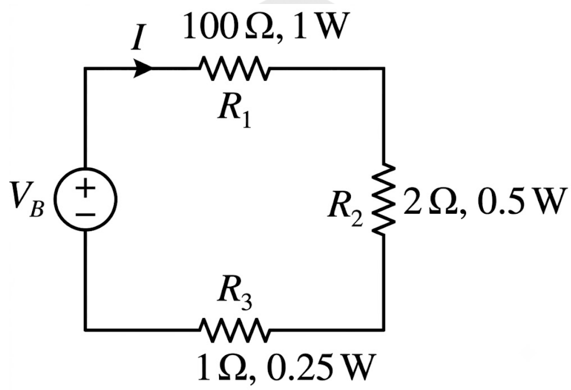


Fig. Solution Q1.20.1

As all resistors are in series, the same current flows through each resistor. The limiting resistor is R_1 since its maximum allowable current is only 0.1 A.

Hence, the maximum safe current in the circuit is

$$I_{\text{max}} = 0.1\ \text{A}$$

$$\boxed{\text{A) } 0.1}$$

Q1.20.2 MCQ 2020 2M Ans: C ● ● ○

Given circuit is shown below.

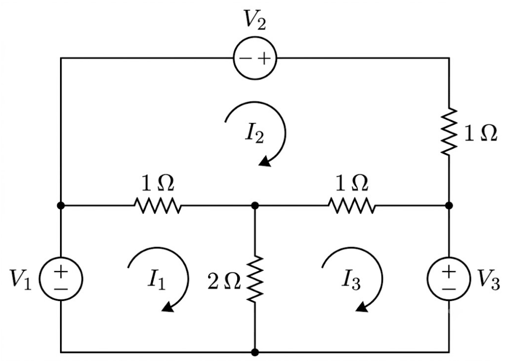


Fig. Solution Q1.20.2

Writing KVL in Loop–1,

$$-V_1 + 1(I_1 - I_2) + 2(I_1 - I_3) = 0$$

$$V_1 = 3I_1 - I_2 - 2I_3 \tag{i}$$

Applying KVL in Loop–2,

$$-V_2 + I_2 + 1(I_2 - I_3) + 1(I_2 - I_1) = 0$$

$$V_2 = -I_1 + 3I_2 - I_3 \tag{ii}$$

Applying KVL in Loop–3,

$$V_3 + 2(I_3 - I_1) + 1(I_3 - I_2) = 0$$

$$-V_3 = -2I_1 - I_2 + 3I_3 \tag{iii}$$

Writing (i), (ii) and (iii) in matrix form,

$$\begin{bmatrix} 3 & -1 & -2 \\ -1 & 3 & -1 \\ -2 & -1 & 3 \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \\ I_3 \end{bmatrix} = \begin{bmatrix} V_1 \\ V_2 \\ -V_3 \end{bmatrix}$$

Hence,

$$\boxed{\text{(C)} \begin{bmatrix} 3 & -1 & -2 \\ -1 & 3 & -1 \\ -2 & -1 & 3 \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \\ I_3 \end{bmatrix} = \begin{bmatrix} V_1 \\ V_2 \\ -V_3 \end{bmatrix}}$$

Q1.19.1 NAT | 2019 | 1M | Ans: 1000 | ● ○ ○

Given circuit consists of constant resistance and ideal independent DC voltage sources.

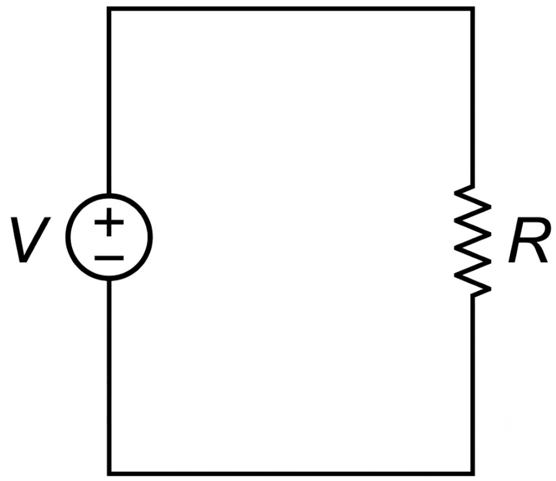


Fig. Solution Q1.19.1

$$P_{\text{abs},1} = \frac{V^2}{R}$$

If all resistances are scaled down by a factor 10 and all source voltages are scaled up by a factor 10,

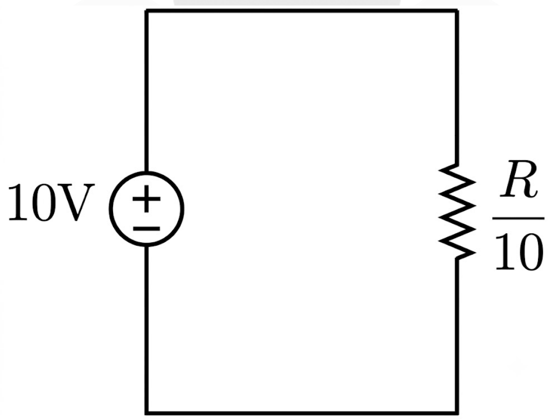


Fig. Solution Q1.19.1

$$P_{\text{abs},2} = \frac{(10V)^2}{R/10} = 1000\frac{V^2}{R} = 1000\,P_{\text{abs},1}$$

Hence, the power dissipated in the circuit scales up by a factor of

$$\boxed{1000}$$

Q1.18.1 MCQ 2018 2M Ans: A ● ○ ○

Given circuit is shown below.

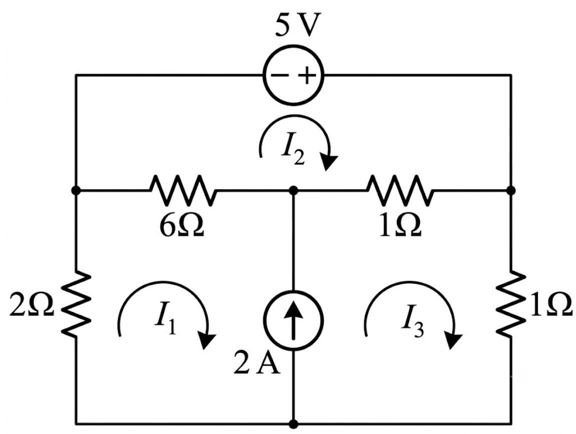


Fig. Solution Q1.18.1

Applying KVL in loop $ABCD$,

$$2I_1 + 6(I_1 - I_2) + (I_3 - I_2) + I_3 = 0$$

$$8I_1 - 7I_2 + 2I_3 = 0 \tag{i}$$

Applying KVL in loop 2,

$$-5 + (I_2 - I_3) + 6(I_2 - I_1) = 0$$

$$-6I_1 + 7I_2 - I_3 = 5 \tag{ii}$$

From the concept of supermesh,

$$I_3 - I_1 = 2 \tag{iii}$$

From (i), (ii) and (iii): $I_1 = 1\text{ A}, \; I_2 = 2\text{ A}, \; I_3 = 3\text{ A}$

$$\boxed{\text{A) } I_1 = 1\text{ A}, I_2 = 2\text{ A and } I_3 = 3}$$

Q1.17.1 NAT 2017 1M Ans: 2 ● ● ○

Given:

$$V_1 = V_{R_1} = -1\text{ V} \tag{i}$$

Given circuit is shown in figure.

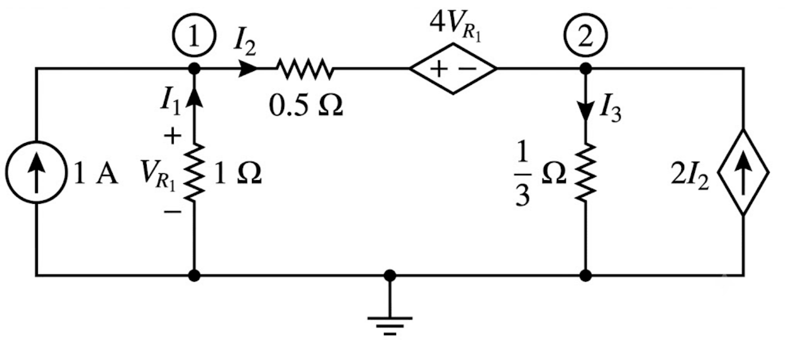


Fig. Solution Q1.17.1

Applying KCL at Node-1,

$$\frac{V_{R_1}}{1} - 1 + \frac{V_{R_1} - 4V_{R_1} - V_2}{0.5} = 0$$

Using (i),

$$-1 - 1 + \frac{-1 + 4 - V_2}{0.5} = 0$$

$$V_2 = 2\,\text{V}$$

$\boxed{2}$

Q1.16.1 NAT 2016 2M Ans: 10

Given circuit is shown below.

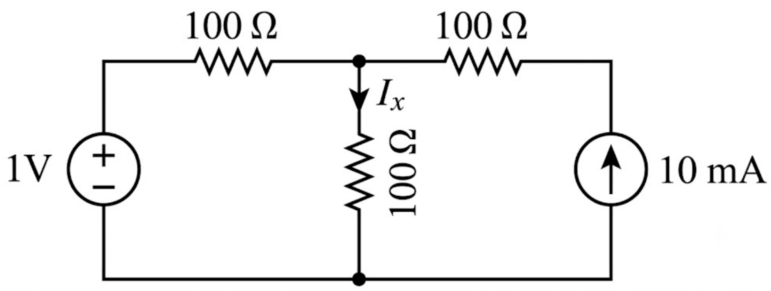


Fig. Solution Q1.16.1

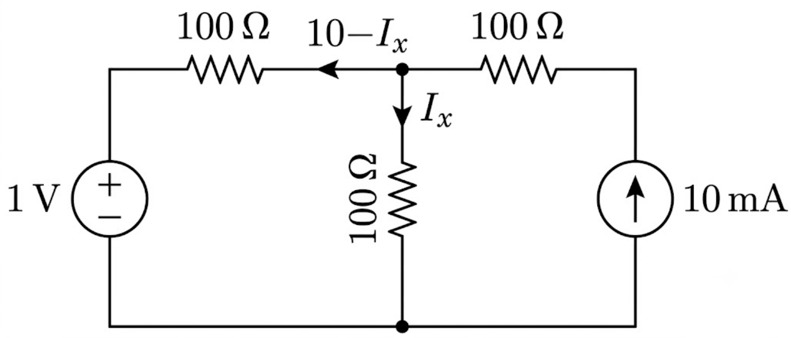


Fig. Solution Q1.16.1

Applying KVL,

$$-1 + 0.1\left[-(10 - I_x) \right] + 0.1 I_x = 0$$

$$0.2 I_x = 2$$

$$I_x = 10\,\text{mA}$$

$$\boxed{10}$$

Q1.15.1 NAT 2015 2M Ans: 1

Given circuit is shown below.

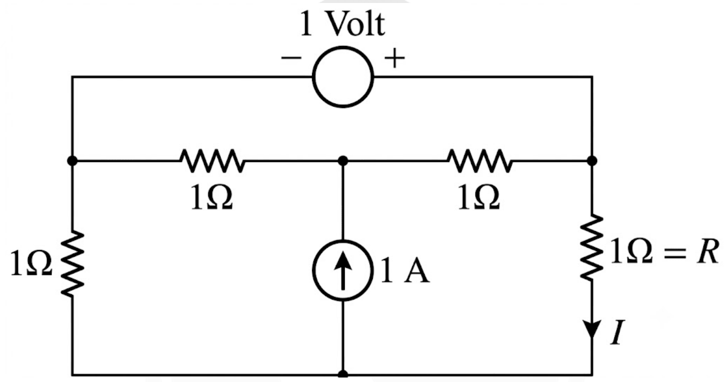


Fig. Solution Q1.15.1

This forms a supernode.

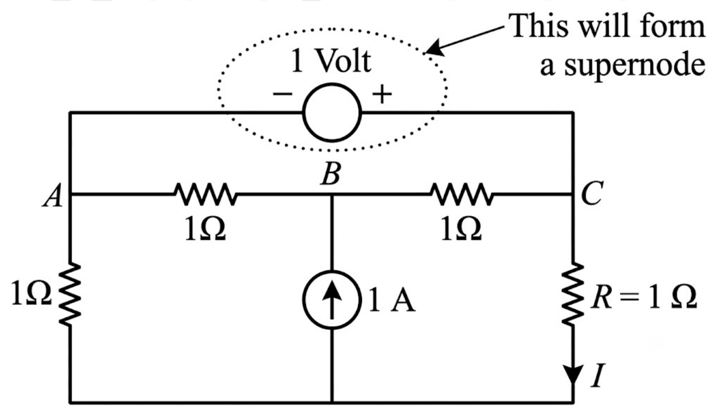


Fig. Solution Q1.15.1

KCL of supernode:

$$\frac{V_A - V_B}{1} + \frac{V_A}{1} + \frac{V_C - V_B}{1} + \frac{V_C}{1} = 0$$

$$2V_A - 2V_B + 2V_C = 0 \tag{i}$$

KVL of supernode:

$$V_C - V_A = 1 \quad \text{(ii)}$$

Applying KCL at node B:

$$\frac{V_B - V_A}{1} + \frac{V_B - V_C}{1} - 1 = 0$$

$$2V_B - V_A - V_C = 1 \quad \text{(iii)}$$

From (i), (ii) and (iii),

$$V_A = 0\,\text{V}, \qquad V_B = 1\,\text{V}, \qquad V_C = 1\,\text{V}$$

The current through resistor R is

$$I = \frac{V_C}{R} = \frac{1}{1} = 1\,\text{A}$$

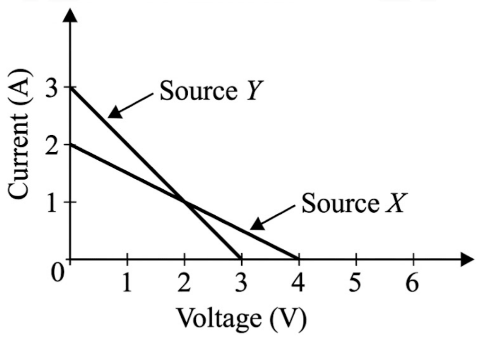
$\boxed{1}$

Key Points

- If an ideal voltage source is connected between two non-reference nodes, those nodes form a supernode.
- A supernode requires both KCL and KVL equations.

Q1.15.2 NAT 2015 2M Ans: 1.75

Given circuit is shown below.



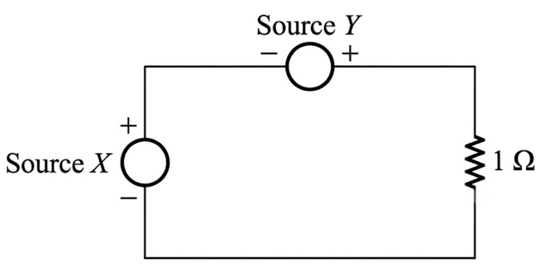


Fig. Solution Q1.15.2

For source X:

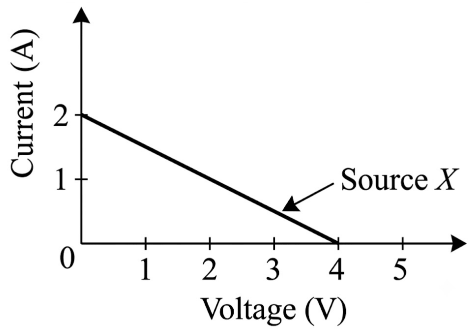


Fig. Solution Q1.15.2

From the figure,

$$V_s = 4\text{ V}, \qquad \frac{V_s}{R_s} = 2\text{ A}$$

$$R_s = \frac{4}{2} = 2\,\Omega$$

For source Y:

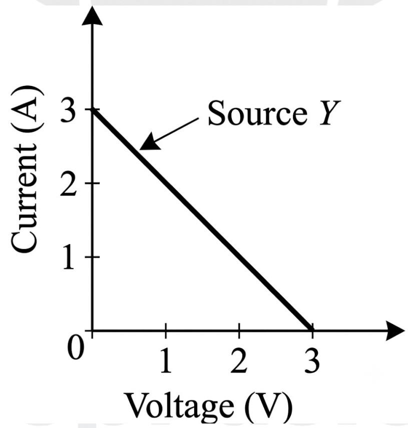


Fig. Solution Q1.15.2

From the figure,

$$V_s = 3\text{ V}, \qquad \frac{V_s}{R_s} = 3\text{ A}$$

$$R_s = \frac{3}{3} = 1\,\Omega$$

The equivalent circuit is shown below.

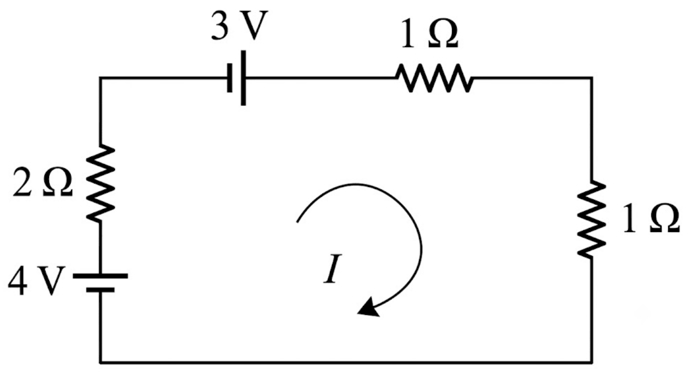


Fig. Solution Q1.15.2

Applying KVL,

$$-4 + 2I - 3 + I + I = 0$$

$$I = \frac{7}{4} = 1.75\text{ A}$$

$$\boxed{1.75}$$

Key Points

- A practical voltage source can be modeled as an ideal voltage source in series with its internal resistance.
- From the I–V characteristic, the voltage-axis intercept gives V_s and the current-axis intercept gives $\frac{V_s}{R_s}$.

Q1.13.1

MCQ | 2013 | 1M | Ans: B

Given delta to star conversion is shown below.

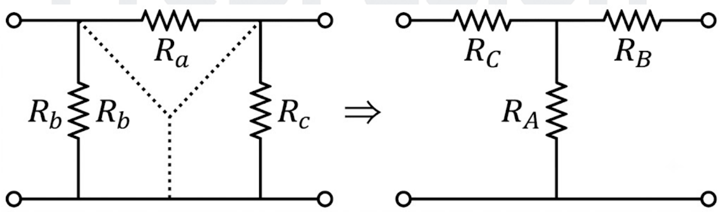


Fig. Solution Q1.13.1

Before scaling,

$$R_A = \frac{R_b R_c}{R_a + R_b + R_c}$$

If

$$R'_a = kR_a, \qquad R'_b = kR_b, \qquad R'_c = kR_c$$

then after scaling,

$$R'_A = \frac{R'_b R'_c}{R'_a + R'_b + R'_c} = \frac{(kR_b)(kR_c)}{kR_a + kR_b + kR_c}$$

$$R'_A = \frac{k^2 R_b R_c}{k(R_a + R_b + R_c)} = kR_A$$

Hence, each star resistance scales by a factor of k.

$\boxed{\text{B) k}}$

Key Points

- If all elements of a Δ network are multiplied by a factor k, all corresponding Y elements are also multiplied by k.

Q1.13.2 MCQ 2013 2M Ans: C ● ● ○

Given:

$$C_1 = 10\,\mu\text{F}, \quad V_1 = 10\,\text{V}$$

$$C_2 = 5\,\mu\text{F}, \quad V_2 = 5\,\text{V}$$

$$C_3 = 2\,\mu\text{F}, \quad V_3 = 2\,\text{V}$$

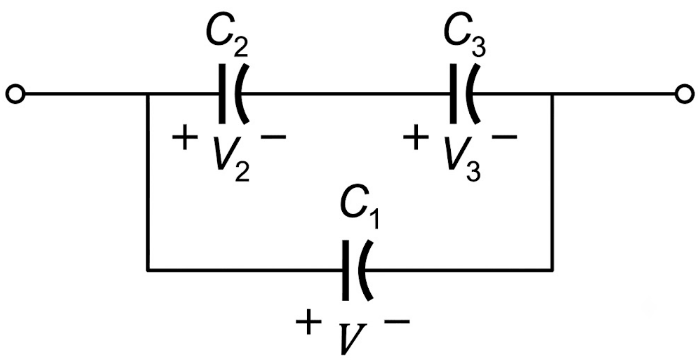


Fig. Solution Q1.13.2

Since C_3 has the minimum breakdown voltage, it determines the maximum safe voltage across the network.

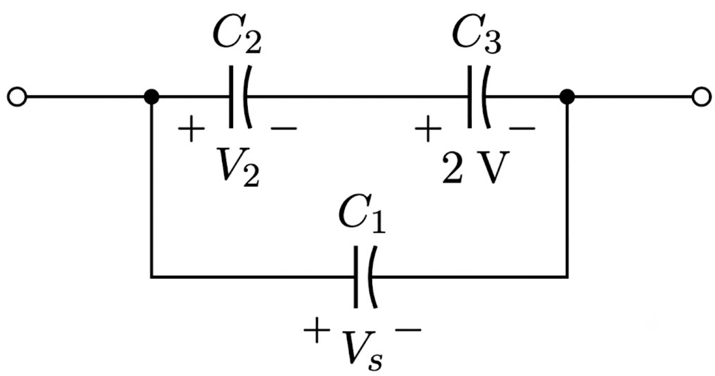


Fig. Solution Q1.13.2

Using voltage division for series capacitors,

$$V_3 = \frac{C_2}{C_2 + C_3} V_s$$

$$2 = \frac{5}{5+2} V_s$$

$$V_s = 2.8\ \text{V}$$

The equivalent capacitance is

$$C_{eq} = C_1 + \frac{C_2 C_3}{C_2 + C_3} = 10 + \frac{5 \times 2}{5+2} = \frac{80}{7}\ \mu\text{F}$$

Hence,

$$Q = C_{eq} V_s = \frac{80}{7} \times 2.8 = 32\ \mu\text{C}$$

$$\boxed{\text{C) 2.8 and 32}}$$

Q1.12.1 MCQ | 2012 | 2M | Ans: A | ● ● ○

Given:

$$V_A - V_B = 6\ \text{V}$$

The circuit is shown below.

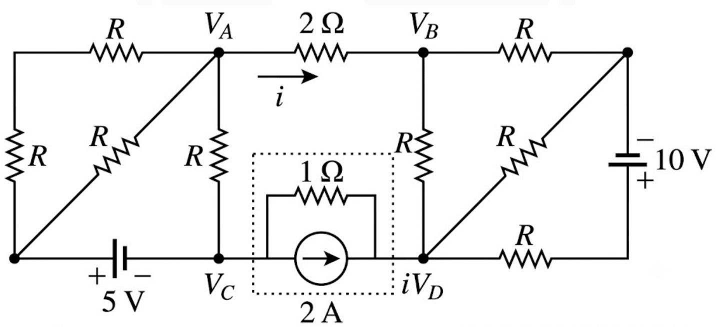


Fig. Solution Q1.12.1

From the circuit,

$$i = \frac{V_A - V_B}{2} = \frac{6}{2} = 3\ \text{A}$$

The same current returns through the branch between nodes D and C.

Method 1

(Using source transformation)
Converting the 2 A current source into a voltage source,

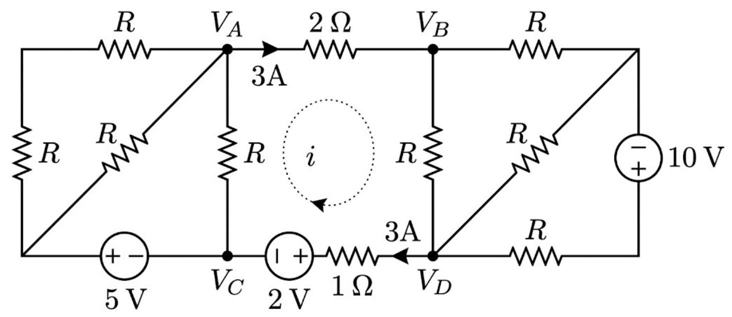


Fig. Solution Q1.12.1

Applying KVL between nodes D and C,

$$V_{DC} = 3 \times 1 + 2 = 5\text{ V}$$

$$V_{CD} = V_C - V_D = -5\text{ V}$$

Method 2

(Without source transformation)

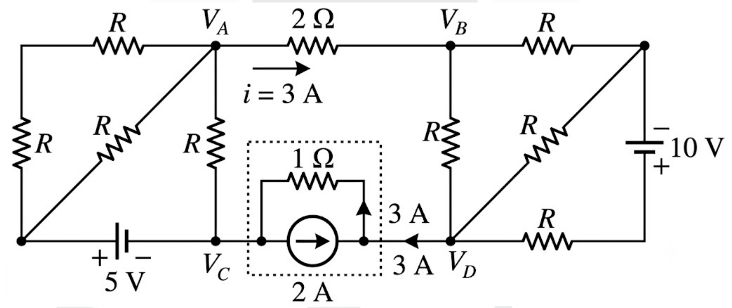


Fig. Solution Q1.12.1

Current through the 1 Ω resistor between nodes D and C is

$$5\text{ A}$$

Hence,

$$V_{DC} = 5 \times 1 = 5\text{ V}$$

$$V_{CD} = V_C - V_D = -5\text{ V}$$

$$\boxed{\text{A) -5}}$$

Q1.11.1 MCQ 2011 1M Ans: A ● ○ ○

Given circuit is shown below.

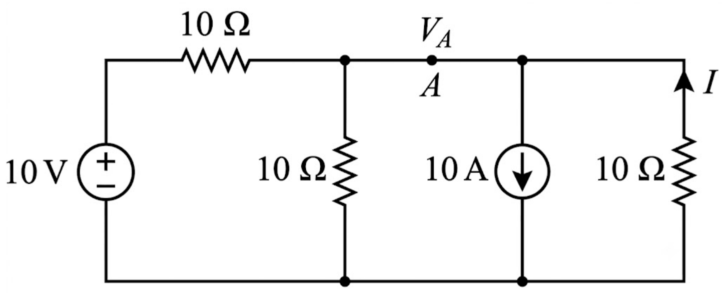


Fig. Solution Q1.11.1

Applying KCL at node A,

$$\frac{V_A}{10} + \frac{V_A - 10}{10} + \frac{V_A}{10} + 10 = 0$$

$$V_A + V_A - 10 + V_A + 100 = 0$$

$$3V_A = -90$$

$$V_A = -30\text{ V}$$

From the figure,

$$I = -\frac{V_A}{10} = -\frac{-30}{10} = 3\text{ A}$$

$$\boxed{\text{A) 3}}$$

Q1.10.1 MCQ | 2010 | 1M | Ans: C | ● ○ ○

According to the question,

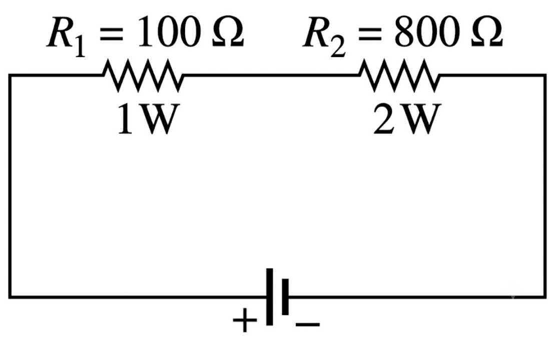


Fig. Solution Q1.10.1

Maximum power dissipated in the 100 Ω resistor is

$$P_{max} = 1\text{ W} = I_{max}^2(100)$$

$$I_{max} = \sqrt{\frac{1}{100}} = 0.1\text{ A}$$

Maximum power dissipated in the 800 Ω resistor is

$$P_{max} = 2\,\text{W} = I_{max}^2(800)$$

$$I_{max} = \sqrt{\frac{2}{800}} = 0.05\,\text{A}$$

Since the resistors are connected in series, the same current flows through both. Hence, the 800 Ω resistor limits the circuit current to

$$I_{max} = 0.05\,\text{A}$$

Therefore,

$$V_{max} = I_{max}(R_1 + R_2)$$

$$V_{max} = 0.05(100 + 800) = 45\,\text{V}$$

$$\boxed{\text{C) } 45}$$

Q1.09.1 | MCQ | 2009 | 2M | Ans: C | ● ○ ○

Given,

$$x(t) = 3 + 2\sin t\cos 2t$$

Using

$$2\sin A\cos B = \sin(A+B) + \sin(A-B)$$

$$x(t) = 3 + \sin 3t - \sin t$$

Therefore,

$$x_{rms} = \sqrt{3^2 + \frac{1^2}{2} + \frac{(-1)^2}{2}} = \sqrt{10}$$

$$\boxed{\text{C) } \sqrt{10}}$$

Q1.08.1 | MCQ | 2008 | 2M | Ans: D | ● ○ ○

From the figure, the potentials at points A, B and C are the same, i.e.,

$$V_A = V_B = V_C$$

Hence, the resistance between A and C can be neglected.
The modified circuit is shown below.

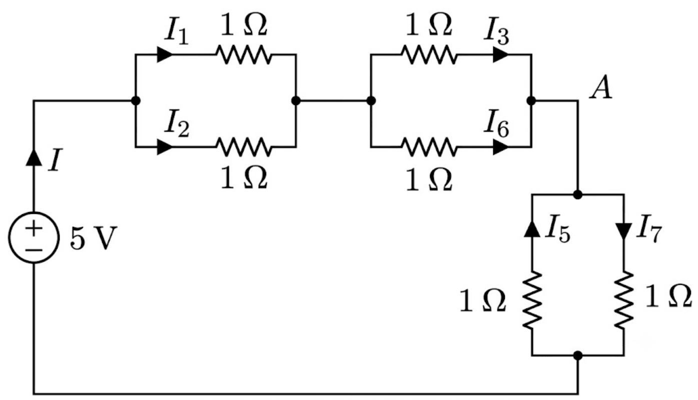


Fig. Solution Q1.08.1

Applying KCL at node A,

$$I_3 + I_5 + I_6 - I_7 = 0$$

$$\boxed{\text{D) } I_3 + I_5 + I_6 - I_7 = 0}$$

Q1.08.2 MCQ 2008 1M Ans: D

Given circuit is shown below.

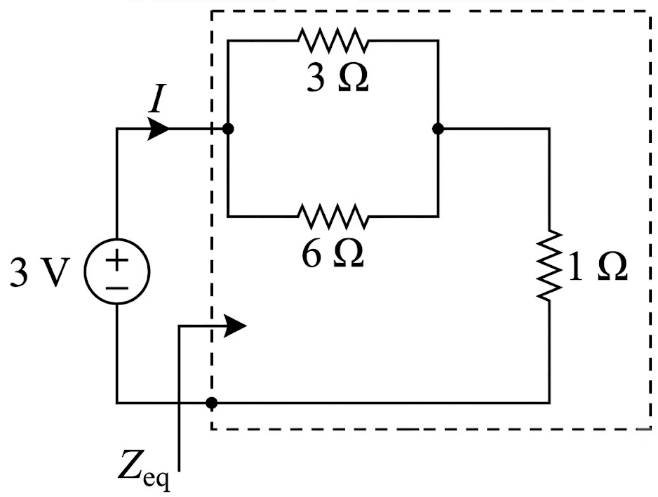


Fig. Solution Q1.08.2

The equivalent resistance is

$$Z_{eq} = (3 \parallel 6) + 1 = 2 + 1 = 3\,\Omega$$

Hence,

$$I = \frac{V}{Z_{eq}} = \frac{3}{3} = 1\,\text{A}$$

The power supplied by the 3 V source is

$$P = VI = 3 \times 1 = 3\,\text{W}$$

D) 3.0

Q1.08.3 MCQ 2008 1M Ans: A

Given circuit is shown below.

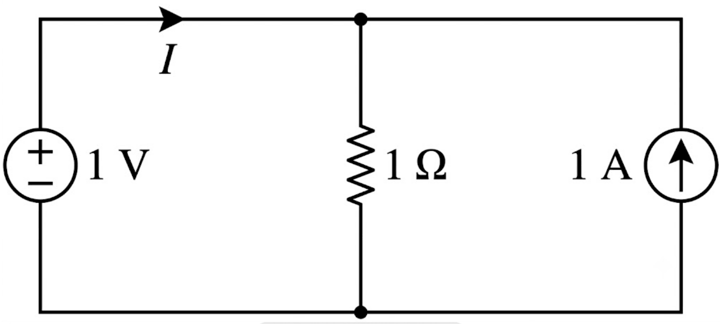


Fig. Solution Q1.08.3

Using Nodal Analysis,

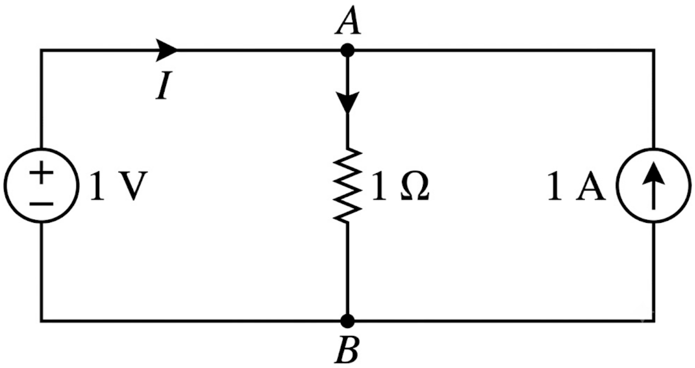


Fig. Solution Q1.08.3

Applying KCL at node A,

$$\frac{V_A}{1} - 1 - I = 0$$

From the figure,

$$V_A = 1\,\text{V}$$

Hence,

$$1 - 1 - I = 0$$

$$I = 0\,\text{A}$$

$$\boxed{\text{A) } 0}$$

Q1.08.4 MCQ | 2008 | 2M | Ans: D

Given circuit is shown below.

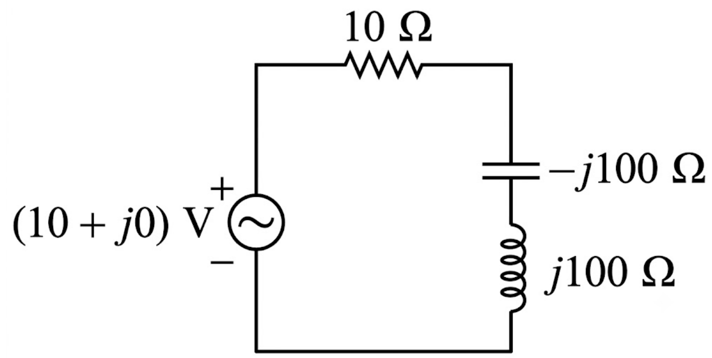
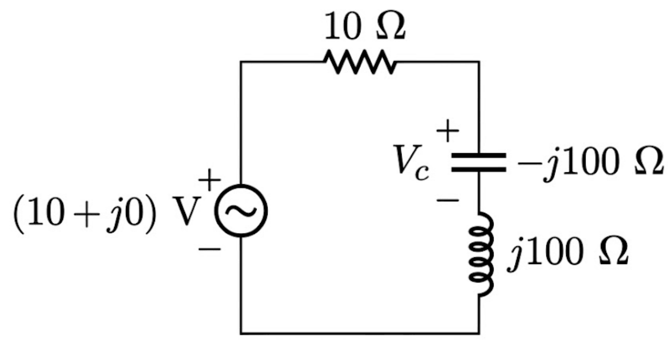


Fig. Solution Q1.08.4

Voltage division rule



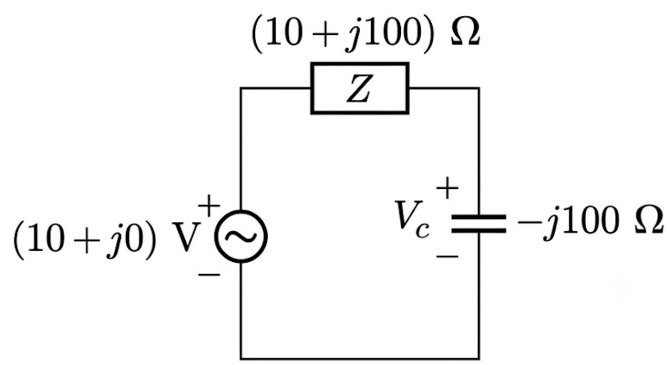


Fig. Solution Q1.08.4

From the figure,

$$V_C = (10 + j0)\left(\frac{-j100}{10 + j100 - j100}\right)$$

$$V_C = (10 + j0)\left(\frac{-j100}{10}\right)$$

$$V_C = 0 - j100\,\text{V}$$

$$\boxed{\text{D) } (0 - j100)\text{ V}}$$

Q1.06.1 MCQ | 2006 | 2M | Ans: B

According to the question,
Before stretching,

$$A = A_1, \qquad l = l_1, \qquad R = R_1$$

Since volume remains constant,

$$V = A\,l = A_1 l_1 \tag{i}$$

After stretching,

$$l_2 = \alpha l$$

Let the new area be A_2. Then,

$$V = A_2 l_2 = A_2(\alpha l) \tag{ii}$$

From (i) and (ii),

$$A\,l = A_2(\alpha l)$$

$$A_2 = \frac{A}{\alpha}$$

Since

$$R = \rho\frac{l}{A}$$

the original resistance is

$$R_1 = R = \rho\frac{l}{A} \tag{iii}$$

The new resistance is

$$R_2 = \rho\frac{l_2}{A_2} = \rho\frac{\alpha l}{A/\alpha} = \alpha^2\frac{\rho l}{A}$$

Using (iii),

$$R_2 = \alpha^2 R$$

$$\boxed{\text{B) } \alpha^2 R}$$

Key Points

- For uniform stretching of a wire with constant volume and unchanged resistivity, resistance varies as the square of its length, i.e., $R \propto l^2$.

Q1.06.2 MCQ 2006 1M Ans: C ● ● ○

According to the question,

$$v(t) = v_1(t) + v_2(t)$$

where,

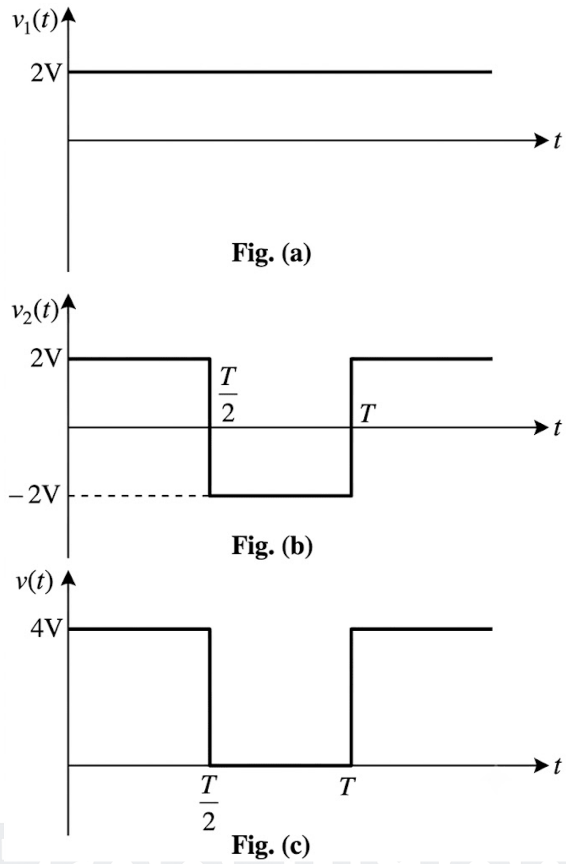


Fig. Solution Q1.06.2

The resultant waveform alternates between 4 V and 0 V with equal time duration.

$$V_{\text{rms}} = \sqrt{\frac{1}{T}\left[\int_0^{T/2} (4)^2\, dt + \int_{T/2}^{T} 0\, dt\right]}$$

$$V_{\text{rms}} = \sqrt{\frac{16}{T}\int_0^{T/2} dt} = \sqrt{\frac{16}{T}\left[\frac{T}{2}\right]} = \sqrt{8}\text{ V}$$

$$\boxed{\text{C) } \sqrt{8}\text{ V}}$$

Q1.04.1 MCQ 2004 2M Ans: C ● ● ○

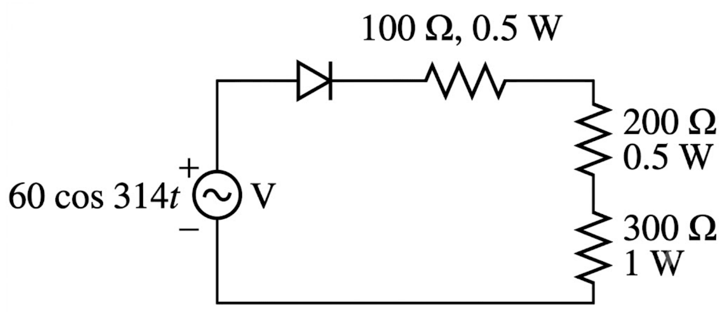


Fig. Solution Q1.04.1

During the positive half cycle, the diode is ON and during the negative half cycle, it is OFF. Hence, the circuit acts as a half-wave rectifier.

Therefore,

$$V_{\text{rms}} = \frac{V_m}{2}$$

where V_m is the peak value of the supply voltage.

$$V_{\text{rms}} = \frac{60}{2} = 30\,\text{V}$$

The total resistance is

$$R_T = 100 + 200 + 300 = 600\,\Omega$$

Hence,

$$I_{\text{rms}} = \frac{V_{\text{rms}}}{R_T} = \frac{30}{600} = \frac{1}{20}\,\text{A}$$

Power dissipated in the 300 Ω resistor is

$$P_{300} = I_{\text{rms}}^2(300) = \left(\frac{1}{20}\right)^2(300)$$

$$P_{300} = \frac{300}{400} = \frac{3}{4} = 0.75\,\text{W}$$

$$\boxed{\text{C) } 0.75\,\text{W}}$$

Q1.01.1 | MCQ | 2001 | 1M | Ans: B | ● ○ ○

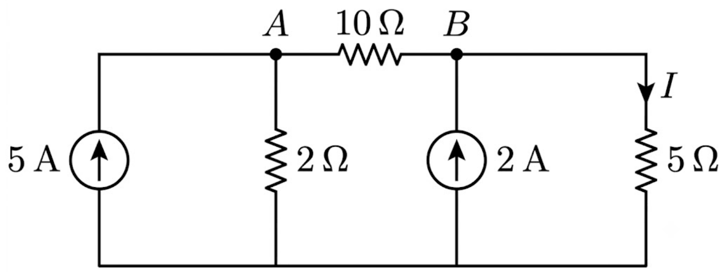


Fig. Solution Q1.01.1

Applying KCL at node A,

$$\frac{V_A}{2} + \frac{V_A - V_B}{10} = 5$$

$$5V_A + V_A - V_B = 50$$

$$6V_A - V_B = 50 \tag{i}$$

Applying KCL at node B,

$$\frac{V_B}{5} + \frac{V_B - V_A}{10} = 2$$

$$2V_B + V_B - V_A = 20$$

$$-V_A + 3V_B = 20 \tag{ii}$$

Solving (i) and (ii),

$$V_A = 10\,\text{V}, \qquad V_B = 10\,\text{V}$$

Hence, the current through the $5\,\Omega$ resistor is

$$I = \frac{V_B}{5} = \frac{10}{5} = 2\,\text{A}$$

$$\boxed{\text{B) } 2\text{ A}}$$

Q1.99.1 MCQ | 1999 | 1M | Ans: A | ● ○ ○

Given,

$$v(t) = 2 + \cos\left(\omega t + \frac{\pi}{6}\right)$$

The average value is equal to the DC component:

$$V_{\text{avg}} = 2\,\text{V} \tag{i}$$

The rms value is

$$V_{\text{rms}} = \sqrt{2^2 + \left(\frac{1}{\sqrt{2}}\right)^2} = \sqrt{4 + \frac{1}{2}} = \sqrt{\frac{9}{2}} \tag{ii}$$

From (i) and (ii),

$$\frac{V_{\text{rms}}}{V_{\text{avg}}} = \frac{\sqrt{9/2}}{2} = \frac{3}{2\sqrt{2}}$$

$$\boxed{\text{A) } \frac{3}{2\sqrt{2}}}$$

Q1.99.2 MCQ 1999 1M Ans: D ● ● ○

Given that the bridge is balanced.

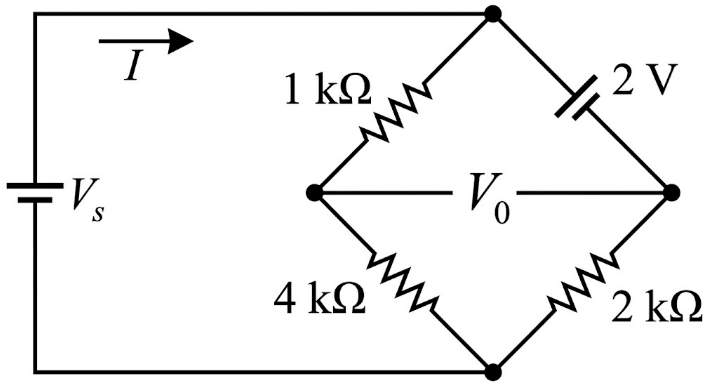


Fig. Solution Q1.99.2

For a balanced bridge,

$$Z_{AD} Z_{BC} = Z_{AB} Z_{CD}$$

$$1\,\mathrm{k\Omega} \times 2\,\mathrm{k\Omega} = Z_{AB} \times 4\,\mathrm{k\Omega}$$

$$Z_{AB} = 0.5\,\mathrm{k\Omega}$$

Current through branch AB is

$$I_{AB} = \frac{V_{AB}}{Z_{AB}} = \frac{2}{0.5 \times 10^3} = 4\,\mathrm{mA} \qquad \text{(i)}$$

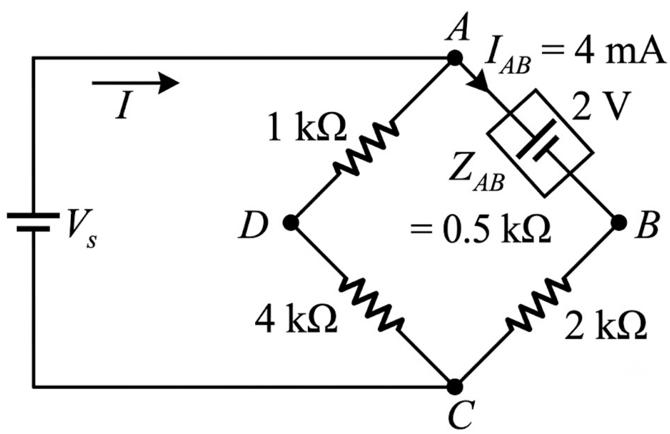


Fig. Solution Q1.99.2

Using current division,

$$I_{AB} = I \frac{1\,\mathrm{k\Omega} + 4\,\mathrm{k\Omega}}{1\,\mathrm{k\Omega} + 4\,\mathrm{k\Omega} + 0.5\,\mathrm{k\Omega} + 2\,\mathrm{k\Omega}}$$

$$I_{AB} = I\left(\frac{5}{7.5}\right)$$

Using (i),

$$4\,\text{mA} = I\left(\frac{5}{7.5}\right)$$

$$I = 6\,\text{mA}$$

$$\boxed{\text{D) 6 mA}}$$

Q1.96.1 MCQ 1996 1M Ans: D

Given,

$$i(t) = \sin t \qquad \text{(i)}$$

$$v = i + 2i^2 \qquad \text{(ii)}$$

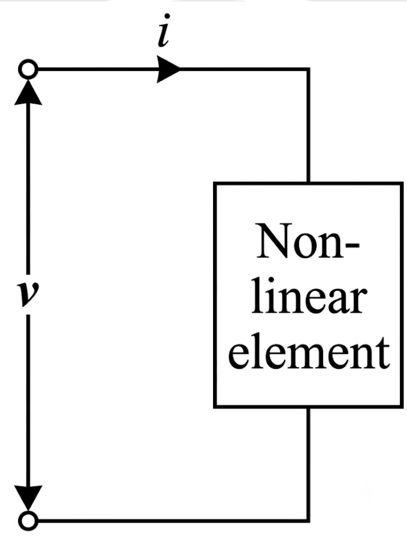


Fig. Solution Q1.96.1

From (i) and (ii),

$$v(t) = \sin t + 2\sin^2 t$$

The instantaneous power consumed is

$$p(t) = v(t)i(t)$$

$$p(t) = \left(\sin t + 2\sin^2 t\right)\sin t$$

$$p(t) = \sin^2 t + 2\sin^3 t$$

Using

$$\sin^2 t = \frac{1-\cos 2t}{2}, \qquad \sin^3 t = \frac{3\sin t - \sin 3t}{4}$$

$$p(t) = \frac{1 - \cos 2t}{2} + \frac{3 \sin t - \sin 3t}{2}$$

$$p(t) = \frac{1}{2} + \frac{3}{2} \sin t - \frac{1}{2} \cos 2t - \frac{1}{2} \sin 3t$$

The average power is the DC component:

$$P_{\text{avg}} = \frac{1}{2} = 0.5\ \text{W}$$

$$\boxed{\text{D) } 0.5\ \text{W}}$$

DEBUG INFO

Chapter I

Basic Concepts of Networks

Total Questions : 28

MCQ : 22

MSQ : 0

NAT : 6

SOLUTIONS

11

Transient Analysis

Topics

1. Introduction to Transient Analysis
2. Basics of Signals for Transient Analysis
3. RC Circuits — Introduction and Examples
4. Impulse Current in Capacitor and Basics of Laplace in Circuit Analysis
5. Slope at $t = 0^+$, Switched Capacitor and Impulse Response of RC Circuits
6. RC Circuits with Pulse Wave Input
7. Initial and Final Value Theorems
8. RL Circuits

Q2.25.1 NAT | 2025 | 2M | Ans: 4.82 | ● ● ○

Given: The circuit is shown below.

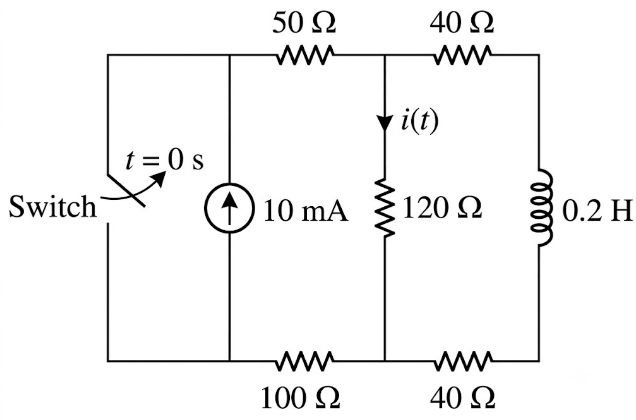


Fig. Solution Q2.25.1

(i) At $t = 0^-$, switch is closed

In steady state, the inductor acts as a short circuit.

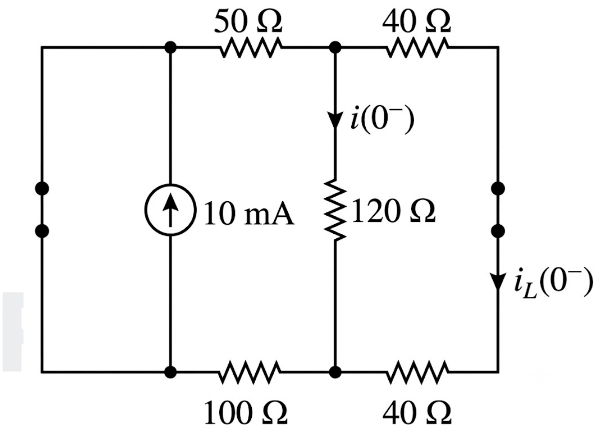


Fig. Solution Q2.25.1

From the circuit,

$$i_L(0^-) = 0$$

Since the current through an inductor cannot change instantaneously,

$$i_L(0^+) = i_L(0^-) = 0$$

At $t = 0^+$, the switch is opened and the inductor acts as an open circuit.

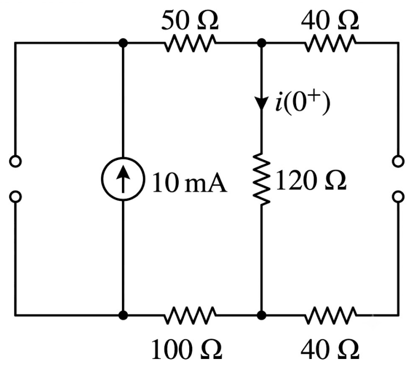


Fig. Solution Q2.25.1

$$i(0^+) = 10 \text{ mA}$$

At $t = \infty$, the switch is open and the circuit is in steady state. Hence, the inductor acts as a short circuit.

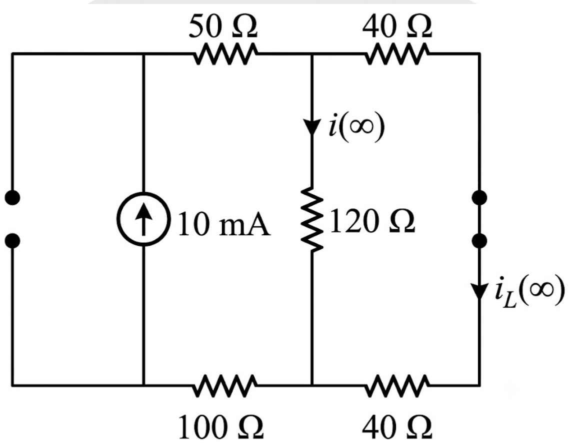


Fig. Solution Q2.25.1

$$i(\infty) = 10\left(\frac{40 + 40}{120 + 40 + 40}\right)$$

$$i(\infty) = 10\left(\frac{80}{200}\right) = \frac{800}{200} = 4 \text{ mA}$$

Time constant of the circuit:

$$L_{eq} = 0.2 \text{ H}$$

For finding R_{eq}, suppress the independent current source.

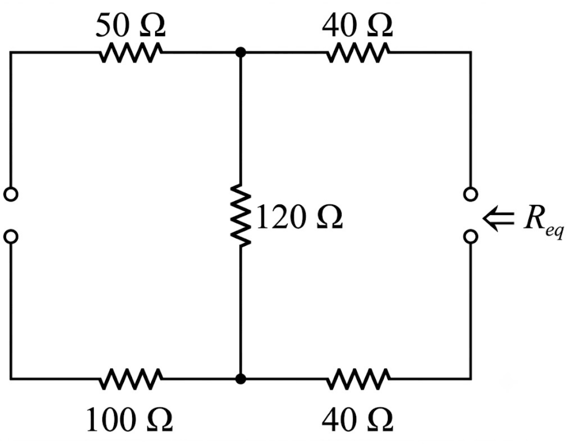


Fig. Solution Q2.25.1

$$R_{eq} = 120 + 40 + 40 = 200\ \Omega$$

$$\tau = \frac{L_{eq}}{R_{eq}} = \frac{0.2}{200} = 1\text{ ms}$$

The current $i(t)$ is

$$i(t) = i(\infty) + \left[i(0^+) - i(\infty)\right]e^{-t/\tau}$$

$$i(t) = 4 + \left[10 - 4\right]e^{-t/\tau} = 4 + 6e^{-t/\tau}$$

$$i(t) = 4 + 6e^{-t/(1\times10^{-3})} = 4 + 6e^{-1000t}$$

At $t = 2$ ms,

$$i(2\text{ ms}) = 4 + 6e^{-1000(2\times10^{-3})}$$

$$i(2\text{ ms}) = 4 + 6e^{-2} = 4.82\text{ mA}$$

Hence,

$$\boxed{4.82}$$

Q2.24.1 NAT | 2024 | 1M | Ans: 15 | ● ● ○

Given circuit is shown below.

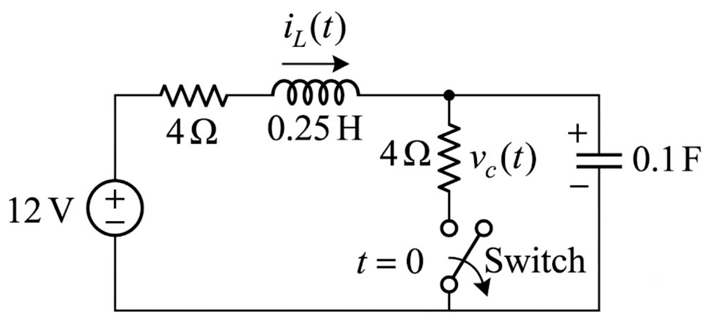


Fig. Solution Q2.24.1

(i) At $t = 0^-$ / $t < 0$ / steady state
Initially, the switch was closed.
In steady state, the capacitor behaves as an open circuit and the inductor behaves as a short circuit.

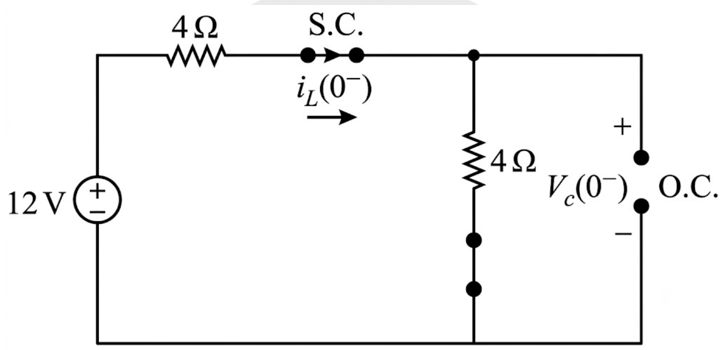


Fig. Solution Q2.24.1

Applying KVL,

$$(4 + 4)i_L(0^-) = 12$$

$$i_L(0^-) = \frac{12}{8} = 1.5 \text{ A}$$

$$V_C(0^-) = 4 \times i_L(0^-) = 4 \times 1.5 = 6 \text{ V}$$

(ii) At $t = 0^+$

- Inductor is replaced by a current source with initial value.

$$i_L(0^-) = i_L(0^+) = 1.5 \text{ A}$$

- Capacitor is replaced by a voltage source with initial value.

$$V_C(0^-) = V_C(0^+) = 6 \text{ V}$$

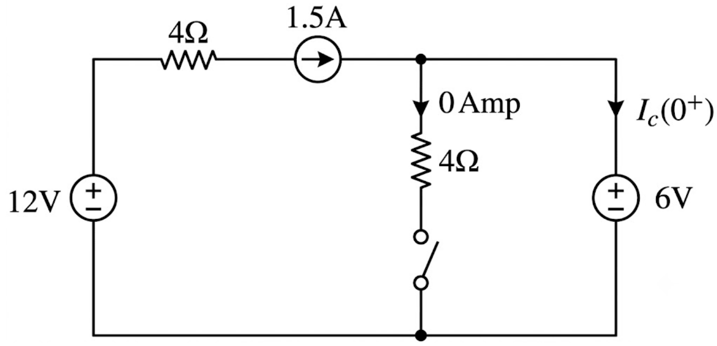


Fig. Solution Q2.24.1

$$I_C(0^+) = 1.5 \text{ A}$$

Current through capacitor is given by

$$I_C(t) = C\frac{dV_C(t)}{dt}$$

At $t = 0^+$,

$$I_C(0^+) = C\frac{d}{dt}\left[V_C(0^+)\right]$$

$$\frac{dV_C(0^+)}{dt} = \frac{I_C(0^+)}{C} = \frac{1.5}{0.1} = 15 \text{ V/s}$$

Hence,

$$\boxed{15}$$

Q2.23.1 NAT 2023 2M Ans: 0.44 ● ● ○

Given:

$$I_L(0^-) = \frac{I_0}{5}, \qquad R = 10 \text{ k}\Omega, \qquad L = 1 \text{ mH}$$

At $t \to \infty$, the inductor acts as a short circuit.

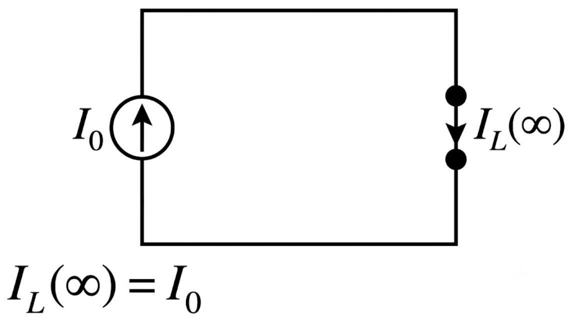


Fig. Solution Q2.23.1

$$I_L(\infty) = I_0$$

Time constant,

$$\tau = \frac{L}{R} = \frac{10^{-3}}{10^4} = 10^{-7}\ \text{s}$$

Current through the inductor is given by

$$I_L(t) = I_L(\infty) + \left[I_L(0^-) - I_L(\infty)\right]e^{-t/\tau}$$

$$I_L(t) = I_0 + \left[\frac{I_0}{5} - I_0\right]e^{-t/10^{-7}}$$

$$I_L(t) = I_0 - \frac{4I_0}{5}e^{-10^7 t}$$

For minimum time taken by current to reach 99% of its final value,

$$I_L(t) = 0.99I_0$$

$$0.99I_0 = I_0 - \frac{4I_0}{5}e^{-10^7 t}$$

$$\frac{4}{5}e^{-10^7 t} = 0.01$$

$$e^{-10^7 t} = 0.0125$$

$$t = \frac{\ln(0.0125)}{-10^7} = 0.438\ \mu\text{s}$$

Hence,

$$\boxed{0.44}$$

Q2.22.1 NAT | 2022 | 2M | Ans: -10.0 | ● ● ●

Given,

$$V_C(0^-) = V_0 = 10\ \text{V}, \qquad i_L(0^-) = 0$$

Switch is closed at $t = 0$.

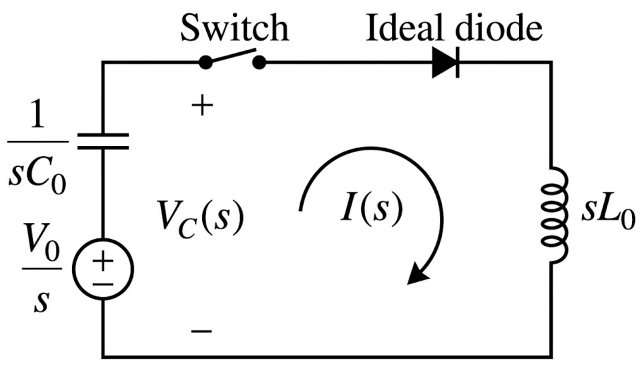


Fig. Solution Q2.22.1

Assuming diode is forward biased, KVL gives

$$\frac{V_0}{s} - sL_0 I(s) - \frac{I(s)}{C_0 s} = 0$$

$$I(s)\left(sL_0 + \frac{1}{C_0 s}\right) = \frac{V_0}{s}$$

$$I(s) = \frac{V_0 C_0}{L_0 C_0 s^2 + 1} = \frac{V_0}{L_0\left(s^2 + \frac{1}{L_0 C_0}\right)} \tag{i}$$

The capacitor voltage is

$$V_C(s) = \frac{V_0}{s} - \frac{I(s)}{C_0 s} = \frac{V_0}{s} - \frac{V_0}{L_0 C_0}\left[\frac{1}{s\left(s^2 + \frac{1}{L_0 C_0}\right)}\right] \tag{ii}$$

Using

$$\frac{1}{s\left(s^2 + \frac{1}{L_0 C_0}\right)} = \frac{L_0 C_0}{s} - \frac{sL_0 C_0}{s^2 + \frac{1}{L_0 C_0}},$$

equation (ii) becomes

$$V_C(s) = \frac{V_0 s}{s^2 + \frac{1}{L_0 C_0}} = V_0\left[\frac{s}{s^2 + \left(\frac{1}{\sqrt{L_0 C_0}}\right)^2}\right].$$

Hence,

$$V_C(t) = V_0 \cos\left(\frac{t}{\sqrt{L_0 C_0}}\right) = V_0 \cos(\omega_0 t), \qquad \omega_0 = \frac{1}{\sqrt{L_0 C_0}}.$$

Since the diode conducts only during the positive half cycle,

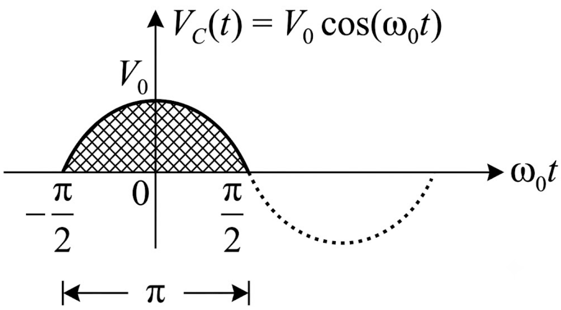


Fig. Solution Q2.22.1

$$\omega_0 t = \pi \;\Rightarrow\; t = \pi\sqrt{L_0 C_0} = \pi\sqrt{(1 \times 10^{-3})(10 \times 10^{-6})} = 0.1\pi \text{ ms}.$$

Therefore,

$$V_C = V_0 \cos\pi = -V_0 = -10 \text{ V}.$$

The diode now becomes reverse biased.

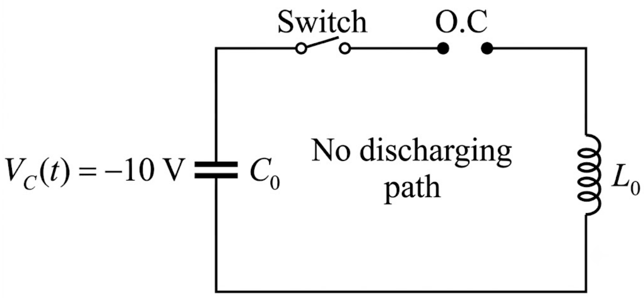


Fig. Solution Q2.22.1

Since there is no discharge path, the capacitor continues to hold -10 V.
As

$$0.5\text{ s} \gg 0.1\pi\text{ ms},$$

$$V_C(0.5\text{ s}) = -10\text{ V}.$$

$$\boxed{-10.0}$$

Q2.19.1 NAT 2019 1M Ans: 4

Given,

$$Q_1 = 6\text{ C}, \qquad Q_2 = 0\text{ C}, \qquad C_1 = 1\text{ F}, \qquad C_2 = 2\text{ F}$$

In steady state, the voltage across both capacitors is the same.

$$V_{C_1} = V_{C_2} = \frac{V_1C_1 + V_2C_2}{C_1 + C_2} \tag{i}$$

From the given data,

$$Q_1 = C_1V_1$$

$$6 = 1 \times V_1$$

$$V_1 = 6\text{ V} \tag{ii}$$

Also,

$$Q_2 = C_2V_2$$

$$0 = 2 \times V_2$$

$$V_2 = 0 \tag{iii}$$

Substituting (ii) and (iii) into (i),

$$V_{C_1} = V_{C_2} = \frac{6 \times 1 + 0 \times 2}{1 + 2} = \frac{6}{3} = 2\text{ V}$$

Hence,

$$Q_{2(\text{steady state})} = C_2 V_{C_2} = 2 \times 2 = 4\ \text{C}$$

Therefore,

$$\boxed{4}$$

Q2.19.2 NAT 2019 2M Ans: 1.90

Given: Step input = 5 V, $V_C(0^-) = 0$ V, Find: $V_P(6\ \mu s)$

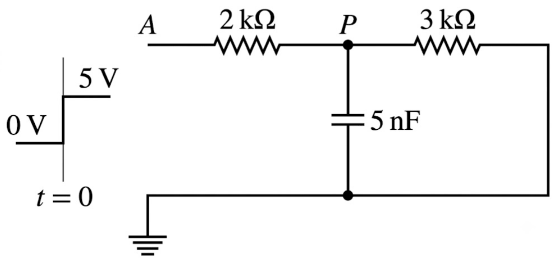


Fig. Solution Q2.19.2

For an RC circuit,

$$V_C(t) = V(\infty) + \left[V(0^+) - V(\infty)\right] e^{-t/\tau}$$

Since the capacitor is initially uncharged,

$$V_C(0^+) = 0\ \text{V}$$

At $t \to \infty$, the capacitor acts as an open circuit.

$$V_C(\infty) = 5\left(\frac{3}{2+3}\right) = 3\ \text{V}$$

Thus,

$$V_C(\infty) = 3\ \text{V}$$

To find the time constant, deactivate the source and find the resistance seen by the capacitor.

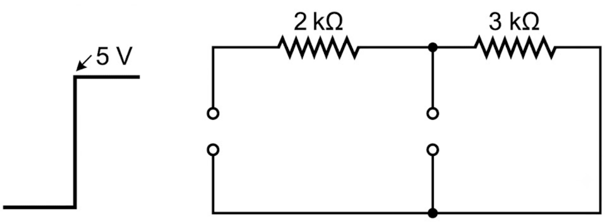


Fig. Solution Q2.19.2

$$R_{eq} = 2 \parallel 3 = \frac{6}{5}\ \text{k}\Omega$$

$$\tau = RC = \frac{6}{5} \times 10^3 \times 5 \times 10^{-9} = 6\ \mu s$$

Therefore,

$$V_C(t) = 3 + (0-3)e^{-t/(6\times10^{-6})} = 3\left(1 - e^{-t/(6\times10^{-6})}\right)$$

At $t = 6\ \mu s$,

$$V_C(6\ \mu s) = 3\left(1 - e^{-1}\right)$$

$$V_C(6\ \mu s) \approx 1.90\ \text{V}$$

Hence, the voltage at node P at $t = 6\ \mu s$ is

$$\boxed{1.90}$$

Q2.17.1 NAT | 2017 | 2M | Ans: 1.527

Given circuit is shown below.

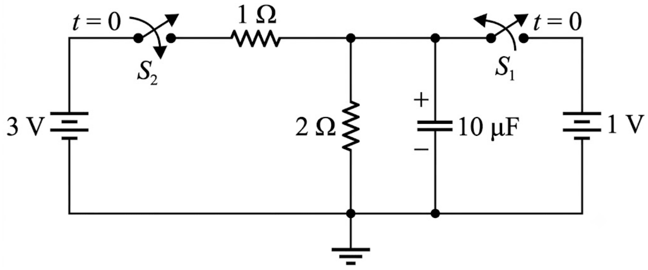


Fig. Solution Q2.17.1

The voltage across the capacitor is

$$V_C(t) = V_C(\infty) + \left[V_C(0^+) - V_C(\infty)\right]e^{-t/\tau}, \qquad t \geq 0 \tag{i}$$

(i) At $t = 0^-$ / $t < 0$ / steady state

In steady state, the capacitor behaves as an open circuit.

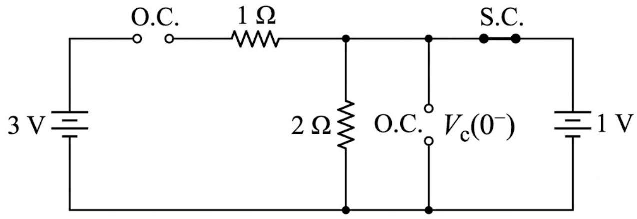


Fig. Solution Q2.17.1

$$V_C(0^-) = 1\ \text{V}$$

From the property of capacitor,

$$V_C(0^-) = V_C(0^+) = 1\ \text{V}$$

(ii) At $t \geq 0$ (transient)

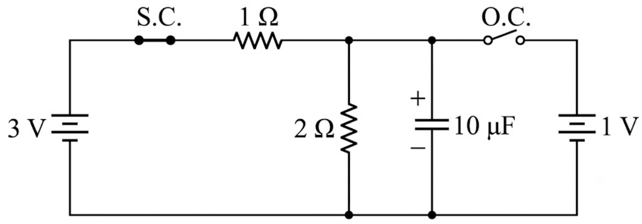


Fig. Solution Q2.17.1

For the RC network,

$$\tau = R_{Th}C$$

where R_{Th} is the Thevenin resistance seen by the capacitor when all independent voltage sources are short-circuited.

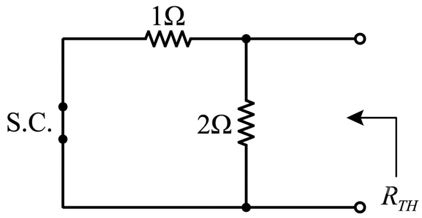


Fig. Solution Q2.17.1

$$R_{Th} = 1 \parallel 2 = \frac{1 \times 2}{1+2} = \frac{2}{3}\ \Omega$$

$$\tau = \frac{2}{3} \times 10 \times 10^{-6} = \frac{20}{3}\ \mu s$$

(iii) At $t = \infty$ / steady state

In steady state, the capacitor behaves as an open circuit.

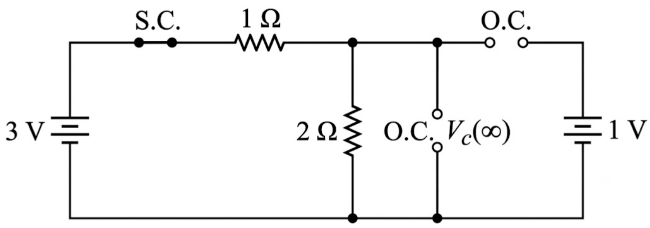


Fig. Solution Q2.17.1

$$V_C(\infty) = 3\left(\frac{2}{2+1}\right) = 2\ \text{V}$$

Using (i),

$$V_C(t) = 2 + (1-2)e^{-t/\tau} = 2 - e^{-t/\tau}$$

Substituting $\tau = \dfrac{20}{3}\ \mu s$,

$$V_C(t) = 2 - e^{-\frac{3\times10^6}{20}t}$$

At $t = 5\ \mu s$,

$$V_C(5\ \mu s) = 2 - e^{-\frac{3\times10^6}{20}(5\times10^{-6})}$$

$$V_C(5\ \mu s) = 1.527\ \text{V}$$

Hence,

$$\boxed{1.527}$$

Q2.17.2 NAT 2017 2M Ans: 1

Given,

t in s	$i(t)$ in A
0	0
0.25	0.197
0.5	0.316
0.75	0.388
1.0	0.432
⋮	⋮
∞	0.5

According to the question,

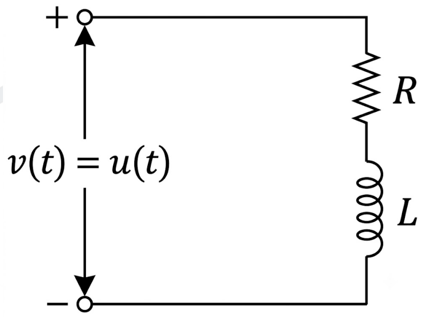


Fig. Solution Q2.17.2

The current through the inductor is

$$i(t) = i(\infty) + \left[i(0^+) - i(\infty)\right]e^{-t/\tau} \quad \text{(i)}$$

(i) At $t = 0^-$ / steady state
In steady state, the inductor behaves as a short circuit.

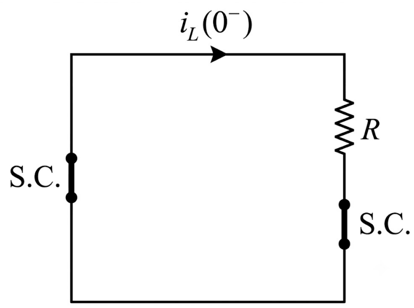


Fig. Solution Q2.17.2

$$i_L(0^-) = 0 \text{ A}$$

From the property of inductor,

$$i_L(0^-) = i_L(0^+) = 0 \text{ A}$$

(ii) At $t > 0$ (transient)
For an RL network,

$$\tau = \frac{L}{R}$$

(iii) At $t = \infty$ / steady state
The inductor behaves as a short circuit.

Fig. Solution Q2.17.2

$$i(\infty) = \frac{1}{R} \text{ A}$$

Substituting $i(\infty)$, $i(0^+)$ and τ into (i),

$$i(t) = \frac{1}{R} - \frac{e^{-t/\tau}}{R} = \frac{1 - e^{-Rt/L}}{R}$$

From the table,

$$i(\infty) = \frac{1}{R} = 0.5$$

$$R = 2\ \Omega$$

Also,

$$i(0.25) = 0.197$$

$$0.197 = \frac{1 - e^{-\frac{2}{L}(0.25)}}{2}$$

Solving,

$$L = 1\text{ H}$$

Hence,

$$\boxed{1}$$

Q2.16.1 NAT 2016 2M Ans: 1.632

Given circuit is shown below.

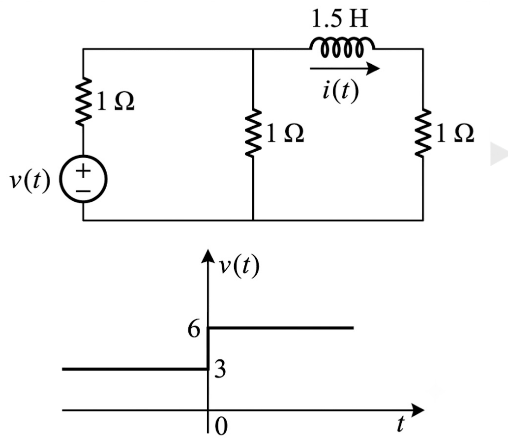


Fig. Solution Q2.16.1

The current through the inductor is

$$i(t) = i(\infty) + \left[i(0^+) - i(\infty)\right]e^{-t/\tau} \qquad \text{(i)}$$

(i) At $t = 0^-$ / $t < 0$ / steady state

In steady state, the inductor behaves as a short circuit.

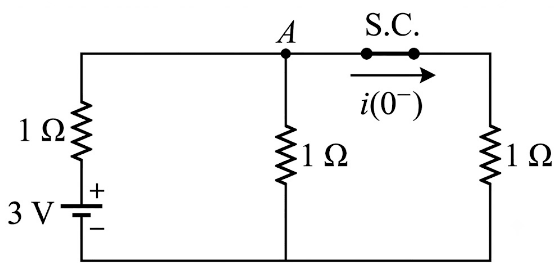


Fig. Solution Q2.16.1

Applying KCL at node A,

$$\frac{V_A}{1} + \frac{V_A - 3}{1} + \frac{V_A}{1} = 0$$

$$V_A = 1\text{ V}$$

From the figure,

$$i(0^-) = \frac{V_A}{1} = 1\text{ A}$$

Using the property of an inductor,

$$i(0^-) = i(0^+) = 1\text{ A}$$

(ii) At $t \geq 0$ (transient)
For an RL network,

$$\tau = \frac{L}{R_{Th}}$$

where R_{Th} is the Thevenin resistance seen by the inductor when the voltage source is short-circuited.

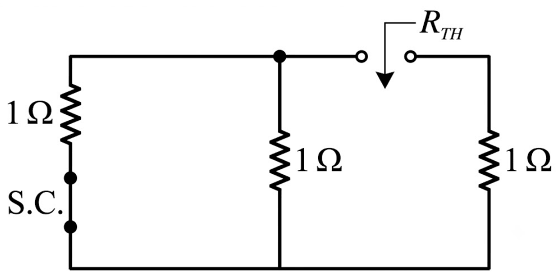


Fig. Solution Q2.16.1

$$R_{Th} = 1 \parallel 1 + 1 = \frac{1 \times 1}{1 + 1} + 1 = \frac{3}{2}\ \Omega$$

$$\tau = \frac{L}{R_{Th}} = \frac{3/2}{3/2} = 1\text{ s}$$

(iii) At $t = \infty$ / steady state
In steady state, the inductor behaves as a short circuit.

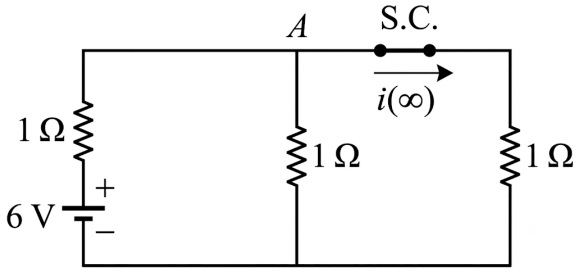


Fig. Solution Q2.16.1

Applying KCL at node A,

$$\frac{V_A}{1} + \frac{V_A - 6}{1} + \frac{V_A}{1} = 0$$

$$V_A = 2\text{ V}$$

Therefore,

$$i(\infty) = \frac{V_A}{1} = 2 \text{ A}$$

Substituting $i(\infty)$, $i(0^+)$ and τ into (i),

$$i(t) = 2 + (1-2)e^{-t} = 2 - e^{-t}$$

At $t = 1$ s,

$$i(1) = 2 - e^{-1}$$

$$i(1) = 1.632 \text{ A}$$

Hence,

$$\boxed{1.632}$$

Q2.16.2 NAT 2016 1M Ans: 8

According to the question,

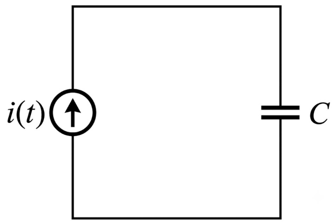


Fig. Solution Q2.16.2

where,

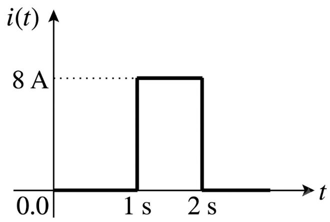


Fig. Solution Q2.16.2

$$i(t) = 8\left[u(t-1) - u(t-2)\right] \text{ A}$$

Transform domain

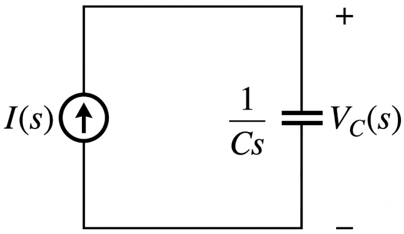


Fig. Solution Q2.16.2

where,

$$I(s) = \frac{8}{s}\left[e^{-s} - e^{-2s}\right]$$

Since $C = 1$ F,

$$V_C(s) = I(s)\frac{1}{Cs} = I(s)\frac{1}{s}$$

$$V_C(s) = \frac{8}{s^2}\left[e^{-s} - e^{-2s}\right]$$

Applying inverse Laplace transform,

$$V_C(t) = 8(t-1)u(t-1) - 8(t-2)u(t-2)$$

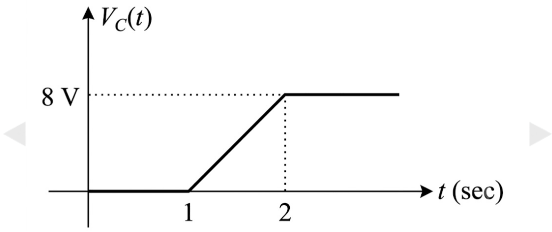


Fig. Solution Q2.16.2

At $t = 2$,

$$V_C(2) = 8 \times 1 = 8 \text{ V}$$

Hence, the voltage across the capacitor for $t > 2$ s is

$$\boxed{8}$$

Q2.15.1 NAT | 2015 | 1M | Ans: 3.678 | ● ○ ○

Given,

$$V_C(0^-) = 10 \text{ V}$$

Given circuit is shown below.

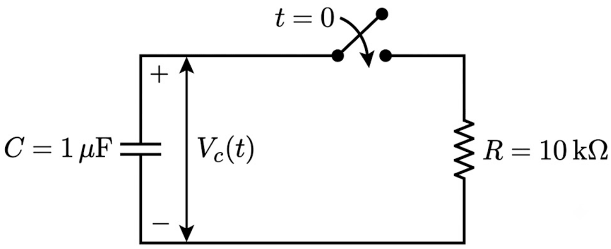


Fig. Solution Q2.15.1

The voltage across the capacitor is

$$V_C(t) = V_C(\infty) + \left[V_C(0^+) - V_C(\infty)\right]e^{-t/\tau} \qquad \text{(i)}$$

From the property of capacitor,

$$V_C(0^-) = V_C(0^+) = 10 \text{ V}$$

(i) Calculation of time constant
For an RC network,

$$\tau = RC$$

$$\tau = 10 \times 10^3 \times 1 \times 10^{-6} = 10 \text{ ms}$$

(ii) At $t = \infty$ / steady state
In steady state, the capacitor behaves as an open circuit.

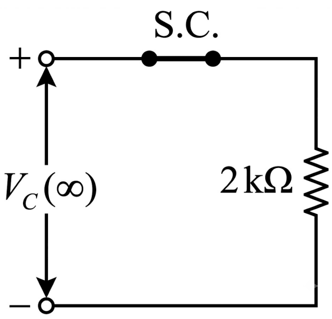


Fig. Solution Q2.15.1

$$V_C(\infty) = 0 \text{ V}$$

Substituting $V_C(\infty)$, $V_C(0^+)$ and τ into (i),

$$V_C(t) = 0 + (10 - 0)e^{-t/(10\times10^{-3})}$$

$$V_C(t) = 10e^{-t/(10\times10^{-3})} \text{ V}$$

At $t = 10$ ms,

$$V_C(10 \text{ ms}) = 10e^{-\frac{10\times10^{-3}}{10\times10^{-3}}}$$

$$V_C(10 \text{ ms}) = \frac{10}{e} = 3.678 \text{ V}$$

Hence,

$$\boxed{3.678}$$

Q2.14.1 | NAT | 2014 | 2M | Ans: 1.516 | ● ● ○

Given circuit is shown below.

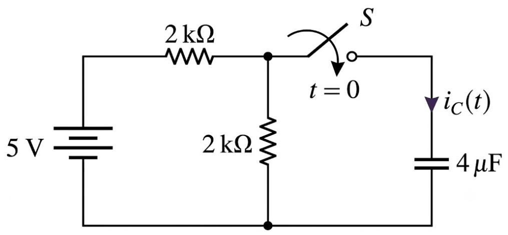


Fig. Solution Q2.14.1

The current through the capacitor is

$$i_C(t) = i_C(\infty) + \left[i_C(0^+) - i_C(\infty)\right]e^{-t/\tau} \quad \text{(i)}$$

(i) At $t = 0^-$ / $t < 0$ / steady state

In steady state, the capacitor behaves as an open circuit.

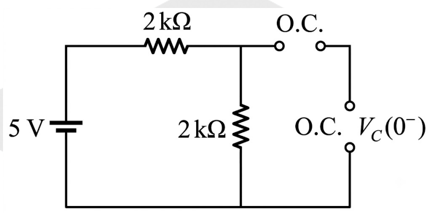


Fig. Solution Q2.14.1

From the figure,

$$V_C(0^-) = 0$$

Using the property of capacitor,

$$V_C(0^-) = V_C(0^+) = 0$$

(ii) At $t = 0^+$

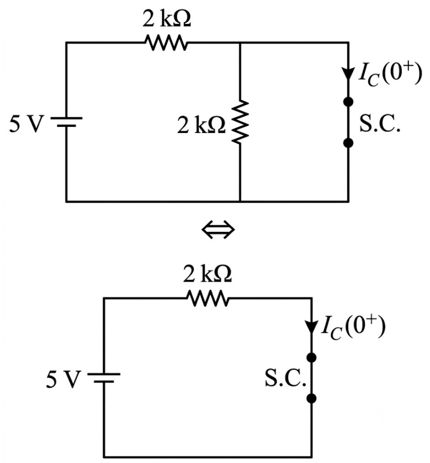


Fig. Solution Q2.14.1

At $t = 0^+$, the capacitor behaves as a short circuit.

$$i_C(0^+) = \frac{5}{2\text{ k}\Omega} = 2.5\text{ mA}$$

(iii) At $t \geq 0$ (transient)

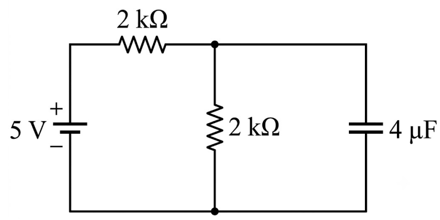


Fig. Solution Q2.14.1

For an RC network,

$$\tau = R_{Th}C$$

where R_{Th} is the Thevenin resistance seen by the capacitor with the voltage source short-circuited.

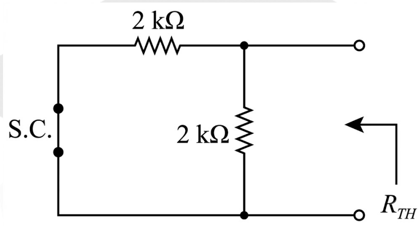


Fig. Solution Q2.14.1

$$R_{Th} = 2 \parallel 2 = 1\text{ k}\Omega$$

$$\tau = RC = 1\times10^{3}\times4\times10^{-6} = 4\text{ ms}$$

(iv) At $t = \infty$ / steady state
In steady state, the capacitor behaves as an open circuit.

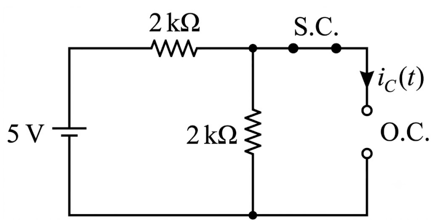


Fig. Solution Q2.14.1

$$i_C(\infty) = 0$$

Substituting $i_C(\infty)$, $i_C(0^+)$ and τ into (i),

$$i_C(t) = 2.5\,e^{-t/(4\times10^{-3})}\text{ mA}$$

At $t = 2$ ms,

$$i_C(2\text{ ms}) = 2.5\,e^{-\frac{2\times10^{-3}}{4\times10^{-3}}}$$

$$i_C(2\text{ ms}) = 1.516\text{ mA}$$

Hence,

$$\boxed{1.516}$$

Q2.14.2 NAT 2014 1M Ans: -30

Given circuit is shown below.

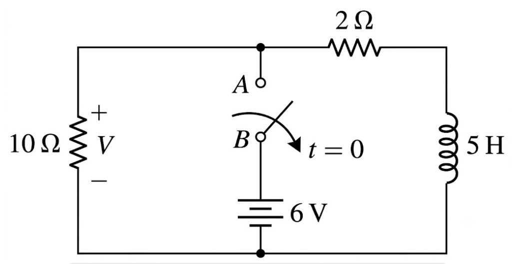


Fig. Solution Q2.14.2

(i) At $t = 0^-$ / $t < 0$ / steady state

In steady state, the inductor behaves as a short circuit.

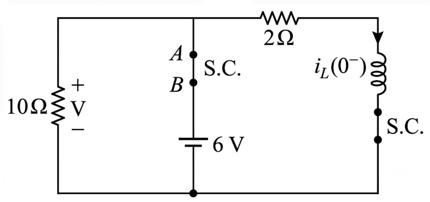


Fig. Solution Q2.14.2

$$i_L(0^-) = \frac{6}{2} = 3\text{ A}$$

From the property of inductor,

$$i_L(0^-) = i_L(0^+) = 3\text{ A}$$

(ii) At $t = 0^+$

The inductor is replaced by a current source of value 3 A.

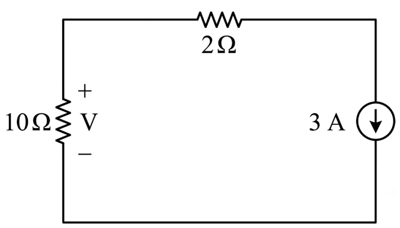


Fig. Solution Q2.14.2

$$V = -3 \times 10 = -30\text{ V}$$

Hence, the voltage across the 10 Ω resistor at $t = 0^+$ is

$$\boxed{-30}$$

Q2.11.1 MCQ 2011 1M Ans: D

Given circuit is shown below.

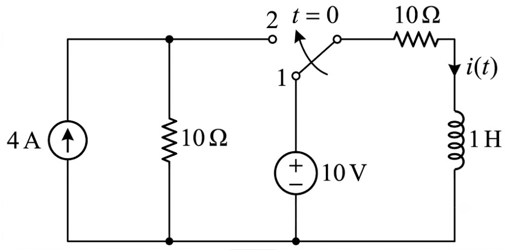


Fig. Solution Q2.11.1

The current through the inductor is

$$i_L(t) = i(\infty) + \left[i(0^+) - i(\infty)\right] e^{-t/\tau} \tag{i}$$

(i) At $t = 0^-$ / $t < 0$ / steady state
In steady state, the inductor behaves as a short circuit.
At $t = 0^-$, the switch is in position 1.

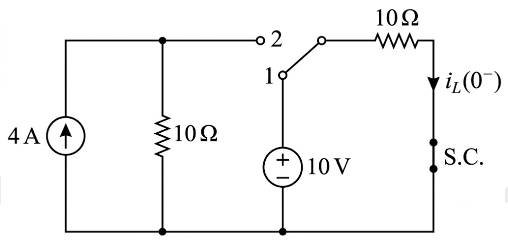


Fig. Solution Q2.11.1

$$i_L(0^-) = \frac{10}{10} = 1 \text{ A}$$

From the property of inductor,

$$i_L(0^-) = i_L(0^+) = 1 \text{ A}$$

(ii) At $t \geq 0$ (transient)
For an RL network,

$$\tau = \frac{L}{R_{Th}}$$

where R_{Th} is the Thevenin resistance seen across L when the voltage source is short-circuited and the current source is open-circuited.

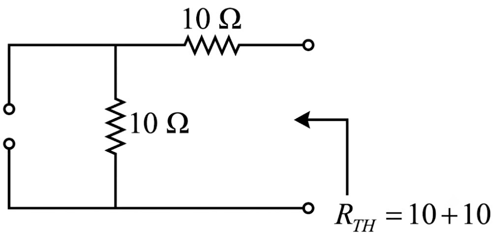


Fig. Solution Q2.11.1

$$R_{Th} = 10 + 10 = 20\ \Omega$$

$$\tau = \frac{1}{20}\ \text{s}$$

(iii) At $t = \infty$ / steady state
In steady state, the inductor behaves as a short circuit.

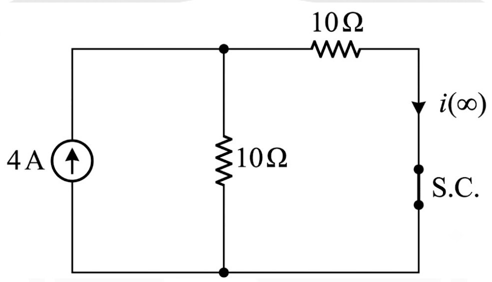


Fig. Solution Q2.11.1

Using current division,

$$i(\infty) = 4\left(\frac{10}{10+10}\right) = 2\ \text{A}$$

Substituting $i(0^+)$, $i(\infty)$ and τ into (i),

$$i_L(t) = 2 + (1-2)e^{-t/(1/20)}$$

$$i_L(t) = 2 - e^{-20t}\ \text{A}$$

Hence,

$$\boxed{\text{(D) } 2 - e^{-20t}\ \text{A}}$$

Q2.10.1 MCQ 2010 1M Ans: B ● ○ ○

At $t = \infty$ / steady state
In steady state, capacitors behave as open circuits.
The equivalent circuit is

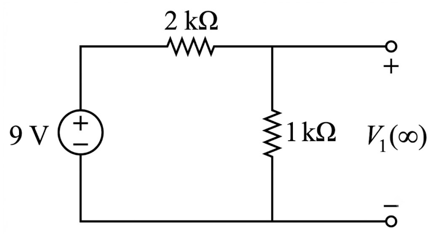


Fig. Solution Q2.10.1

From the figure,

$$V_1(\infty) = 9\left(\frac{1}{1+2}\right)$$

$$V_1(\infty) = 3\text{ V}$$

Considering the capacitor network,

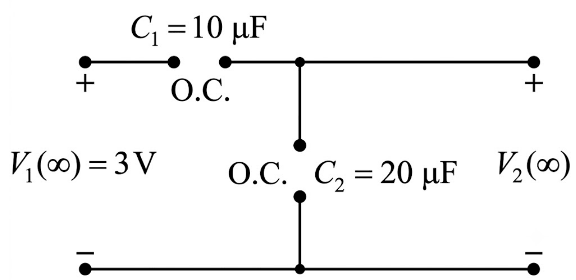


Fig. Solution Q2.10.1

Using capacitive voltage division,

$$V_2 = V_1\left(\frac{C_1}{C_1 + C_2}\right)$$

At $t = \infty$,

$$V_2(\infty) = V_1(\infty)\left(\frac{10}{10+20}\right)$$

$$V_2(\infty) = 3\left(\frac{10}{30}\right)$$

$$V_2(\infty) = 1\text{ V}$$

Hence,

$$\boxed{\text{(B) } 1}$$

Q2.08.1 MCQ 2008 2M Ans: A ● ● ○

Given circuit is shown below.

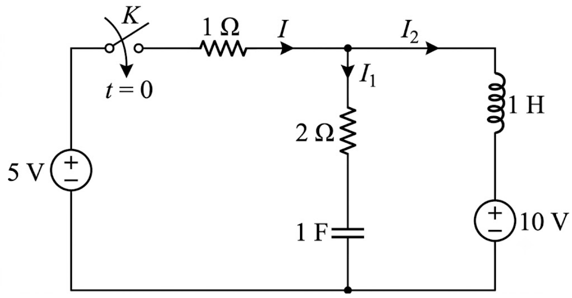


Fig. Solution Q2.08.1

(i) At $t = 0^-$ / $t < 0$ / steady state

In steady state, the inductor behaves as a short circuit and the capacitor behaves as an open circuit.

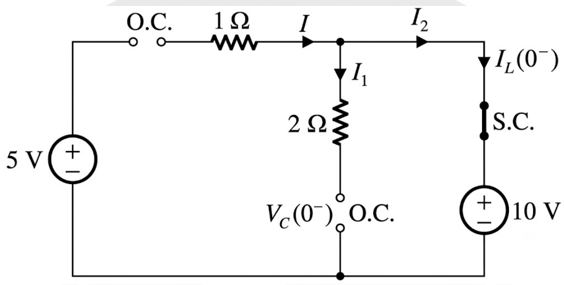


Fig. Solution Q2.08.1

From the figure,

$$V_C(0^-) = 10 \text{ V}, \qquad I_L(0^-) = 0 \text{ A}$$

From the properties of capacitor and inductor,

$$V_C(0^-) = V_C(0^+) = 10 \text{ V}$$

$$I_L(0^-) = I_L(0^+) = 0 \text{ A}$$

(ii) At $t = 0^+$

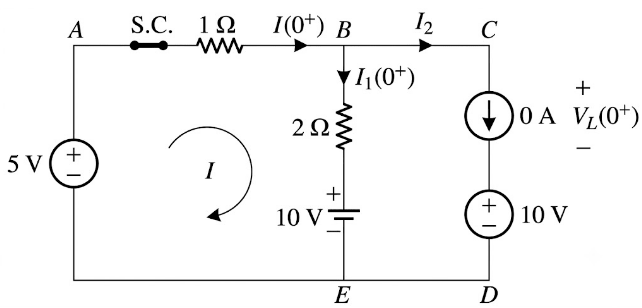


Fig. Solution Q2.08.1

Applying KVL in the loop,

$$I(0^+) = I_1(0^+) = \frac{5-10}{1+2} = -\frac{5}{3}\text{ A}$$

Hence,

$$\boxed{\text{(A) } -\frac{5}{3}\text{ A}}$$

Q2.08.2 MCQ 2008 2M Ans: B

From the previous question,

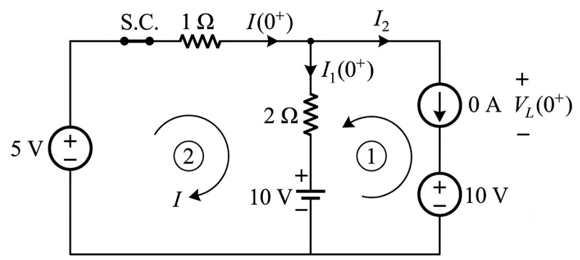


Fig. Solution Q2.08.2

$$I(0^+) = I_1(0^+) = -\frac{5}{3}$$

Since,

$$V_L(t) = L\frac{dI_2}{dt}$$

$$\left.\frac{dI_2}{dt}\right|_{t=0^+} = \frac{V_L(0^+)}{L} \qquad \text{(i)}$$

Applying KVL in loop 1,

$$-10 - V_L(0^+) + 2I_1(0^+) + 10 = 0$$

$$V_L(0^+) = 2I_1(0^+)$$

Substituting $I_1(0^+)$,

$$V_L(0^+) = 2\left(-\frac{5}{3}\right) = -\frac{10}{3}\text{ V}$$

From equation (i),

$$\left.\frac{dI_2}{dt}\right|_{t=0^+} = \frac{-10/3}{1} = -\frac{10}{3}\text{ A/s}$$

Hence,

$$\boxed{\text{(B) } -\frac{10}{3}\text{ A/s}}$$

Q2.07.1 MCQ 2007 2M Ans: C ● ● ○

Given circuit is shown below.

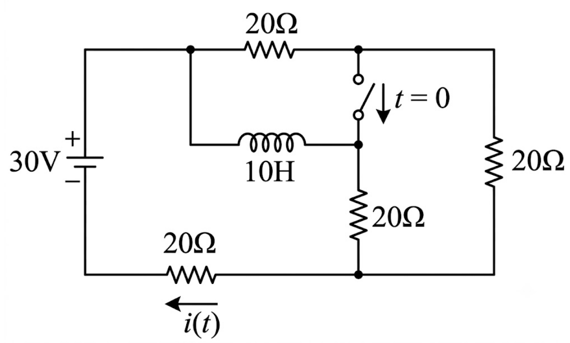


Fig. Solution Q2.07.1

(i) At $t = 0^-$ / $t < 0$ / steady state
In steady state, the inductor behaves as a short circuit.

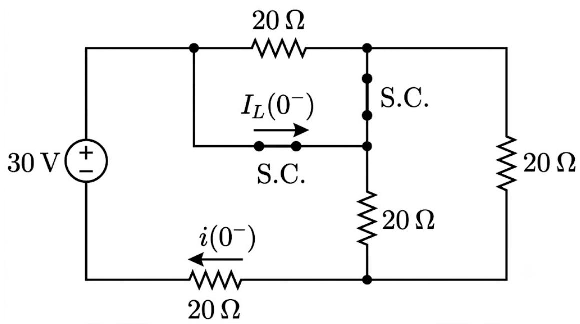


Fig. Solution Q2.07.1

The simplified circuit is

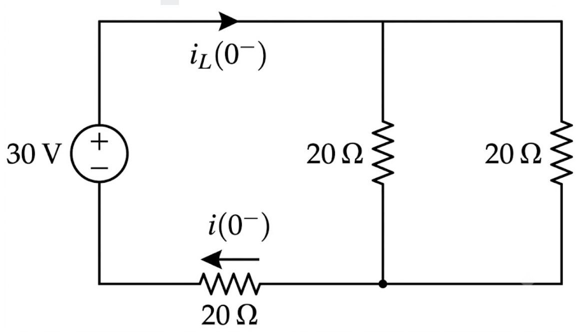


Fig. Solution Q2.07.1

From the figure,

$$i_L(0^-) = i(0^-) = \frac{30}{(20 \parallel 20) + 20} = 1\text{ A}$$

From the property of inductor,

$$i_L(0^-) = i_L(0^+) = 1\text{ A}$$

(ii) At $t = 0^+$

The inductor is replaced by a current source of value 1 A.

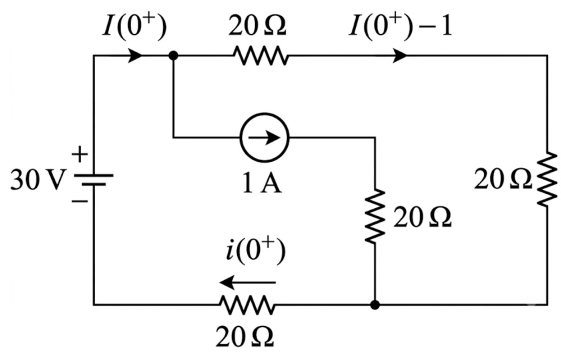


Fig. Solution Q2.07.1

Applying KVL,

$$-30 + 20\left[i(0^+) - 1\right] + 20\left[i(0^+) - 1\right] + 20i(0^+) = 0$$

$$-30 + 60i(0^+) - 40 = 0$$

$$60i(0^+) = 70$$

$$i(0^+) = \frac{7}{6}\text{ A}$$

Hence,

$$\boxed{\text{(C) 1 A and } \frac{7}{6}\text{ A}}$$

Q2.06.1 MCQ | 2006 | 2M | Ans: A

Given circuit is shown below.

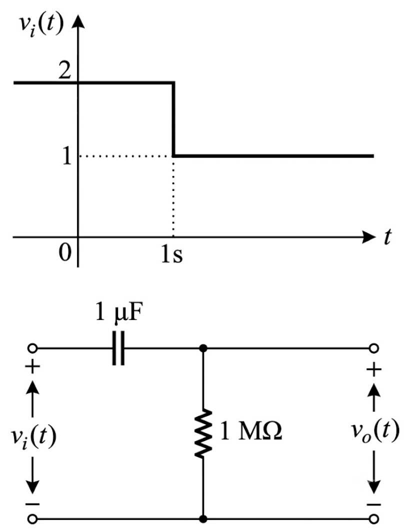


Fig. Solution Q2.06.1

(i) At $t = 1^-$
Due to the supply voltage of 2 V, the capacitor gets fully charged to 2 V.

$$V_C(1^-) = 2 \text{ V}$$

From the property of capacitor,

$$V_C(1^-) = V_C(1^+) = 2 \text{ V}$$

(ii) Calculation of time constant ($t \geq 1$)
For an RC network,

$$R_{Th} = R = 1 \text{ M}\Omega$$

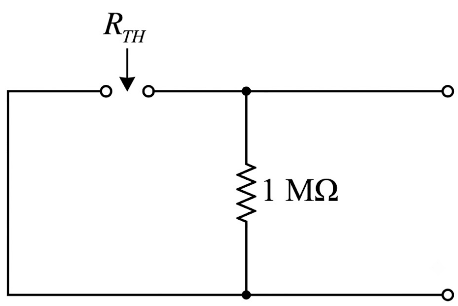


Fig. Solution Q2.06.1

$$\tau = RC = (1 \times 10^6)(1 \times 10^{-6}) = 1 \text{ s}$$

(iii) At $t = 1^+$

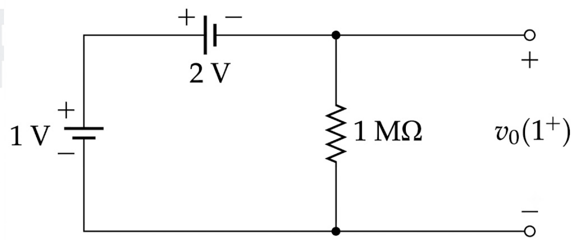


Fig. Solution Q2.06.1

Applying KVL,

$$-1 + 2 + v_o(1^+) = 0$$

$$v_o(1^+) = -1 \text{ V}$$

(iv) At $t = \infty$ / steady state
In steady state, the capacitor behaves as an open circuit.

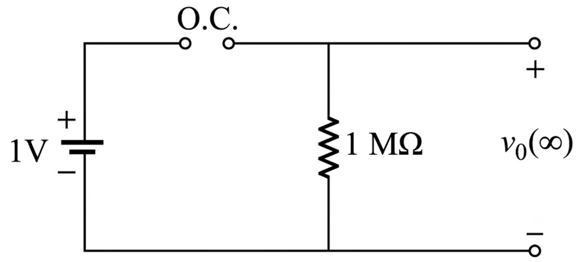


Fig. Solution Q2.06.1

$$v_o(\infty) = 0 \text{ V}$$

The transient equation is

$$v_o(t) = v_o(\infty) + \left[v_o(1^+) - v_o(\infty)\right]e^{-\frac{t-1}{\tau}} \quad \text{(i)}$$

Substituting $v_o(\infty)$, $v_o(1^+)$ and τ into (i),

$$v_o(t) = 0 + [-1 - 0]e^{-(t-1)}$$

$$v_o(t) = -e^{-(t-1)}$$

The output voltage 2 s after the change in input (i.e. at $t = 3$ s) is

$$v_o(3) = -e^{-(3-1)} = -e^{-2} \text{ V}$$

Hence,

$$\boxed{\text{(A)} - \exp(-2) \text{ Volt}}$$

Q2.05.1 MCQ | 2005 | 2M | Ans: B

Given circuit is shown below.

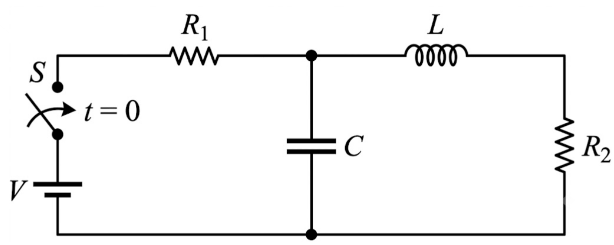


Fig. Solution Q2.05.1

(i) **At $t = 0^-/t < 0$/steady state:**

In steady state, inductor behaves as a short circuit and capacitor behaves as an open circuit.

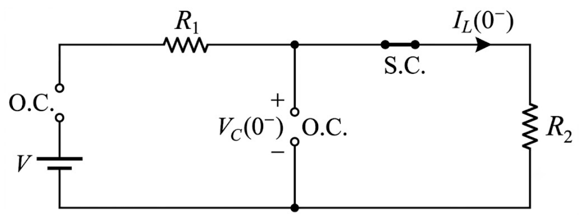


Fig. Solution Q2.05.1

From figure,

$$V_C(0^-) = 0 \text{ V}, \qquad I_L(0^-) = 0 \text{ A}$$

From the property of inductor and capacitor,

$$I_L(0^-) = I_L(0^+) = 0, \qquad V_C(0^-) = V_C(0^+) = 0$$

(ii) **At $t = 0^+$:**

Capacitor behaves as a short circuit and the inductor is replaced by a current source of value $I_L(0^+) = 0$ A.

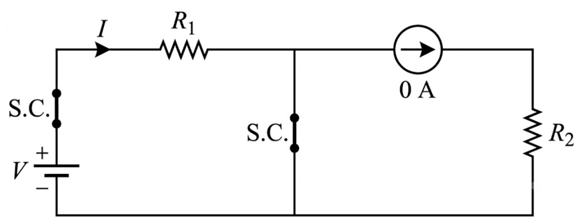


Fig. Solution Q2.05.1

From figure,

$$I(0^+) = \frac{V}{R_1}$$

(iii) **At $t = \infty$/steady state:**

In steady state, inductor behaves as a short circuit and capacitor behaves as an open circuit.

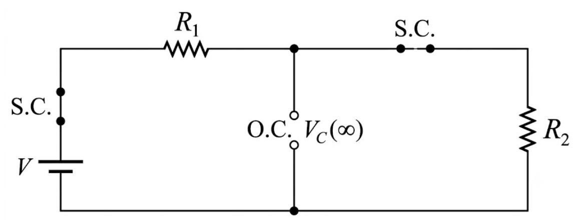


Fig. Solution Q2.05.1

From figure,

$$V_C(\infty) = V\left(\frac{R_2}{R_1 + R_2}\right) \qquad \text{[By VDR]}$$

Hence,

$$\boxed{(B)\ \left(\frac{V}{R_1}, \frac{VR_2}{R_1 + R_2}\right)}$$

Q2.00.1 MCQ | 2000 | 2M | Ans: A | ● ● ○

Given,

$$V_{C_1}(0^-) = 20 \text{ V}, \qquad V_{C_2}(0^-) = 0 \text{ V}$$

Given circuit is shown below.

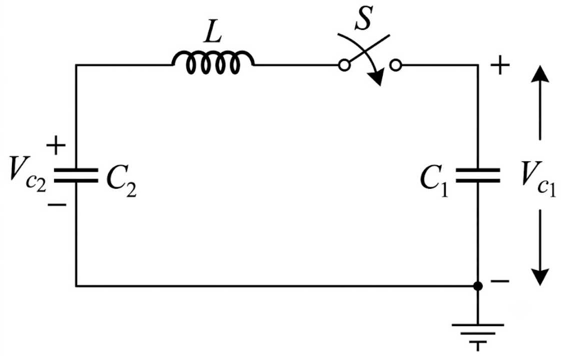


Fig. Solution Q2.00.1

Transform domain:

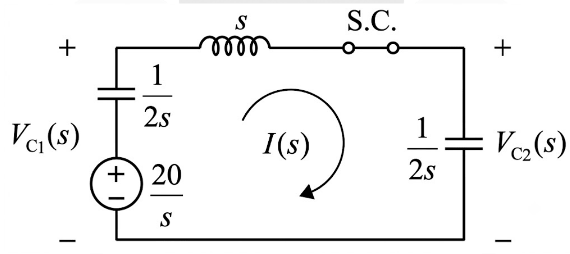


Fig. Solution Q2.00.1

Applying KVL,

$$-\frac{20}{s} + \frac{1}{2s}I(s) + sI(s) + \frac{1}{2s}I(s) = 0$$

$$\frac{20}{s} = I(s)\left[\frac{1}{2s} + s + \frac{1}{2s}\right]$$

$$I(s) = \frac{\frac{20}{s}}{\frac{1}{2s} + \frac{1}{2s} + s} = \frac{20}{s^2 + 1} \tag{i}$$

From figure,

$$-V_{C_1}(s) + \left(s + \frac{1}{2s}\right) I(s) = 0$$

$$V_{C_1}(s) = \left(s + \frac{1}{2s}\right) I(s)$$

Using (i),

$$V_{C_1}(s) = \left(s + \frac{1}{2s}\right)\frac{20}{s^2 + 1} = \frac{10s}{s^2 + 1} + \frac{10}{s}$$

Taking inverse Laplace transform,

$$V_{C_1}(t) = 10(1 + \cos t)\, u(t) \tag{ii}$$

Also,

$$V_{C_2}(s) = I(s)\frac{1}{2s} = \frac{20}{s^2+1} \cdot \frac{1}{2s} = \frac{10}{s} - \frac{10s}{s^2+1}$$

Taking inverse Laplace transform,

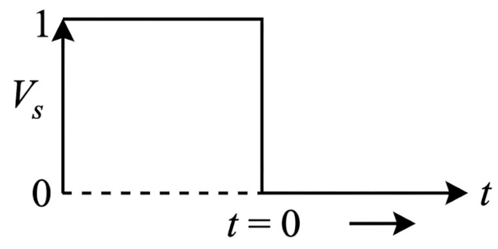
$$V_{C_2}(t) = 10(1 - \cos t)\, u(t) \tag{iii}$$

From (ii) and (iii),

$$\boxed{\text{(A) } V_{C_1}(t) = 10(1+\cos t)u(t), \quad V_{C_2}(t) = 10(1-\cos t)u(t)}$$

Q2.99.1 MCQ | 1999 | 2M | Ans: B

Given circuit is shown below.



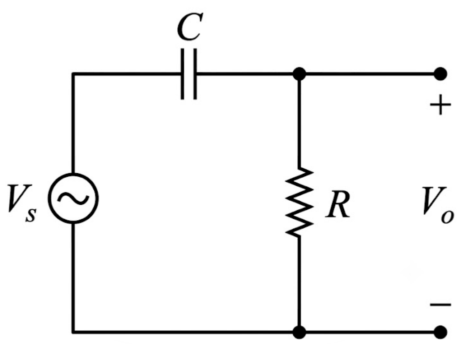


Fig. Solution Q2.99.1

Voltage across R is given by,

$$V_0(t) = V_0(\infty) + \left[V_0(0^+) - V_0(\infty)\right]e^{-t/\tau} \tag{i}$$

(i) **At $t = 0^-/t < 0$/steady state :**

In steady state, capacitor behaves as an open circuit.

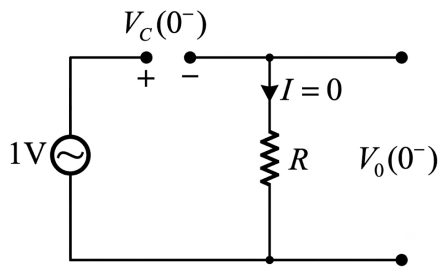


Fig. Solution Q2.99.1

From figure,

$$V_0(0^-) = IR = 0 \times R = 0 \text{ V}$$

$$V_C(0^-) = 1 \text{ V}$$

From the property of capacitor,

$$V_C(0^-) = V_C(0^+) = 1 \text{ V}$$

(ii) **At $t = 0^+$:**

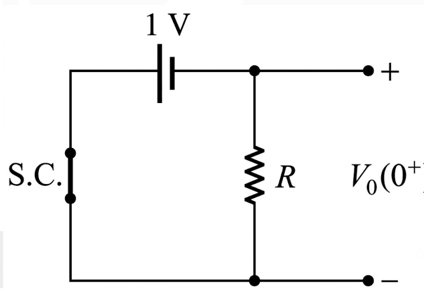


Fig. Solution Q2.99.1

$$V_0(0^+) = -1 \text{ V} \tag{ii}$$

For an RC network,

$$\tau = RC$$

(iii) **At $t = \infty$/steady state :**

In steady state, capacitor behaves as an open circuit.

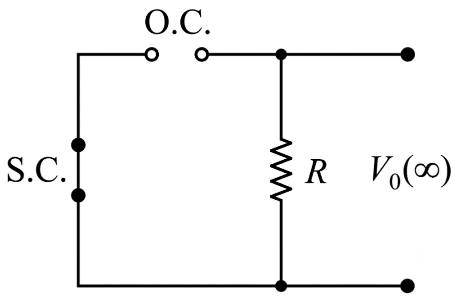


Fig. Solution Q2.99.1

$$V_0(\infty) = 0 \text{ V} \quad \text{(iii)}$$

Substituting (ii) and (iii) into (i),

$$V_0(t) = 0 + \left[-1 - 0\right]e^{-t/RC}$$

$$V_0(t) = -e^{-t/RC}$$

Hence,

$$\boxed{\text{(B)} \ -e^{-t/RC}}$$

Q2.98.1 MCQ 1998 2M Ans: A

Given circuit is shown below.

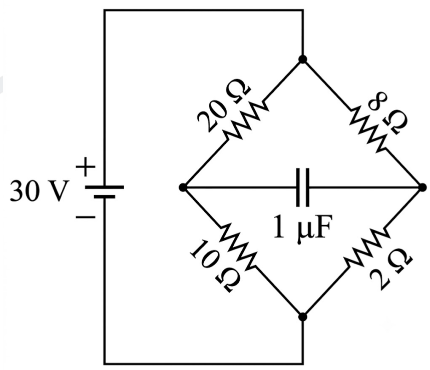


Fig. Solution Q2.98.1

Since the source is DC ($\omega = 0$), the capacitor behaves as an open circuit.

$$X_C = \frac{1}{\omega C} \xrightarrow[\omega \to 0]{} \infty$$

The equivalent circuit is shown below.

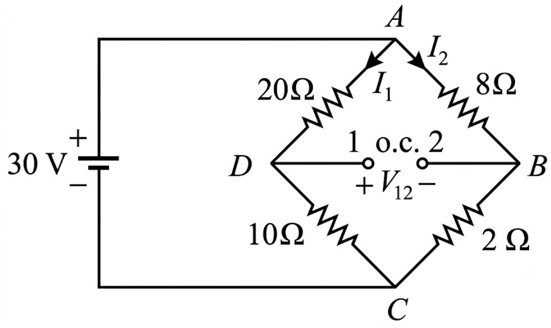


Fig. Solution Q2.98.1

Using current division,

$$I_1 = \frac{30}{20+10} = 1\text{ A} \tag{i}$$

$$I_2 = \frac{30}{8+2} = 3\text{ A} \tag{ii}$$

Voltage across the capacitor,

$$V_{12} = 8I_2 - 20I_1$$

Substituting (i) and (ii),

$$V_{12} = 8 \times 3 - 20 \times 1 = 4\text{ V}$$

Stored energy in the capacitor is,

$$E = \frac{1}{2}CV^2$$

$$E = \frac{1}{2}(1 \times 10^{-6})(4)^2$$

$$E = 8 \times 10^{-6}\text{ J} = 8\ \mu\text{J}$$

Hence,

$$\boxed{\text{(A) } 8\ \mu\text{J}}$$

Q2.95.1 MCQ | 1995 | 2M | Ans: C | ● ○ ○

Given circuit is shown below.

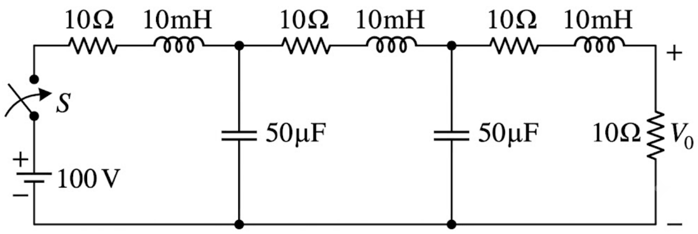


Fig. Solution Q2.95.1

Since the switch is open for $t < 0$,

$$i_{L(10\,\text{mH})}(0^-) = i_{L(10\,\text{mH})}(0^+) = 0$$

$$V_{C(50\,\mu\text{F})}(0^-) = V_{C(50\,\mu\text{F})}(0^+) = 0$$

(i) **At $t = \infty$/steady state :**

In steady state, inductors behave as short circuits and capacitors behave as open circuits.

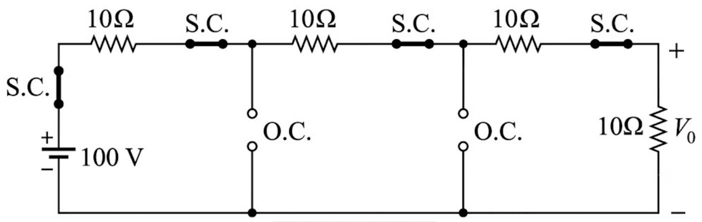


Fig. Solution Q2.95.1

The simplified circuit is shown below.

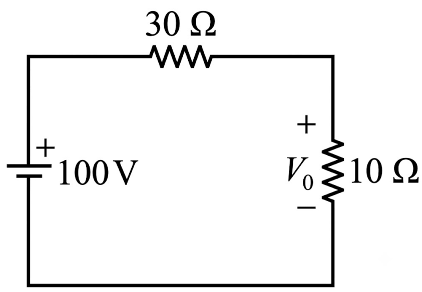


Fig. Solution Q2.95.1

Using voltage division,

$$V_0(\infty) = 100\left(\frac{10}{30+10}\right)$$

$$V_0(\infty) = 25\ \text{V}$$

Hence,

$$\boxed{\text{(C) 25 V}}$$

DEBUG INFO

Chapter II

Transient Analysis

Total Questions : 24

MCQ : 11

MSQ : 0

NAT : 13

SOLUTIONS

III

Network Theorems For DC & Miscellaneous Topics

Topics

1. Source Transformation and Superposition Theorem
2. Thevenin's Theorem
3. Norton's Theorem
4. Concept of Linearity
5. Reciprocity Theorem
6. Maximum Power Transfer Theorem
7. Star-Delta Connections and Conversions
8. Finding Equivalent Resistances
9. Equivalent Resistance of a Cube Across Different Nodes
10. Equivalent Resistance in an Infinite Ladder
11. Bulb-Based Problems

Q3.25.1 NAT | 2025 | 2M | Ans: 9

The Thevenin equivalent seen by the galvanometer is determined first.

(i) Calculation of Thevenin voltage

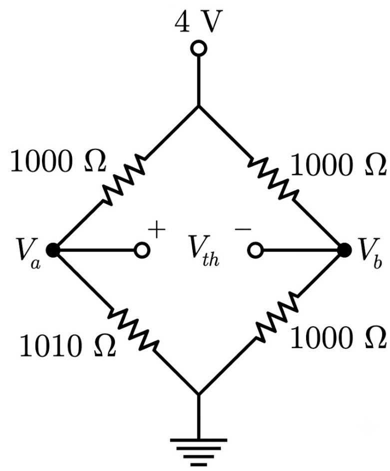


Fig. Solution Q3.25.1

Node voltage V_a is,

$$V_a = \frac{4 \times 1010}{1000 + 1010} = \frac{404}{201}\text{ V}$$

Node voltage V_b is,

$$V_b = \frac{4 \times 1000}{1000 + 1000} = 2\text{ V}$$

Therefore,

$$V_{th} = V_a - V_b = \frac{404}{201} - 2 = \frac{2}{201}\text{ V} \approx 0.01\text{ V}$$

(ii) Calculation of Thevenin resistance

For finding R_{th}, the independent voltage source is short-circuited.

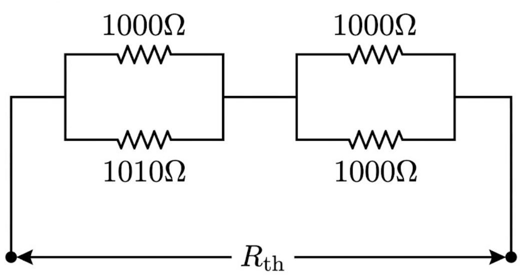


Fig. Solution Q3.25.1

$$R_{th} = (1000 \parallel 1010) + (1000 \parallel 1000)$$

$$R_{th} = \frac{1000 \times 1010}{1000 + 1010} + 500 = 502.49 + 500 = 1002.49\ \Omega$$

The Thevenin equivalent circuit across the galvanometer is

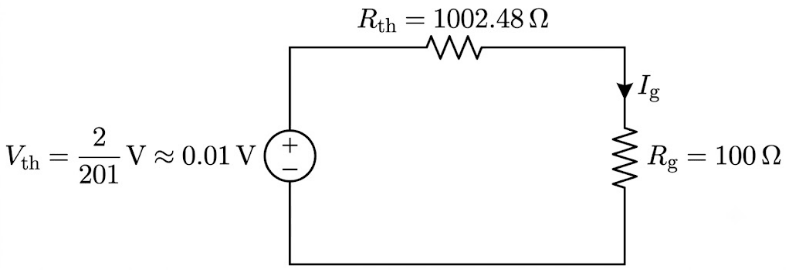


Fig. Solution Q3.25.1

$$I_G = \frac{V_{th}}{R_{th} + R_G} = \frac{\frac{2}{201}}{1002.49 + 100}$$

$$I_G = 9.07 \times 10^{-6}\ \text{A} = 9.07\ \mu\text{A}$$

$$\boxed{I_G \approx 9\ \mu\text{A}}$$

Q3.23.1 NAT 2023 1M Ans: 1

Given circuit is shown below.

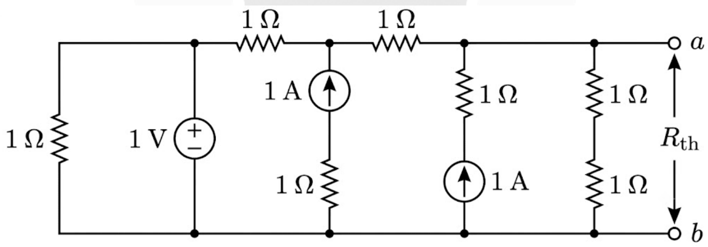


Fig. Solution Q3.23.1

For calculating R_{th}, independent sources are replaced by their internal resistances, i.e., voltage source is replaced by a short circuit and current source by an open circuit.

Hence, the circuit becomes,

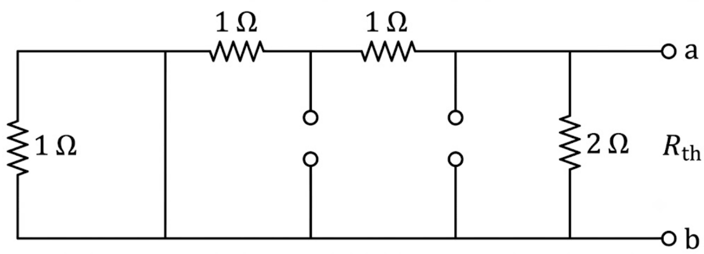


Fig. Solution Q3.23.1

$$R_{th} = 2 \parallel 2$$

$$R_{th} = 1\ \Omega$$

$$\boxed{1}$$

Q3.22.1 NAT 2022 2M Ans: 0

Given circuit A and B is shown in figure below,

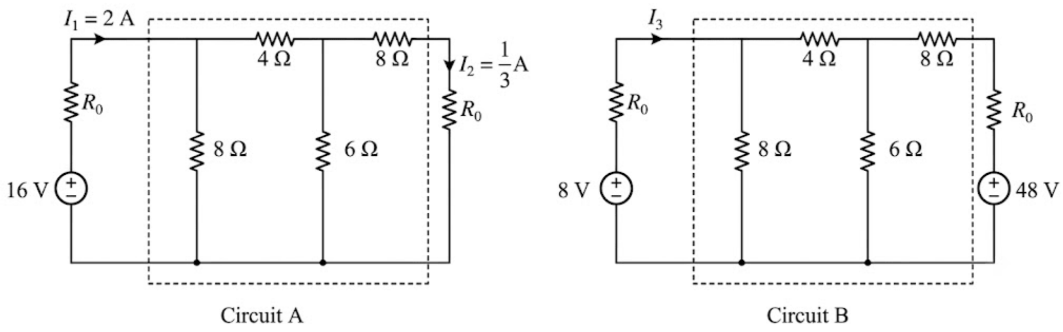


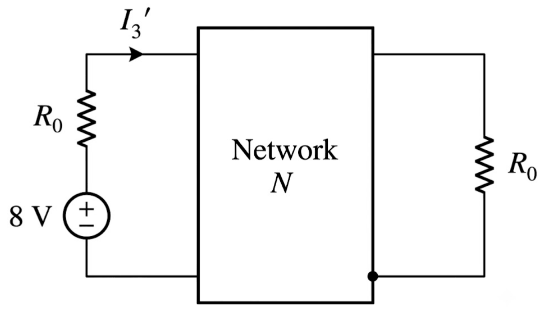
Fig. Solution Q3.22.1

Replacing the dotted part of circuit as shown above by network N and applying superposition theorem to calculate the value of I_3 as,

$$I_3 = I_3 \text{ (due to 8 V source)} + I_3 \text{ (due to 48 V source)}$$

Redrawing circuit B :

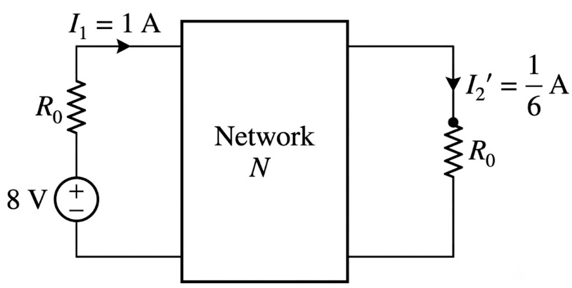
Case 1 : When only 8 V source is active, 48 V source becomes short circuit.



Circuit C

Fig. Solution Q3.22.1

Apply homogeneity property in circuit A (Divide by a factor of 2).



Circuit D

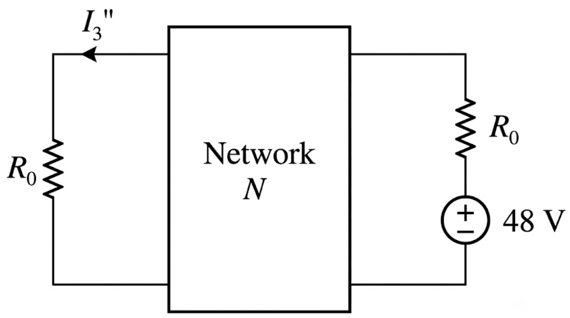
Fig. Solution Q3.22.1

$$I_1' = 1 \text{ A}, \qquad I_2' = \frac{1}{6} \text{ A}$$

Compare circuit D with circuit C, so

$$I_3' = I_1' = 1 \text{ A}$$

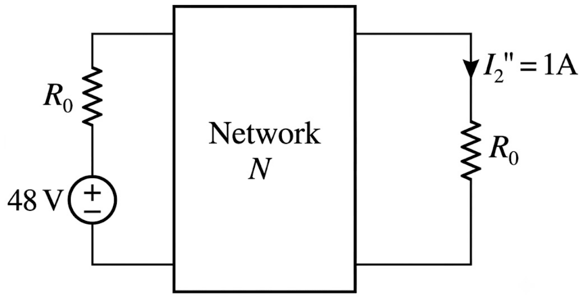
Case 2 : When only 48 V source is active and 8 V source becomes short circuit.



Circuit E

Fig. Solution Q3.22.1

Apply homogeneity property in circuit A (Multiply by a factor of 3).

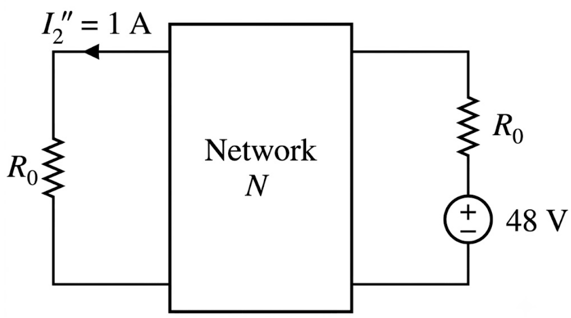


Circuit F

Fig. Solution Q3.22.1

$$I_2'' = 1 \text{ A}$$

Apply reciprocity theorem in above circuit F (Replace excitation and response).



Circuit G

Fig. Solution Q3.22.1

Compare circuit G with circuit E, so

$$I_3'' = -I_2'' = -1 \text{ A}$$

Now, applying superposition theorem to calculate I_3,

$$I_3 = I_3' + I_3''$$

$$I_3 = 1 - 1 = 0 \text{ A}$$

$$\boxed{0}$$

Q3.19.1 NAT | 2019 | 1M | Ans: 5

Given circuit is shown below.

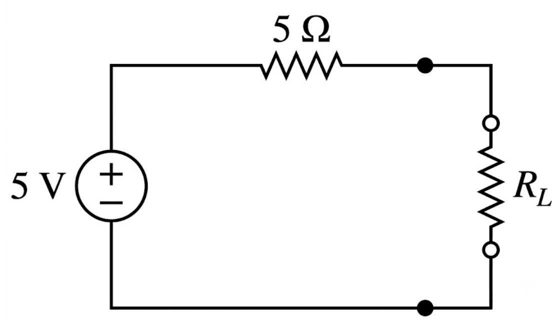


Fig. Solution Q3.19.1

According to the **Maximum Power Transfer Theorem**, for a purely resistive network, maximum power is transferred to the load when the load resistance is equal to the source (Thevenin) resistance.

$$R_L = R_s$$

Here,

$$R_s = 5\ \Omega$$

Therefore,

$$R_L = 5\ \Omega$$

$$\boxed{5}$$

Q3.18.1 MCQ | 2018 | 1M | Ans: C

Given circuit is shown below,

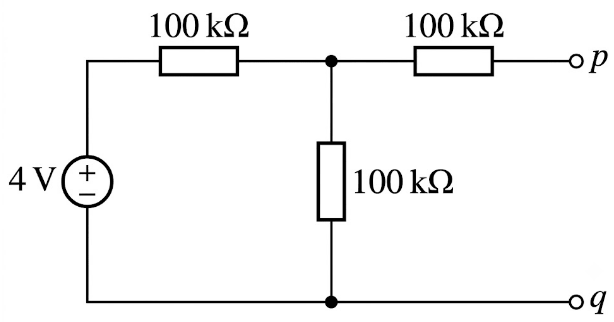


Fig. Solution Q3.18.1

Thevenin's equivalent circuit :

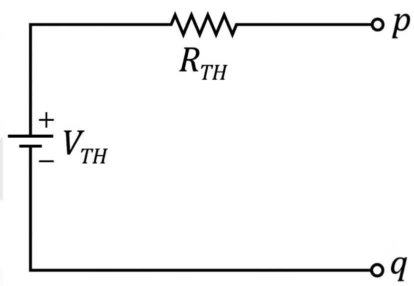


Fig. Solution Q3.18.1

1. **Calculation of V_{TH} :**

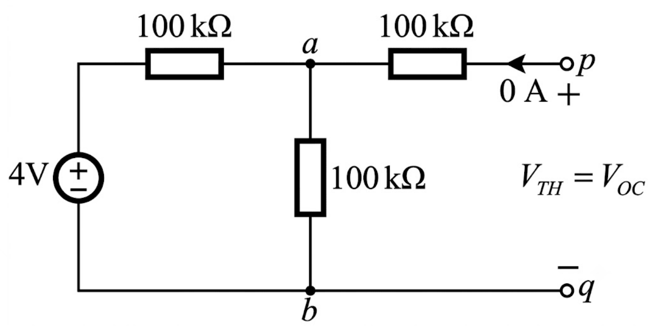


Fig. Solution Q3.18.1

Since the terminal pq is open circuit, there is no current in branch a–p.

$$V_{TH} = V_{ab}$$

$$V_{TH} = V_{pq} = 4 \times \frac{100}{100 + 100}$$

$$V_{TH} = 2 \text{ V}$$

2. **Calculation of R_{TH} :**

Replace all independent sources by their internal resistances, i.e., short circuit all independent voltage sources and open circuit all independent current sources.

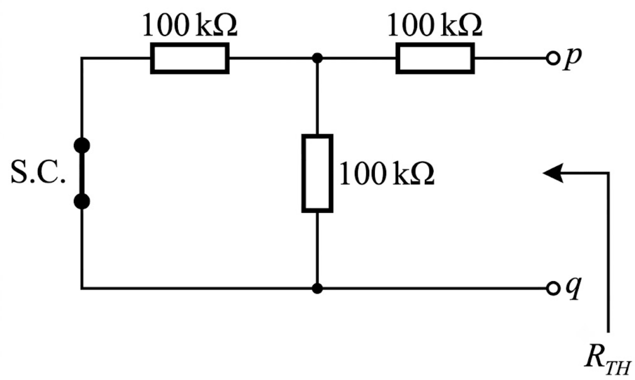


Fig. Solution Q3.18.1

$$R_{TH} = (100 \parallel 100) + 100$$

$$R_{TH} = 50 + 100 = 150 \text{ k}\Omega$$

Therefore, the Thevenin equivalent circuit is,

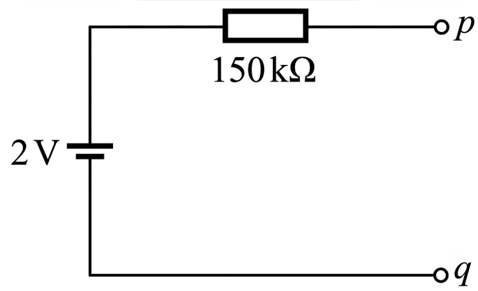


Fig. Solution Q3.18.1

(C) 2 V source in series with 150 kΩ

Q3.18.2 NAT 2018 2M Ans: 1 ● ● ○

Given circuit is shown below,

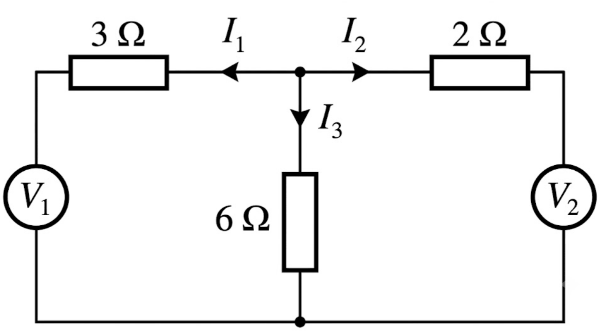


Fig. Solution Q3.18.2

Given :

(i) At $V_1 = 0$, $I_1 = 6$ A

(ii) At $V_2 = 0$, $I_2 = -6$ A

1. **Consider only V_1 voltage source is active :**

   Replace other voltage source by its internal resistance, i.e., $R = 0$ (S.C.).

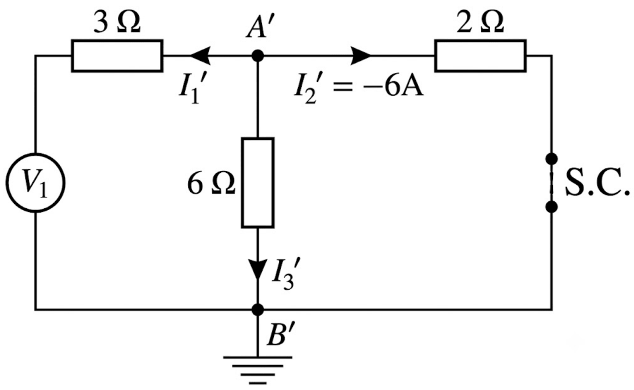


Fig. Solution Q3.18.2

From figure,

$$V_{A'B'} = I_2' \times 2 = -12$$

$$V_{A'B'} = I_3' \times 6$$

$$-12 = I_3' \times 6$$

$$I_3' = -2 \text{ A}$$

2. **Consider only V_2 voltage source is active :**

   Replace other voltage source by its internal resistance, i.e., $R = 0$ (S.C.).

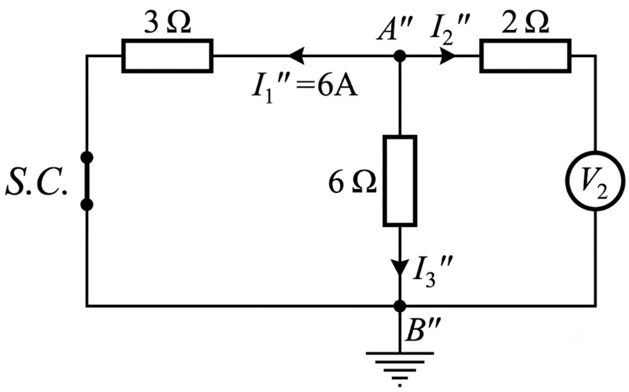


Fig. Solution Q3.18.2

From figure,

$$V_{A''B''} = I_1'' \times 3 = 6 \times 3 = 18$$

$$V_{A''B''} = I_3'' \times 6$$

$$18 = I_3'' \times 6$$

$$I_3'' = 3 \text{ A}$$

3. **When both sources V_1 and V_2 are active :**

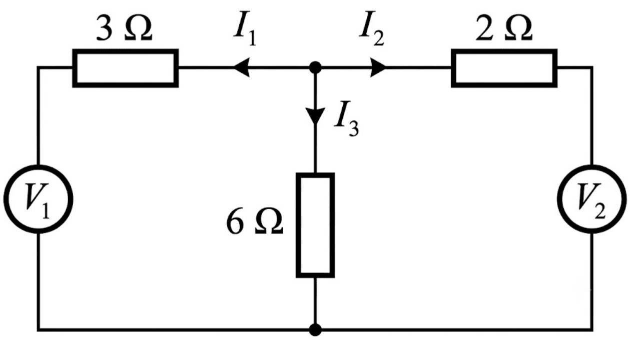


Fig. Solution Q3.18.2

According to superposition theorem,

$$I_3 = I_3' + I_3''$$

$$I_3 = -2 + 3 = 1 \text{ A}$$

$$\boxed{1.00}$$

Q3.15.1 MCQ | 2015 | 1M | Ans: A

Given circuit is shown below,

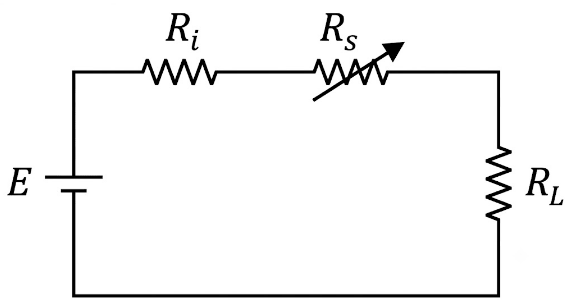


Fig. Solution Q3.15.1

Concept :

- Maximum power transfer theorem is valid only for a variable load.
- In the given network, load is fixed. Therefore, maximum power transfer theorem is not applicable.
- For a fixed load, maximum power is delivered to the load when the internal resistance of the source circuit is minimum.

Given,

$$R_s \geq 0$$

Since R_s can either be 0 or a positive value, the minimum value of R_s is

$$R_s = 0$$

$$\boxed{\text{(A) } 0}$$

Q3.15.2 | NAT | 2015 | 2M | Ans: 1 | ● ● ○

Given circuit is shown below,

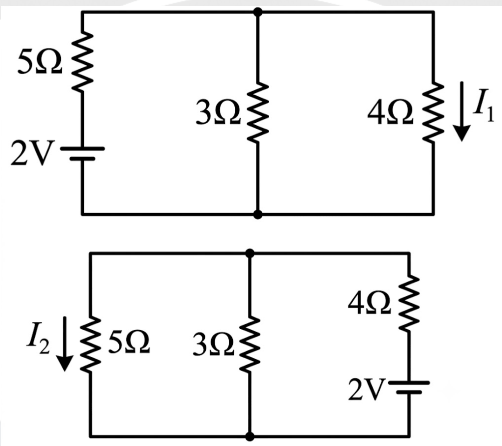


Fig. Solution Q3.15.2

Method Nodal Analysis

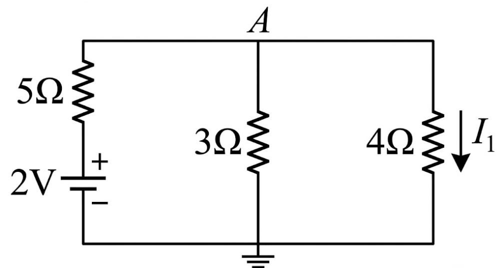


Fig. (a)

Fig. Solution Q3.15.2

Applying KCL at node A,

$$\frac{V_A}{3} + \frac{V_A - 2}{5} + \frac{V_A}{4} = 0$$

$$20V_A + 12V_A - 24 + 15V_A = 0$$

$$47V_A = 24$$

$$V_A = \frac{24}{47} \text{ V}$$

From Fig. (a),

$$I_1 = \frac{V_A}{4} = \frac{24/47}{4} = \frac{6}{47} \text{ A} \quad \text{(i)}$$

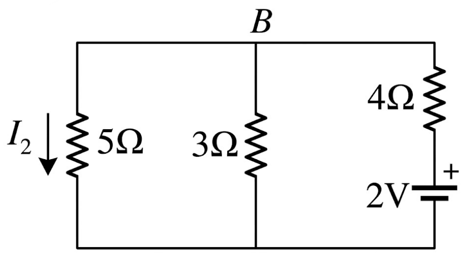


Fig. (b)

Fig. Solution Q3.15.2

Applying KCL at node B,

$$\frac{V_B}{3} + \frac{V_B}{5} + \frac{V_B - 2}{4} = 0$$

$$20V_B + 12V_B + 15V_B - 30 = 0$$

$$V_B = \frac{30}{47} \text{ V}$$

From Fig. (b),

$$I_2 = \frac{V_B}{5} = \frac{30/47}{5} = \frac{6}{47} \text{ A} \quad \text{(ii)}$$

From equations (i) and (ii),

$$\left|\frac{I_1}{I_2}\right| = \frac{6/47}{6/47} = 1$$

Method Reciprocity Theorem

According to reciprocity theorem, in a linear network, ratio of response to excitation remains constant even if the position of source and response are interchanged.

In the given network, the position of source and response are interchanged. Therefore,

$$\frac{I_1}{2} = \frac{I_2}{2}$$

$$I_1 = I_2$$

Hence,

$$\left|\frac{I_1}{I_2}\right| = 1$$

$$\boxed{1}$$

Key Points

(i) Reciprocity theorem is applicable only for linear and reciprocal passive networks.

(ii) Reciprocity theorem is valid only when the network has a single independent source.

(iii) For multiple independent sources, reciprocity theorem is applied along with superposition theorem.

Q3.14.1 NAT 2014 2M Ans: 20

Given circuit is shown below,

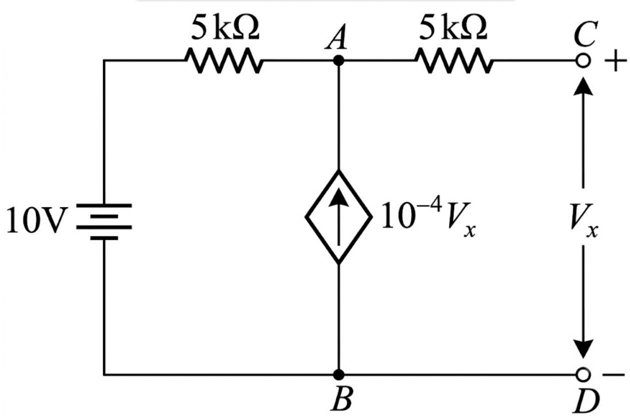


Fig. Solution Q3.14.1

Calculation of R_{TH} :
(When dependent sources are present)

(i) Apply a voltage source V_{dc} between terminals C and D and assume current I_{dc} flowing into the network.

(ii) Replace all independent sources by their internal resistances, i.e., short circuit all independent voltage sources ($R_{in} = 0$) and open circuit all independent current sources ($R_{in} = \infty$).

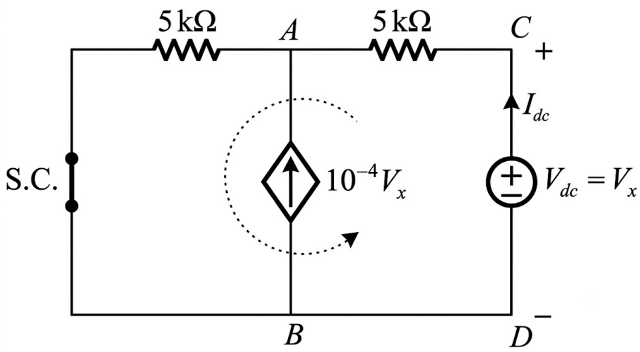


Fig. Solution Q3.14.1

Applying KVL in the loop,

$$-V_{dc} + 5I_{dc} + 5(I_{dc} + 10^{-4}V_x) = 0$$

$$V_{dc} = 5 \times 10^3(I_{dc} + 10^{-4}V_x) + 5 \times 10^3 I_{dc}$$

Also,

$$V_x = V_{dc}$$

$$V_{dc} = 0.5V_{dc} + (10 \times 10^3)I_{dc}$$

$$0.5V_{dc} = (10 \times 10^3)I_{dc}$$

$$R_{TH} = \frac{V_{dc}}{I_{dc}} = 20 \times 10^3\ \Omega$$

$$\boxed{20}$$

Q3.13.1 MCQ | 2013 | 1M | Ans: C | ● ○ ○

According to the question, the circuit is shown below,

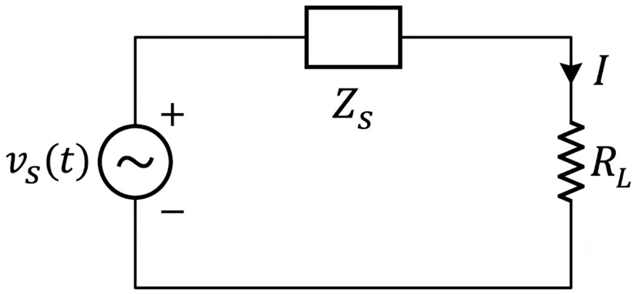


Fig. Solution Q3.13.1

Internal impedance/source impedance is given as,

$$Z_s = R_s + jX_s = (4 + j3)\ \Omega$$

For a purely resistive load connected to a complex source impedance, maximum power is transferred when

$$R_L = |Z_s| = \sqrt{R_s^2 + X_s^2}$$

$$R_L = \sqrt{4^2 + 3^2} = 5\ \Omega$$

$$\boxed{\text{(C) } 5\ \Omega}$$

Q3.13.2 MCQ 2013 2M Ans: C

Given circuit is shown below,

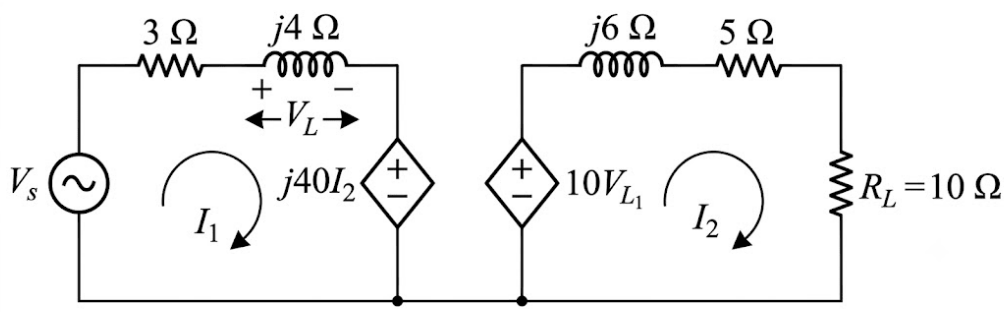


Fig. Solution Q3.13.2

Thevenin's equivalent voltage across load R_L is V_{TH}.

$$V_{TH} = V_{OC}$$

For open circuit condition,

$$I_2 = 0$$

Hence, the load resistance R_L is removed.
Since,

$$j40I_2 = 0$$

the dependent voltage source becomes zero.
Thus, the circuit becomes,

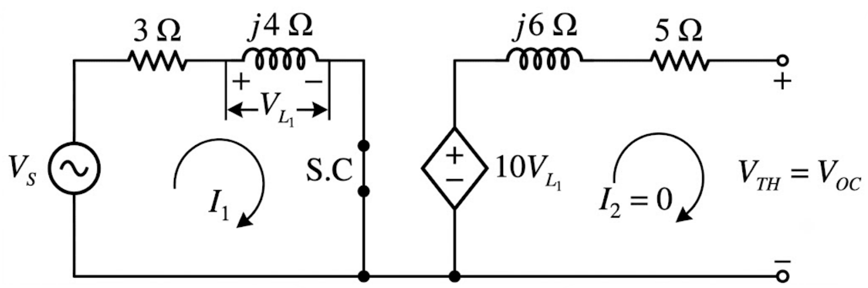


Fig. Solution Q3.13.2

From the circuit,

$$V_{TH} = V_{OC} = 10V_L$$

Applying KVL in the left loop,

$$V_s - 3I_1 - j4I_1 = 0$$

$$I_1 = \frac{V_s}{3 + j4} = \frac{100\angle 53.13°}{5\angle 53.13°}$$

$$I_1 = 20\angle 0°\ \text{A}$$

Therefore,

$$V_L = j4I_1 = j4 \times 20\angle 0^\circ$$

$$V_L = 80\angle 90^\circ \text{ V}$$

Hence,

$$V_{TH} = V_{OC} = 10V_L$$

$$V_{TH} = 800\angle 90^\circ \text{ V}$$

$$\boxed{\text{(C) } 800\angle 90^\circ}$$

Q3.12.1 MCQ | 2012 | 1M | Ans: A | ● ● ○

Given circuit is shown below,

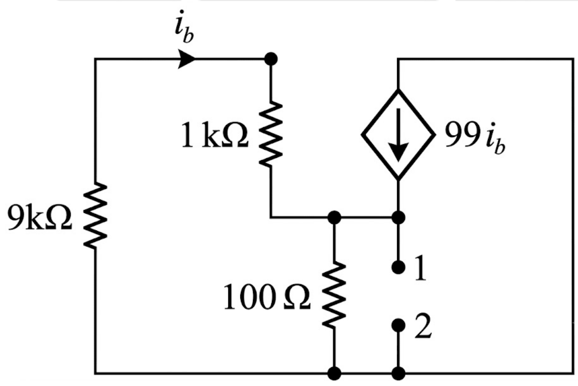


Fig. Solution Q3.12.1

Here, the network contains a dependent source only.

Calculation of R_{TH} :

(When dependent sources are present)

(i) Apply a voltage source V_{dc} between terminals 1 and 2 and assume current I_{dc} flowing into the network.

(ii) Replace all independent sources by their internal resistances, i.e., short circuit all independent voltage sources ($R_{in} = 0$) and open circuit all independent current sources ($R_{in} = \infty$).

(iii) Dependent source remains as it is.

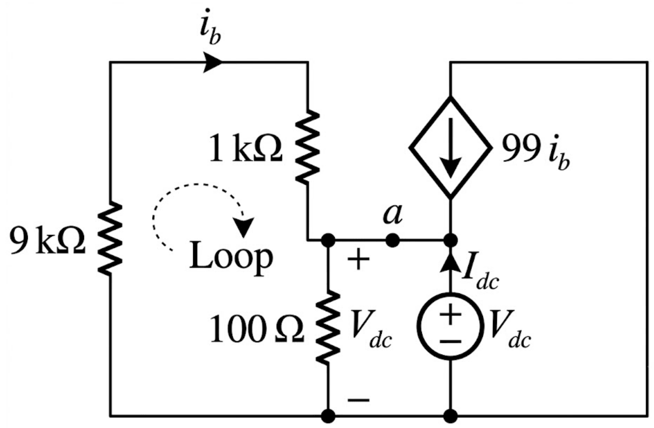


Fig. Solution Q3.12.1

The Thevenin equivalent impedance is given by,

$$Z_{TH} = \frac{V_{dc}}{I_{dc}}$$

Applying KCL at node a,

$$-i_b + \frac{V_{dc}}{100} - 99i_b - I_{dc} = 0$$

$$\frac{V_{dc}}{100} - 100i_b = I_{dc} \qquad \text{(i)}$$

Applying KVL in the loop,

$$10 \times 10^3\, i_b + V_{dc} = 0$$

$$i_b = -\frac{V_{dc}}{10 \times 10^3} \qquad \text{(ii)}$$

Substituting (ii) into (i),

$$\frac{V_{dc}}{100} - 100\left(-\frac{V_{dc}}{10^4}\right) = I_{dc}$$

$$\frac{2V_{dc}}{100} = I_{dc}$$

$$\frac{V_{dc}}{I_{dc}} = 50\ \Omega$$

$$Z_{TH} = 50\ \Omega$$

$$\boxed{\text{(A) } 50\ \Omega}$$

Q3.12.2 MCQ 2012 2M Ans: A ● ● ○

Maximum power is transferred from circuit A to circuit B, if circuit B offers $2\ \Omega$ to circuit A.

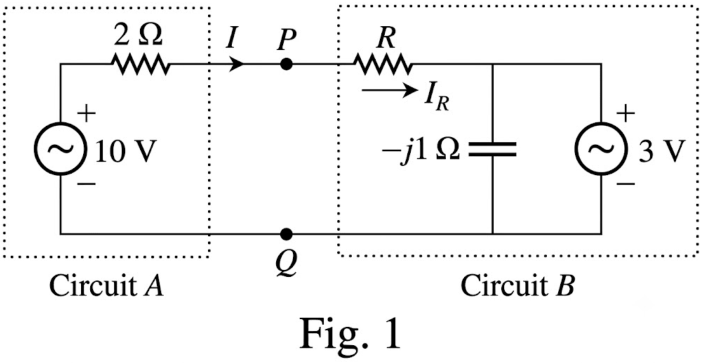


Fig. Solution Q3.12.2

So, replace circuit B by $2\ \Omega$ resistance. The Thevenin equivalent of circuit A is given by,

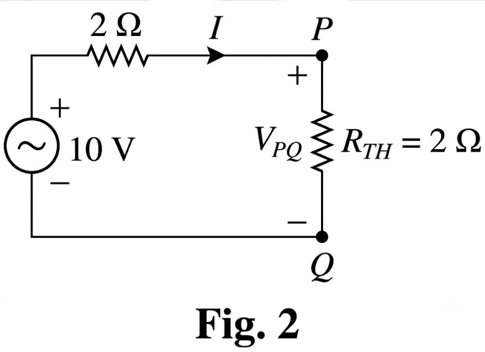


Fig. Solution Q3.12.2

$$I = \frac{10}{4} = 2.5\ \text{A}$$

and

$$V_{PQ} = I \times 2 = 2.5 \times 2 = 5\ \text{V}$$

From Fig. 1,

$$I_R = \frac{V_{PQ} - 3}{R}$$

$$I_R = \frac{5-3}{R} = \frac{2}{R}$$

Also,

$$I_R = I$$

$$\frac{2}{R} = 2.5$$

$$R = \frac{2}{2.5} = 0.8\ \Omega$$

(A) 0.8 Ω

Key Points

- Circuit B is replaced by a resistance of 2 Ω for maximum power transfer from circuit A to circuit B.

Q3.09.1 MCQ 2009 2M Ans: C

Maximum power is transferred from circuit A to circuit B, if circuit B offers a resistance of 2 Ω to circuit A.

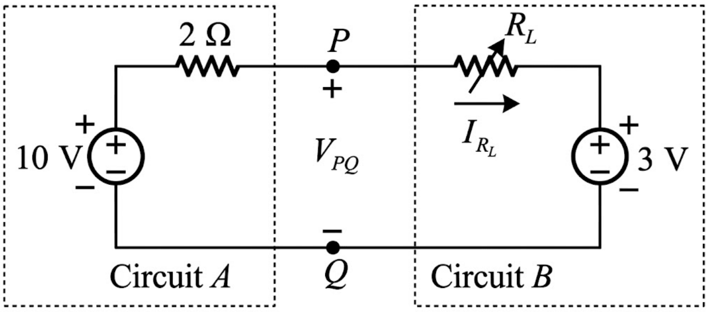


Fig. (a)

Fig. Solution Q3.09.1

Thevenin's equivalent of circuit A is given by,

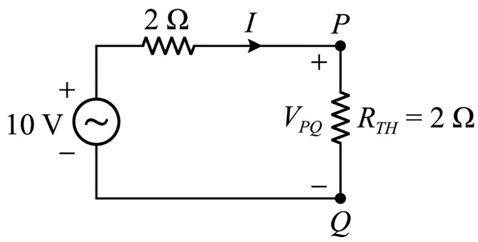


Fig. (b)

Fig. Solution Q3.09.1

From Fig. (b),

$$I = \frac{10}{2+2} = 2.5\text{ A}$$

and

$$V_{PQ} = I \times 2 = 2.5 \times 2 = 5\text{ V}$$

From Fig. (a),

$$I_{R_L} = \frac{V_{PQ} - 3}{R_L}$$

$$I_{R_L} = \frac{5-3}{R_L} = \frac{2}{R_L}$$

Also,

$$I_{R_L} = I$$

$$\frac{2}{R_L} = 2.5$$

$$R_L = \frac{2}{2.5} = 0.8\ \Omega$$

$$\boxed{\text{(C) } 0.8\ \Omega}$$

Q3.08.1 MCQ | 2008 | 2M | Ans: B

Given circuit is shown below,

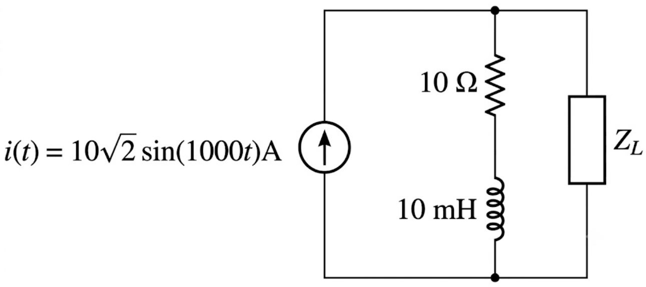


Fig. Solution Q3.08.1

From figure,

$$i_{rms} = \frac{i_{max}}{\sqrt{2}} = \frac{10\sqrt{2}}{\sqrt{2}} = 10\text{ A}$$

$$\omega = 1000\text{ rad/s}$$

$$X_L = j\omega L = j(1000)(10 \times 10^{-3}) = j10\ \Omega$$

1. Calculation of V_{TH} :

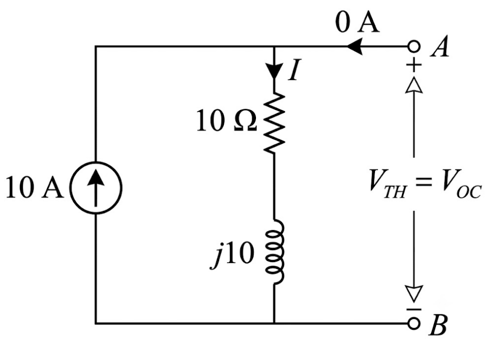


Fig. Solution Q3.08.1

$$V_{TH} = V_{OC}$$

$$V_{TH} = (10 + j10) \times 10$$

$$V_{TH} = 141.42\angle 45^\circ \text{ V}$$

2. Calculation of Z_{TH} :

Replace all independent sources by their internal resistances, i.e., short circuit all independent voltage sources ($R_{in} = 0$) and open circuit all independent current sources ($R_{in} = \infty$).

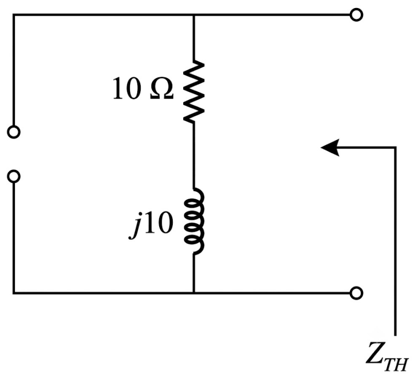


Fig. Solution Q3.08.1

$$Z_{TH} = (10 + j10)\ \Omega$$

Thevenin's equivalent circuit :

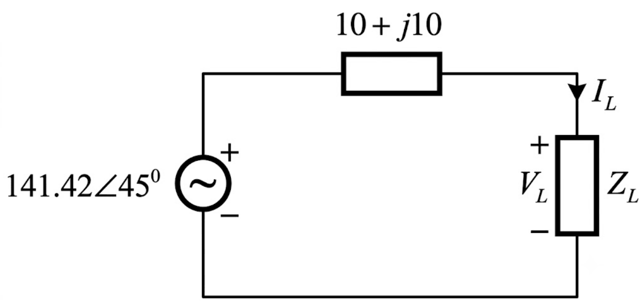


Fig. Solution Q3.08.1

For maximum power transfer in an AC network,

$$Z_L = Z_{TH}^* = (10 - j10)$$

Load current is,

$$I_L = \frac{V_{TH}}{Z_L + Z_{TH}} = \frac{141.42\angle 45^\circ}{(10 - j10) + (10 + j10)}$$

$$I_L = 7.071\angle 45^\circ \text{ A}$$

$$|I_L| = 7.071 \text{ A}$$

$$P_{max} = |I_L|^2 \times \Re(Z_L)$$

$$P_{max} = (7.071)^2 \times 10 = 500 \text{ W}$$

$$\boxed{\text{(B) 500 W}}$$

Q3.07.1 MCQ | 2007 | 1M | Ans: C | ● ● ○

Given :

(i) Short circuit current $(I_{sc}) = 75$ mA

(ii) For a terminal voltage of 0.6 V, current $(I) = 70$ mA

Let, the given load be Z_L.
The equivalent circuit of solar cell is represented as Norton equivalent circuit.

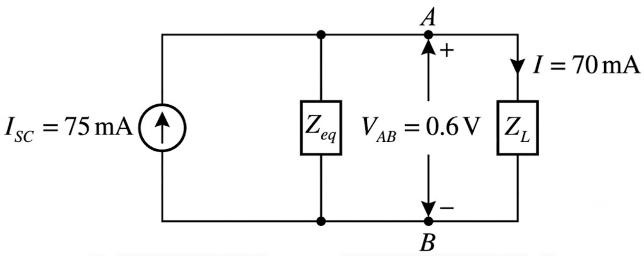


Fig. Solution Q3.07.1

$$I = I_{sc} \times \frac{Z_{eq}}{Z_L + Z_{eq}}$$

$$70 \text{ mA} = 75 \text{ mA} \times \frac{Z_{eq}}{Z_L + Z_{eq}}$$

$$\frac{75}{70} = 1 + \frac{Z_L}{Z_{eq}}$$

$$\frac{Z_L}{Z_{eq}} = \frac{75}{70} - 1 = \frac{5}{70} = \frac{1}{14}$$

$$Z_{eq} = 14 Z_L \quad \text{(i)}$$

Also,

$$Z_L = \frac{V_{AB}}{I} = \frac{0.6}{70 \text{ mA}}$$

$$Z_L = \frac{60}{7} \ \Omega$$

From (i),

$$Z_{eq} = 14 \times \frac{60}{7} = 120 \ \Omega$$

$$\boxed{\text{(C) 120 } \Omega}$$

Q3.07.2 MCQ 2007 2M Ans: A

Given circuit is shown below,

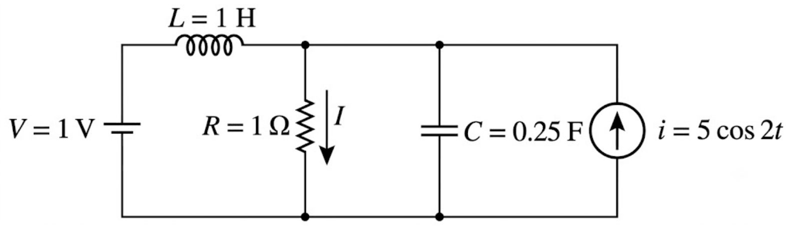


Fig. Solution Q3.07.2

(i) **Consider only** 1 **V voltage source :**

For DC, the capacitor is open circuited and the inductor is short circuited.

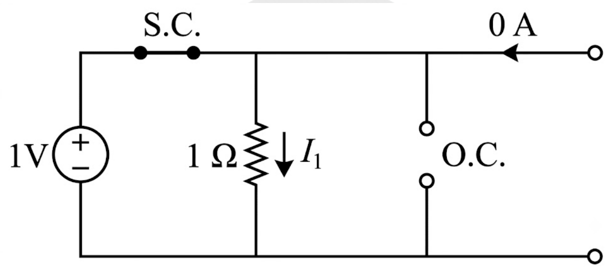


Fig. Solution Q3.07.2

Key Points

For DC source (at steady state)

$$\omega = 0 \Rightarrow f = 0$$

$$X_C = \frac{1}{\omega C} = \infty \qquad \text{(Open circuit)}$$

$$X_L = \omega L = 0 \qquad \text{(Short circuit)}$$

The value of current I_1 is,

$$I_1 = \frac{1}{1} = 1 \text{ A}$$

(ii) **Consider only current source i :**

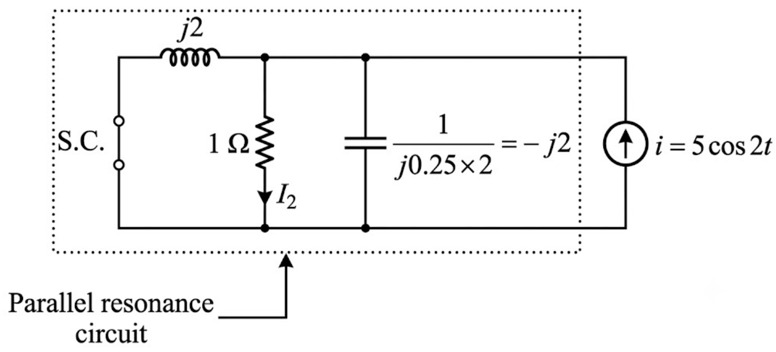


Fig. Solution Q3.07.2

Here,

$$X_L = j\omega L = j(2)(1) = j2$$

$$X_C = \frac{1}{j\omega C} = \frac{1}{j(2)(0.25)} = -j2$$

Imaginary part of admittance,

$$Y = \frac{1}{j2} + \frac{1}{(-j2)} = 0$$

Hence, it is a parallel resonance circuit and the entire current passes through the 1 Ω resistor.

$$I_2 = (5\cos 2t) \text{ A}$$

According to superposition theorem,

$$I = I_1 + I_2$$

$$I = (1 + 5\cos 2t) \text{ A}$$

$$\boxed{\text{(A) } (1 + 5\cos 2t) \text{ A}}$$

Q3.03.1 MCQ 2003 1M Ans: B ● ● ○

Given circuit is shown below,

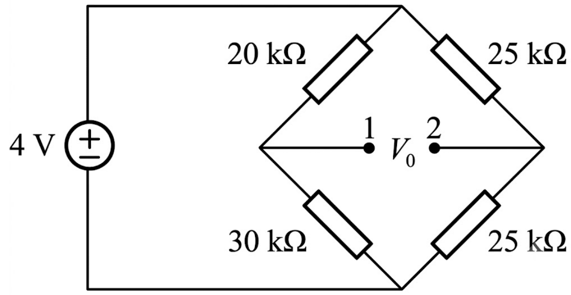


Fig. Solution Q3.03.1

The output resistance across terminals 1 and 2, i.e.,

$$R_{12} = R_{TH}$$

Calculation of R_{TH} :

(When dependent sources are not present)

Replace all independent sources by their internal resistances, i.e., short circuit all independent voltage sources ($R_{in} = 0$) and open circuit all independent current sources ($R_{in} = \infty$).

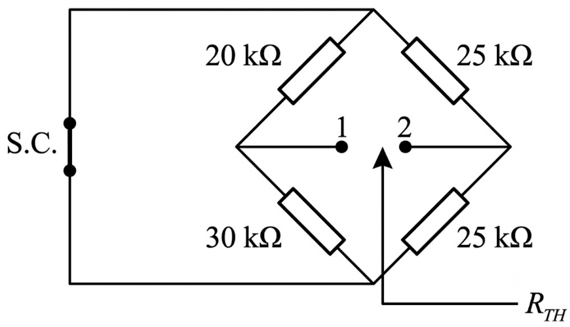


Fig. Solution Q3.03.1

The above figure can be represented as,

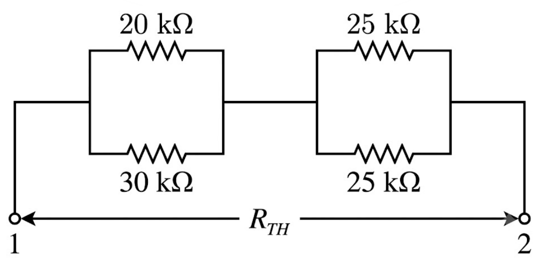


Fig. Solution Q3.03.1

$$R_{TH} = R_{12} = (20 \parallel 30) + (25 \parallel 25)$$

$$R_{TH} = R_{12} = \frac{20 \times 30}{20 + 30} + \frac{25 \times 25}{25 + 25}$$

$$R_{TH} = R_{12} = (12 + 12.5)\ \text{k}\Omega$$

$$R_{TH} = R_{12} = 24.5\ \text{k}\Omega$$

$$\boxed{\text{(B) } 24.5\ \text{k}\Omega}$$

Key Points

(i) Parallel combination of resistors decreases the equivalent resistance of the combination.

(ii) Series combination of resistors increases the equivalent resistance of the combination.

Q3.00.1 MCQ 2000 1M Ans: A ● ● ○

Given circuit is shown below,

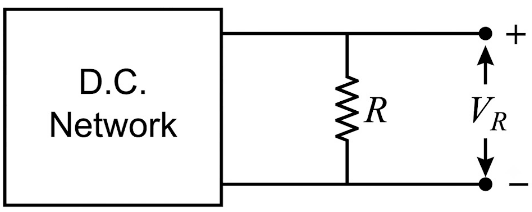


Fig. Solution Q3.00.1

(i) $V_R = 20$ V when $R = 10\ \Omega$

(ii) $V_R = 30$ V when $R = 20\ \Omega$

Thevenin's equivalent circuit :

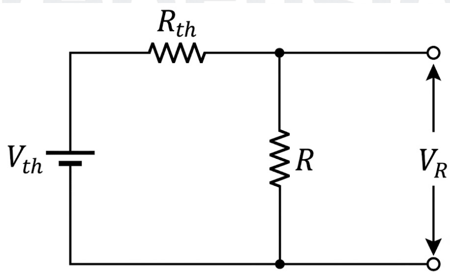


Fig. Solution Q3.00.1

$$V_R = V_{TH} \times \frac{R}{R + R_{TH}} \quad \text{(i)}$$

From the given data,

$$20 = V_{TH} \times \frac{10}{10 + R_{TH}}$$

$$10V_{TH} - 20R_{TH} = 200 \quad \text{(ii)}$$

Also,

$$30 = V_{TH} \times \frac{20}{20 + R_{TH}}$$

$$20V_{TH} - 30R_{TH} = 600 \qquad \text{(iii)}$$

From (ii) and (iii),

$$V_{TH} = 60 \text{ V}$$

$$R_{TH} = 20\ \Omega$$

The equivalent circuit becomes,

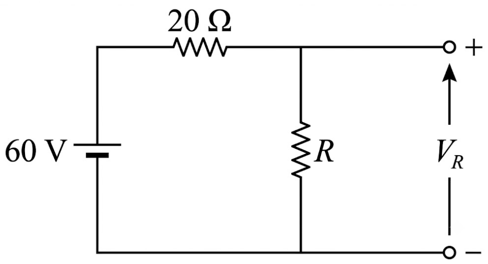


Fig. Solution Q3.00.1

From (i),

$$V_R = 60 \times \frac{R}{R + 20}$$

When $R = 80\ \Omega$,

$$V_R = 60 \times \frac{80}{80 + 20} = 48 \text{ V}$$

$$\boxed{\text{(A) 48 V}}$$

Q3.00.2 MCQ | 2000 | 1M | Ans: C

Given circuit is shown below,

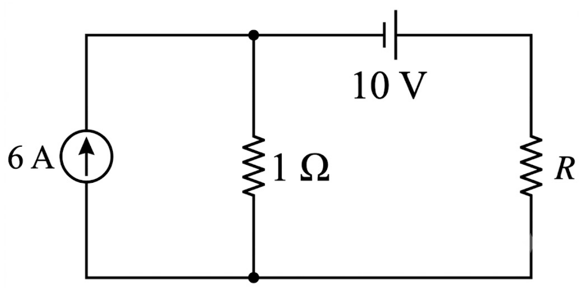


Fig. Solution Q3.00.2

Thevenin's equivalent circuit :

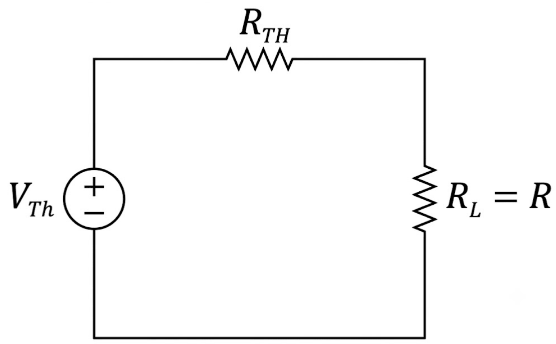


Fig. Solution Q3.00.2

For a resistive network, maximum power is transferred to R_L when

$$R_L = R_{TH} \tag{i}$$

and

$$P_{\max} = \frac{V_{TH}^2}{4R_{TH}}$$

Calculation of V_{TH} :

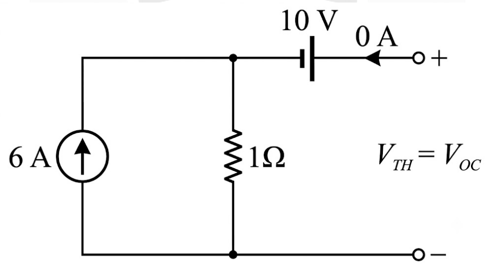


Fig. Solution Q3.00.2

From the figure,

$$-V_{TH} + 10 + (1)(6) = 0$$

$$V_{TH} = 6 \times 1 + 10 = 16 \text{ V}$$

Calculation of R_{TH} :

(When dependent sources are not present)

Replace all independent sources by their internal resistances, i.e., short circuit all independent voltage sources ($R_{in} = 0$) and open circuit all independent current sources ($R_{in} = \infty$).

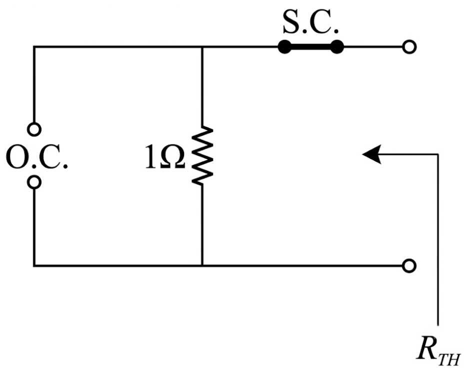


Fig. Solution Q3.00.2

$$R_{TH} = R = 1\ \Omega$$

Thevenin's equivalent circuit :

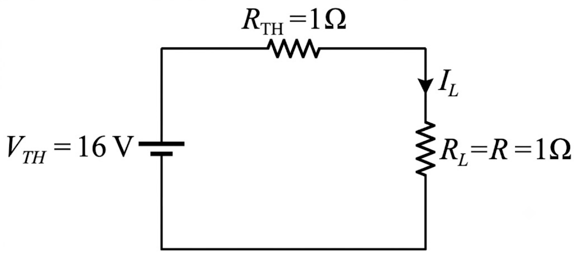


Fig. Solution Q3.00.2

Load current is,

$$I_L = \frac{V_{TH}}{R_{TH} + R} = \frac{16}{1+1} = 8\text{ A}$$

$$P_{max} = |I_L|^2 R_L = 8^2 \times 1 = 64\text{ W}$$

or

$$P_{max} = \frac{V_{TH}^2}{4R_{TH}} = \frac{16^2}{4 \times 1} = 64\text{ W}$$

$$\boxed{\text{(C) 64 W}}$$

Q3.00.3 MCQ | 2000 | 2M | Ans: B | ● ○ ○

Superposition theorem is not applicable to networks containing non-linear elements such as diodes.
The Superposition theorem is applicable to :

(i) Linear networks.

(ii) Time-variant or time-invariant networks.

(iii) Networks containing independent sources.

(iv) Networks containing linear dependent sources (voltage or current).

(v) Networks containing linear passive elements R, L, C and linear transformers.

(vi) Networks containing different frequency sources.

$\boxed{\text{(B) networks containing diodes and R-C element}}$

Key Points

- Superposition theorem is applicable only to **linear networks**.
- It is not applicable to circuits containing non-linear elements such as diodes, transistors operating in non-linear regions, etc.
- Superposition can be used in networks containing both independent and linear dependent sources.

DEBUG INFO

Chapter III

Network Theorems For DC & Miscellaneous Topics

Total Questions : 21

MCQ : 14

MSQ : 0

NAT : 7

SOLUTIONS

IV

2nd Order Dynamics and Complex Impedance

Topics

Q4.01.1 NAT 2001 5M Ans: *

The model of the circuit in s-domain is as follows:

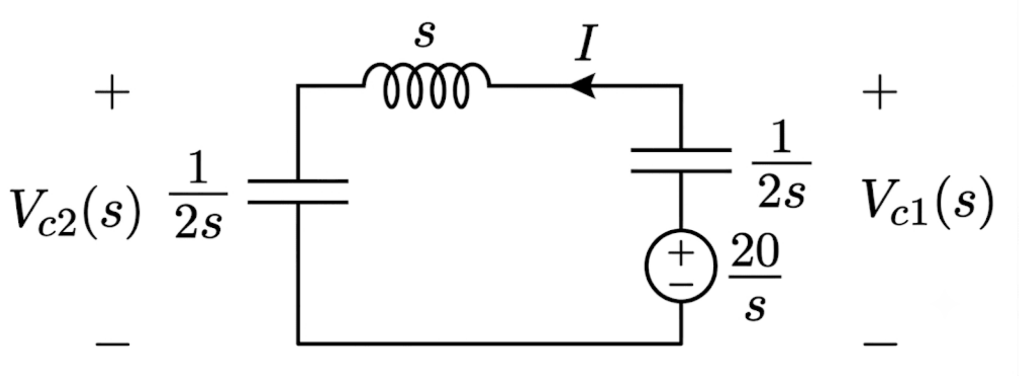


Fig. Solution Q4.01.1

The total current $I(s)$ flowing in the series loop can be calculated as:

$$I = \frac{\frac{20}{s}}{\frac{1}{2s} + \frac{1}{2s} + s} = \frac{\frac{20}{s}}{\frac{1}{s} + s} = \frac{20}{s^2 + 1}$$

The voltage across the second capacitor $V_{c2}(s)$ is given by:

$$V_{c2}(s) = \frac{1}{2s} \cdot I = \frac{20}{2s(s^2+1)} = \frac{10}{s(s^2+1)}$$

Using partial fraction expansion:

$$\frac{10}{s(s^2+1)} = \frac{A}{s} + \frac{Bs + C}{s^2 + 1}$$

$$\frac{10}{s(s^2+1)} = \frac{A(s^2+1) + (Bs + C)s}{s(s^2+1)} = \frac{(A+B)s^2 + Cs + A}{s(s^2+1)}$$

Comparing the coefficients on the left-hand side and right-hand side:

$$A + B = 0 \implies B = -A$$

$$C = 0$$

$$A = 10 \implies B = -10$$

Substituting these constants back into the equation:

$$V_{c2}(s) = \frac{10}{s} - \frac{10}{s^2 + 1}$$

Performing the inverse Laplace transform, we obtain the time-domain response for $v_{c2}(t)$:

$$v_{c2}(t) = 10(1 - \sin t)\text{ V}$$

Similarly, the voltage across the first capacitor $V_{c1}(s)$ is computed as:

$$V_{c1}(s) = \frac{20}{s} - I\left(\frac{1}{2s}\right) = \frac{20}{s} - \frac{10}{s(s^2+1)}$$

$$V_{c1}(s) = \frac{20}{s} - \left(\frac{10}{s} - \frac{10}{s^2+1}\right) = \frac{10}{s} + \frac{10}{s^2+1}$$

Performing the inverse Laplace transform, we get the time-domain response for $v_{c1}(t)$:

$$v_{c1}(t) = 10(1 + \sin t)\text{ V}$$

Key Points

- The s-domain impedance of an inductor L is sL, and for a capacitor C is $\frac{1}{sC}$.
- Initial conditions for energy storage elements can be modeled as independent sources in the Laplace transformed equivalent circuit.

$$\boxed{v_{c1}(t) = 10(1 + \sin t)\text{ V},\ v_{c2}(t) = 10(1 - \sin t)\text{ V}}$$

Q4.00.1

MCQ | 2000 | 2M | Ans: B

We know that capacitive impedance is inversely proportional to frequency:

$$X_c = \frac{1}{sC}$$

Thus, as $s \to \infty$, the capacitive reactance approaches zero ($X_c \to 0$), meaning all capacitors behave as short circuits. Shorting the capacitors in the given network simplifies the circuit diagram to a parallel combination of resistors.

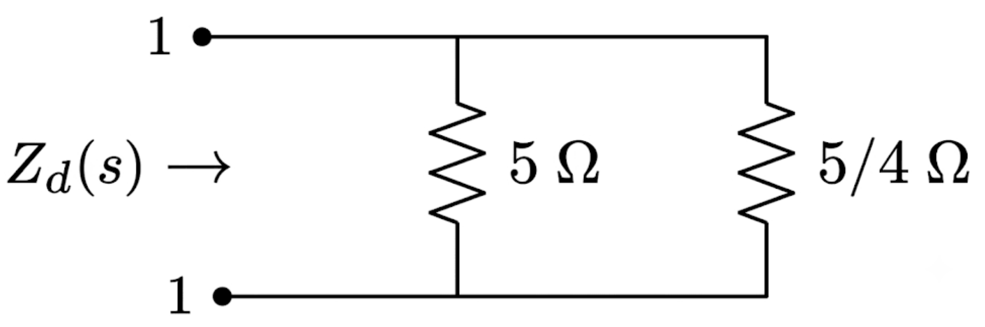


Fig. Solution Q4.00.1

The equivalent driving-point impedance $Z_d(s)$ as $s \to \infty$ is calculated as the parallel combination of the remaining $5\,\Omega$ and $\frac{5}{4}\,\Omega$ resistors:

$$Z_d(s) = 5 \parallel \frac{5}{4}$$

$$Z_d(s) = \frac{5 \times \frac{5}{4}}{5 + \frac{5}{4}} = \frac{\frac{25}{4}}{\frac{25}{4}} = 1\,\Omega$$

$$\boxed{(b)1}$$

DEBUG INFO

Chapter IV

2nd Order Dynamics and Complex Impedance

Total Questions : 2

MCQ : 1

MSQ : 0

NAT : 1

SOLUTIONS

V

AC Circuit Analysis

Topics

1. Basics of Phasor Diagrams
2. Phasor Diagram for Series RL, RC and RLC Circuits
3. Phasor Diagram for Parallel RL, RC and RLC Circuits
4. Locus Diagram of Current, Admittance and Impedance
5. Introduction to Power in AC Circuits
6. Active, Reactive and Apparent Power in Complex Networks
7. Series RLC Resonance Circuit
8. Quality Factor and Stored Energy in Series Resonance
9. Parallel RLC Resonance Circuit
10. Application of Network Theorems in AC Circuits
11. Maximum Power Transfer in AC Circuits
12. Three Phase Circuits

Q5.26.1 MCQ 2026 1M Ans: D

To find the type of filter we need to investigate the circuit under 2 conditions : $\omega = 0\,\text{rad/s}$ and $\omega = \infty\,\text{rad/s}$
We know that inductive reactance is proportional to frequency and capacitive reactance inversely proportional to it.
Thus at $\omega = 0\,\text{rad/s}$, inductive reactance is zero (or L appears shorted) and capacitive reactance is ∞ or (C appears open). The circuit is thus open and no current flows through it. The output voltage is equal to the input voltage.

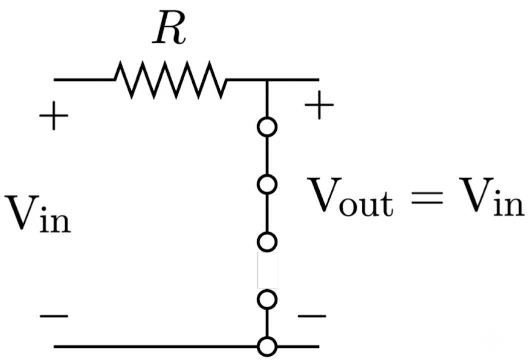


Fig. Solution Q5.26.1

Thus at $\omega = \infty\,\text{rad/s}$, inductive reactance is ∞ (or L appears open) and capacitive reactance is zero or (C appears shorted). The circuit is thus open and no current flows through it. The output voltage is equal to the input voltage.

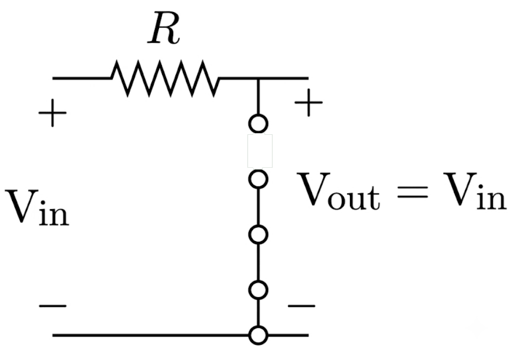


Fig. Solution Q5.26.1

Since output is high at both corner cases the filter is either a band reject (or band stop) or an all pass filter. Judging by the options band reject filter is the most suitable option, and we do not investigate further.

Key Points

- When concluding whether a filter is band reject or all pass we must check the pole zero locations of the system, which if symmetric, the filter is an all pass one else it is a band reject one.

Q5.26.2 NAT 2026 1M Ans: 25 to 25

Let the AC source voltage be V_s. The current I in the series RL circuit is given by:

$$I = \frac{V_s}{R + j\omega L}$$

Taking V_s as the reference phasor ($V_s = |V_s|\angle 0^\circ$), the phase angle of the current ($\angle I$) is:

$$\angle I = \angle V_s - \tan^{-1}\left(\frac{\omega L}{R}\right) = -\tan^{-1}\left(\frac{\omega L}{R}\right)$$

Given that the current I lags the voltage V_s by 45°:

$$\tan^{-1}\left(\frac{\omega L}{R}\right) = 45^\circ \quad \text{(where the negative sign denotes lagging)}$$

$$\frac{\omega L}{R} = \tan(45^\circ) = 1$$

$$\omega = \frac{R}{L}$$

Since $\omega = 2\pi f$, the frequency f is:

$$f = \frac{R}{2\pi L}$$

Substituting the given circuit values, $R = 100\,\Omega$ and $L = 2\,\text{H}$:

$$f = \frac{100}{2\pi \times 2} = 25\,\text{Hz}$$

$$\boxed{25}$$

Q5.25.1 MCQ 2025 1M Ans: A ● ● ○

We know the relation between voltage, admittance, and current is given by:

$$VY = I$$

Since all three admittances are in parallel, they have the same voltage across them. Therefore, we can write:

$$VY_i = I_i \quad \text{where } i = a, b, c$$

Given the parallel branch admittance values and a common reference voltage $V = 10\angle 45^\circ$ V, the individual branch currents are determined as follows:

For branch a:

$$I_a = VY_a = 10\angle 45^\circ \times (-j0.2) = 10\angle 45^\circ \times 0.2\angle -90^\circ = 2\angle -45^\circ \text{ A}$$

For branch b:

$$I_b = VY_b = 10\angle 45^\circ \times 0.3 = 3\angle 45^\circ \text{ A}$$

For branch c:

$$I_c = VY_c = 10\angle 45^\circ \times j0.4 = 10\angle 45^\circ \times 0.4\angle 90^\circ = 4\angle 135^\circ \text{ A}$$

The combined current I flowing through branches b and c is:

$$I = I_b + I_c = 3\angle 45^\circ + 4\angle 135^\circ = 5\angle 98.13^\circ \text{ A}$$

The total source current I_s is the sum of all individual branch currents:

$$I_s = I_a + I_b + I_c = I_a + I = 2\angle -45^\circ + 5\angle 98.13^\circ = 3.6\angle 78.69^\circ \text{ A}$$

From the phase angle information of the calculated currents:

$$\theta_I = 98.13^\circ \quad \text{and} \quad \theta_{I_s} = 78.69^\circ$$

The phase difference shows that I leads I_s by:

$$98.13^\circ - 78.69^\circ = 19.44^\circ$$

$$\boxed{(a) I \textit{ leads } I_s \textit{ by } 19.44^\circ}$$

Q5.25.2 NAT 2025 2M Ans: 590 to 610 ● ● ○

Let us calculate the equivalent impedance seen by the voltage source. The expression for the equivalent impedance Z_{eq} is:

$$Z_{eq} = -j10 + 10 + ((5 - j5) \parallel j5)$$

Simplifying the parallel combination of $(5 - j5)$ and $j5$:

$$(5 - j5) \parallel j5 = \frac{(5 - j5)(j5)}{(5 - j5) + j5} = \frac{j25 - j^2 25}{5} = \frac{j25 + 25}{5} = 5 + j5$$

Substituting this back into the expression for Z_{eq}:

$$Z_{eq} = -j10 + 10 + 5 + j5 = 15 - j5\,\Omega$$

The magnitude of the equivalent impedance $|Z_{eq}|$ is:

$$|Z_{eq}| = \sqrt{15^2 + (-5)^2} = \sqrt{225 + 25} = \sqrt{250} \approx 15.811\,\Omega$$

The real part of the equivalent impedance is:

$$\text{Re}(Z_{eq}) = 15\,\Omega \quad \text{(This value is used to find the active power)}$$

Thus, the equivalent impedance can be represented by a resistance of value $15\,\Omega$ and a reactance of $-5\,\Omega$.

Assuming the source voltage magnitude is $|V| = 100$ V, the magnitude of the current $|I|$ delivered by the source is:

$$|I| = \frac{100}{15.811}\text{ A}$$

The active power P supplied by the source is calculated using:

$$P = |I|^2 \times \text{Re}(Z_{eq}) = \left(\frac{100}{\sqrt{15^2 + 5^2}}\right)^2 \times 15 = \frac{100^2}{15^2 + 5^2} \times 15 = \frac{10000}{250} \times 15 = 40 \times 15 = 600\text{ W}$$

Key Points

- The equivalent impedance $Z_{eq} = R + jX$ consists of a real part representing net resistance (R) and an imaginary part representing net reactance (X).
- Active power is dissipated only across the resistive elements of the network and is given by $P = |I|^2 R$.

$$\boxed{590\ to\ 610}$$

Q5.24.1 NAT 2024 1M Ans: 200 to 200 ● ○ ○

Average power absorbed by a circuit containing non-sinusoidal voltage and current waveforms is given by:

$$P = V_o I_o + \sum_{n=1}^{\infty} \frac{V_{mn} I_{mn}}{2} \cos(\theta_{vn} - \theta_{in})$$

where:

m : maximum value

n : harmonic order

θ_{in}, θ_{vn} : phase angle of current and voltage respectively for the n^{th} harmonic

Thus, we can find the average active power in this case as:

$$P = 95 \times 0 + \frac{200 \times 4}{2} \times \cos(60^\circ)$$

Note that other harmonics are not used because active power can only be developed by voltage and current components of the identical frequency/harmonic order. Here, the remaining current component belongs to the 2nd harmonic while the remaining voltage component belongs to the 3rd harmonic, resulting in zero average power contribution for those terms.

Calculating the non-zero term:

$$P = 0 + \frac{800}{2} \times 0.5 = 400 \times 0.5 = 200\,\text{W}$$

Key Points

- Average active power in non-sinusoidal systems is the sum of the DC power and the AC active powers of corresponding harmonic components.
- A voltage harmonic of order n interacting with a current harmonic of order k (where $n \neq k$) results in an instantaneous power component whose average value over a full cycle is zero.
- The formula uses peak values (V_{mn}, I_{mn}), hence the divisor of 2 converts them to product of RMS values ($V_{\text{rms}} I_{\text{rms}} = \frac{V_m}{\sqrt{2}} \frac{I_m}{\sqrt{2}}$).

Mistakes to be avoided

- Do not multiply voltage and current harmonic magnitudes of different frequencies (e.g., 2nd harmonic current with 3rd harmonic voltage).

$\boxed{200}$

Q5.24.2 NAT 2024 1M Ans: 27.0 to 30.0

The RMS value of a non-sinusoidal periodic current containing a DC component and several harmonic components is given by:

$$I_{\text{rms}} = \sqrt{I_o^2 + \sum_{n=1}^{\infty} \frac{I_{mn}^2}{2}}$$

where:

m : maximum (peak) value
n : harmonic order
I_o : DC component

This is because sine and cosine functions of the same frequency have a phase difference of 90°; they are in quadrature. To find the overall peak magnitude (I_{mn}) of the 1st and 3rd harmonic components, we need to combine their respective orthogonal sine and cosine coefficients.

Since they are in quadrature, we calculate the combined peak magnitudes as follows: For the 1st harmonic component (I_1):

$$I_1 = \sqrt{30^2 + 20^2} = \sqrt{900 + 400} = \sqrt{1300} = 10\sqrt{13}\,\text{A}$$

For the 3rd harmonic component (I_3):

$$I_3 = \sqrt{15^2 + 10^2} = \sqrt{225 + 100} = \sqrt{325} = 5\sqrt{13}\,\text{A}$$

Assuming a DC offset current component $I_o = 4\,\text{A}$, we substitute these peak harmonic magnitudes into the total RMS current formula:

$$I_{\text{rms}} = \sqrt{4^2 + \frac{\left(10\sqrt{13}\right)^2}{2} + \frac{\left(5\sqrt{13}\right)^2}{2}}$$

$$I_{\text{rms}} = \sqrt{4^2 + \frac{10^2 \times 13}{2} + \frac{5^2 \times 13}{2}}$$

$$I_{\text{rms}} = \sqrt{16 + \frac{1300}{2} + \frac{325}{2}} = \sqrt{16 + 650 + 162.5} = \sqrt{828.5} \approx 28.8\,\text{A}$$

Key Points

- When a harmonic order consists of both a sine component ($A_n \sin(n\omega t)$) and a cosine component ($B_n \cos(n\omega t)$), they are in time quadrature. Their net peak amplitude is $I_{mn} = \sqrt{A_n^2 + B_n^2}$.

Mistakes to be avoided

- Remember to divide the squared peak values (I_{mn}^2) by 2 when computing total RMS power quantities, whereas the DC term (I_o^2) is not divided by 2.

$\boxed{27.0 \text{ to } 30.0}$

Q5.23.1 NAT 2023 2M Ans: 1.40 to 1.50 ● ● ○

To find the resonance condition for the given network, we calculate the total equivalent input admittance (Y_{in}) seen by the source. The circuit consists of two parallel branches: a practical inductor branch (R_1 in series with L) and a practical capacitor branch (R_2 in parallel or series context as shown).

Based on the admittance derivation, the expression for the input admittance is given by:

$$Y_{in} = \frac{1}{R_1 + j\omega L} + \frac{j\omega C}{j\omega R_2 C + 1}$$

Rationalizing both terms by multiplying their respective denominators by their complex conjugates:

$$Y_{in} = \frac{R_1 - j\omega L}{R_1^2 + \omega^2 L^2} + j\omega C \left[\frac{1 - j\omega R_2 C}{1 + \omega^2 R_2^2 C^2}\right]$$

Separating the real and imaginary parts:

$$Y_{in} = \frac{R_1}{R_1^2 + \omega^2 L^2} + \frac{\omega^2 C^2 R_2}{1 + \omega^2 R_2^2 C^2} + j\left[\frac{\omega C}{1 + \omega^2 R_2^2 C^2} - \frac{\omega L}{R_1^2 + \omega^2 L^2}\right]$$

At resonance, the circuit operates at a unity power factor, which means the imaginary part of the input admittance must equal zero:

$$\text{Im}(Y_{in}) = 0$$

$$\frac{\omega C}{1 + \omega^2 C^2 R_2^2} - \frac{\omega L}{R_1^2 + \omega^2 L^2} = 0$$

Since $\omega \neq 0$ for a practical AC resonance condition, we can divide by ω:

$$\frac{C}{1 + \omega^2 C^2 R_2^2} - \frac{L}{R_1^2 + \omega^2 L^2} = 0$$

$$C(R_1^2 + \omega^2 L^2) - L(1 + \omega^2 C^2 R_2^2) = 0$$

Expanding and rearranging the terms to form a quadratic equation or an explicit expression for the parameters:

$$CR_1^2 + C\omega^2L^2 - L - L\omega^2C^2R_2^2 = 0$$

$$L\omega^2C^2R_2^2 - C(R_1^2 + \omega^2L^2) + L = 0$$

If we substitute the values L = 7mH, $\omega = 100\pi$ rad/s and $R_1 = R_2 = 2.2\,\text{k}\Omega$, this system for the capacitance C yields:

$$C = 1.45 \times 10^{-3}\,\text{F} = 1.45\,\text{mF}$$

Mistakes to be avoided

- Solve the quadratic equation without making early approximations. Substitute all values and maintain the coefficients as exact fractions to avoid floating-point errors that could incorrectly result in complex conjugate solutions.

$$\boxed{1.40 \text{ to } 1.50}$$

Q5.23.2 NAT | 2023 | 2M | Ans: 99.0 to 101.0

Given the AC source voltage:

$$V_s = 10\sqrt{2}\sin(20000\pi t)\ \text{V}$$

From the given expression, the angular frequency ω of the system is:

$$\omega = 20000\pi\ \text{rad/s}$$

The total current I flowing through the series RLC circuit is given by:

$$I = \frac{V_s}{50 + 25 + 25 + j\omega L + \frac{1}{j\omega C}}$$

Combining the series resistive elements, the total resistance R is:

$$R = 50 + 25 + 25 = 100\,\Omega$$

Substituting the given reactive component values, $L = \frac{10^{-3}}{20\pi}$ H and $C = \frac{10^{-3}}{20\pi}$ F:

$$j\omega L = j\,(20000\pi) \times \frac{10^{-3}}{20\pi} = j1\,\Omega$$

$$\frac{1}{j\omega C} = \frac{1}{j\,(20000\pi) \times \frac{10^{-3}}{20\pi}} = \frac{1}{j1} = -j1\,\Omega$$

Substitute these values back into the expression for current I:

$$I = \frac{V_s}{100 + j1 - j1} = \frac{V_s}{100}$$

Substituting the full time-domain expression for V_s:

$$I = \frac{10\sqrt{2}}{100}\sin(20000\pi t) = 0.1\sqrt{2}\sin(20000\pi t)\ \text{A}$$

The maximum (peak) value of the current is $I_m = 0.1\sqrt{2}$ A. Therefore, the root-mean-square current I_{rms} is:

$$I_{\text{rms}} = \frac{I_m}{\sqrt{2}} = \frac{0.1\sqrt{2}}{\sqrt{2}} = 0.1\ \text{A} = 100\ \text{mA}$$

$$\boxed{99.0 \text{ to } 101.0}$$

Q5.22.1 NAT 2022 1M Ans: 6.22 to 6.42

The quality factor is mathematically defined as:

$$Q = \frac{1}{R}\sqrt{\frac{L_o}{C_o}}$$

Given the following circuit parameters:

$$R = 50\,\Omega, \quad L_o = 1\,\text{mH} = 1\times10^{-3}\,\text{H}, \quad \text{and} \quad C_o = 10\,\text{nF} = 10\times10^{-9}\,\text{F}$$

Substituting these values into the Q-factor formula:

$$Q = \frac{1}{50}\sqrt{\frac{1\times10^{-3}}{10\times10^{-9}}}$$
$$Q \approx 6.32$$
$$\boxed{6.22\ to\ 6.42}$$

Q5.22.2 NAT 2022 2M Ans: 0.8 to 0.8

The load impedance Z_l of the given network is composed of a parallel branch in series with a resistive-inductive branch:

$$Z_l = \left(R_1 \parallel \frac{1}{j\omega C_1}\right) + R_2 + j\omega L_1$$

Let us express the parallel combination in its simplified rectangular form:

$$R_1 \parallel \frac{1}{j\omega C_1} = \frac{R_1\left(\frac{1}{j\omega C_1}\right)}{R_1 + \frac{1}{j\omega C_1}} = \frac{R_1}{1 + j\omega C_1 R_1}$$

Substituting this back into the total load impedance equation:

$$Z_l = \frac{R_1}{1 + j\omega C_1 R_1} + R_2 + j\omega L_1$$

$$Z_l = \frac{R_1 + (R_2 + j\omega L_1)(1 + j\omega C_1 R_1)}{1 + j\omega C_1 R_1}$$

Expanding the numerator:

$$Z_l = \frac{R_1 + R_2 + j\omega C_1 R_1 R_2 + j\omega L_1 - \omega^2 L_1 C_1 R_1}{1 + j\omega C_1 R_1}$$

Grouping the real and imaginary parts in the numerator:

$$Z_l = \frac{(R_1 + R_2 - \omega^2 L_1 C_1 R_1) + j(\omega C_1 R_1 R_2 + j\omega L_1)}{1 + j\omega C_1 R_1}$$

The phase angle of the total load impedance $\angle Z_l$ is given by the difference between the numerator phase angle and the denominator phase angle:

$$\angle Z_l = \tan^{-1}\left(\frac{\omega C_1 R_1 R_2 + \omega L_1}{R_1 + R_2 - \omega^2 L_1 C_1 R_1}\right) - \tan^{-1}(\omega C_1 R_1)$$

Given the following circuit parameters:

$$R_1 = 20\,\Omega$$
$$R_2 = 4\,\Omega$$
$$L_1 = \frac{20}{\pi}\,\text{mH} = \frac{20}{\pi}\times10^{-3}\,\text{H}$$
$$C_1 = \frac{1000}{\pi}\,\mu\text{F} = \frac{1000}{\pi}\times10^{-6}\,\text{F}$$
$$f = 50\,\text{Hz} \implies \omega = 2\pi f = 100\pi\ \text{rad/s}$$

Let us compute the individual intermediate numeric parameters:

$$\omega L_1 = 100\pi \times \frac{20}{\pi} \times 10^{-3} = 2\,\Omega$$

$$\omega C_1 = 100\pi \times \frac{1000}{\pi} \times 10^{-6} = 0.1\,\text{S}$$

$$\omega C_1 R_1 = 0.1 \times 20 = 2$$

$$\omega C_1 R_1 R_2 = 0.1 \times 20 \times 4 = 8\,\Omega$$

$$\omega^2 L_1 C_1 R_1 = (\omega L_1)(\omega C_1 R_1) = 2 \times 2 = 4\,\Omega$$

Substituting these intermediate values into the impedance angle equation:

$$\angle Z_l = \tan^{-1}\left(\frac{8+2}{20+4-4}\right) - \tan^{-1}(2)$$

$$\angle Z_l = \tan^{-1}\left(\frac{10}{20}\right) - \tan^{-1}(2) = \tan^{-1}(0.5) - \tan^{-1}(2)$$

Using values $\tan^{-1}(0.5) \approx 26.57°$ and $\tan^{-1}(2) \approx 63.43°$:

$$\angle Z_l = 26.57° - 63.43° = -36.87°$$

Taking the reference voltage phase as $\angle V_1 = 0°$, the current phasor relative to the source is:

$$I = \frac{V_1}{Z_l} \implies \angle I = \angle V_1 - \angle Z_l = 0° - (-36.87°) = +36.87°$$

Because the angle is positive, the current leads the voltage by 36.87°. Thus, the power factor (pf) is:

$$\text{pf} = \cos(36.87°) = 0.8 \quad \text{(leading)}$$

$$\boxed{0.8}$$

Q5.21.1 NAT 2021 2M Ans: 39 to 40

Active power P consumed by an electrical machine such as a fan is given as 60 W. The apparent power S consumed by the fan can be calculated from the peak (maximum) values of the voltage and current waveforms:

$$S = \frac{|V_{\max}||I_{\max}|}{2}$$

By converting the peak parameters into their corresponding root-mean-square (RMS) equivalents:

$$S = \frac{|V_{\max}|}{\sqrt{2}} \cdot \frac{|I_{\max}|}{\sqrt{2}} = |V_{\text{rms}}||I_{\text{rms}}|$$

Given that $V_{\text{rms}} = 230$ V and $I_{\text{rms}} = 0.3125$ A:

$$S = 230 \times 0.3125 = 71.875\ \text{VA}$$

The reactive power Q can be derived from the power triangle relationship:

$$S^2 = P^2 + Q^2$$

$$Q = \sqrt{S^2 - P^2}$$

Substituting the computed value of apparent power S and the given active power P:

$$Q = \sqrt{71.875^2 - 60^2}$$

$$Q = \sqrt{5166.0156 - 3600} = \sqrt{1566.0156} \approx 39.57\ \text{VAR}$$

Rounding the result to the nearest integer gives:

$$Q \approx 40\ \text{VAR}$$

$$\boxed{40}$$

Key Points

- The power triangle establishes a geometric relationship between active power (P in W), reactive power (Q in VAR), and apparent power (S in VA) such that $S = \sqrt{P^2 + Q^2}$.

Q5.21.2 NAT 2021 2M Ans: 30 to 30 ● ○ ○

The maximum voltage across a capacitor in a variable-parameter series RLC circuit occurs when the circuit is adjusted to satisfy the series resonance condition.
At the resonance condition, the inductive reactance (X_L) balances the capacitive reactance (X_C), which minimizes the total impedance to its purely resistive component ($Z = R$). Therefore, the loop current I achieves its maximum value:

$$I = \frac{V}{R} = \frac{10}{10} = 1\,\text{A}$$

At this resonant condition, the magnitudes of the voltage drops across the inductor (V_L) and the capacitor (V_C) are equal and reach their peak operating values:

$$V_L = V_C = 30\,\text{V}$$

Expressing the voltage drop across the inductor in terms of the line current and its inductive reactance gives:

$$I\omega L = 30\,\text{V}$$

Substituting the computed current $I = 1$ A:

$$1 \times \omega L = 30 \implies \omega L = 30$$

Given that the operating angular frequency of the system is $\omega = 1000\,\text{rad/s}$, we solve for the series inductance L:

$$L = \frac{30}{1000} = 0.03\,\text{H} = 30\,\text{mH}$$

$$\boxed{30}$$

Q5.20.1 NAT 2020 1M Ans: 5 to 5 ● ○ ○

The capacitance of the variable capacitive sensor is defined as:

$$C_x = 1000x\,\text{pF}$$

Given that the input to the sensor is $x = 0.1$:

$$C_x = 1000 \times 0.1\,\text{pF} = 100\,\text{pF}$$

The internal connecting cable capacitance is also given as:

$$C_{\text{cable}} = 100\,\text{pF}$$

The source voltage applied to the measurement system is defined in the time domain as:

$$V_i(t) = 10\sin(100\pi t)\,\text{V}$$

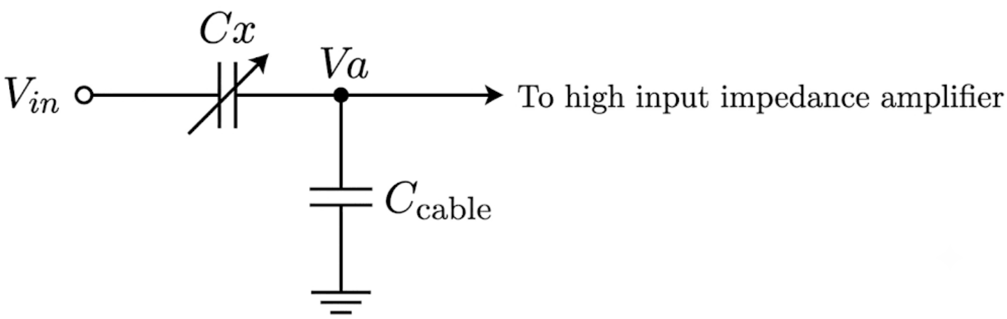


Fig. Solution Q5.20.1

From this time-varying equation, the peak input magnitude is $V_i = 10\,\text{V}$. Since the signal conditioning amplifier is specified to have an extremely high input impedance, there is no loading effect acting upon the transducer network.

The peak value of the output voltage V_A appearing across the amplifier input terminals can be derived using the standard AC capacitive voltage divider configuration:

$$V_A = V_i \times \frac{\frac{1}{j\omega C_{\text{cable}}}}{\frac{1}{j\omega C_{\text{cable}}} + \frac{1}{j\omega C_x}}$$

Multiplying the numerator and denominator by $j\omega C_{\text{cable}} C_x$ simplifies the relationship to:

$$V_A = V_i \times \frac{C_x}{C_x + C_{\text{cable}}}$$

Substituting the computed capacitances and peak input voltage into the simplified divider relation:

$$V_A = 10 \times \frac{100 \times 10^{-12}}{(100 + 100) \times 10^{-12}} = 10 \times \frac{100}{200} = 5\,\text{V}$$

Key Points

- High input impedance buffers isolate capacitive sensors from loading errors, ensuring that the voltage transfer ratio remains purely a function of the internal reactances.

5

Q5.20.2 NAT 2020 2M Ans: 1 to 1

Given parameters from the circuit problem: Operating frequency, $\omega = 1000\,\text{rad/s}$

The inductive reactance is given by:

$$X_L = j\omega L = j \cdot 1000 \cdot 100 \cdot 10^{-3} = j100\,\Omega$$

The capacitive reactance is given by:

$$X_C = \frac{1}{j\omega C} = \frac{1}{j \cdot 1000 \cdot 10 \cdot 10^{-6}} = -j100\,\Omega$$

The total impedance of the series network is:

$$Z = R + X_L + X_C$$

$$Z = 100 + j100 - j100 = 100\,\Omega$$

Since the net reactive component is zero, the circuit is purely resistive at this frequency (resonance condition). Given the voltage source waveform as $v(t) = 100\sqrt{2}\cos(1000t)$ V, the current waveform $i(t)$ is:

$$i(t) = \frac{v(t)}{R} = \frac{100\sqrt{2}\cos(1000t)}{100} = \sqrt{2}\cos(1000t)\text{ A}$$

From the current expression: The peak value of the current is $I_m = \sqrt{2}$ A. The RMS value of the current is:

$$I_{\text{rms}} = \frac{I_m}{\sqrt{2}} = \frac{\sqrt{2}}{\sqrt{2}} = 1\text{ A}$$

$$\boxed{1}$$

Q5.20.3 NAT 2020 1M Ans: 1.7 to 1.8

Given a balanced delta-connected (Δ) load system with the following specifications : Line voltage, $V_{\text{line}} = 400$ V Phase resistance, $R_{\text{ph}} = 400\,\Omega$ In a balanced delta-connected network, the phase voltage V_{ph} is equal to the line voltage V_{line}:

$$V_{\text{ph}} = V_{\text{line}} = 400\text{ V}$$

The phase current I_{ph} through each resistive arm is calculated using Ohm's law:

$$I_{\text{ph}} = \frac{V_{\text{ph}}}{R_{\text{ph}}} = \frac{400}{400} = 1\text{ A}$$

For a balanced delta connection, the relationship between the magnitudes of line current I_{line} and phase current I_{ph} is given by:

$$I_{\text{line}} = \sqrt{3} \cdot I_{\text{ph}}$$

Substituting the value of phase current:

$$I_{\text{line}} = \sqrt{3} \cdot 1 = 1.732\text{ A}$$

$$\boxed{1.732}$$

Q5.20.4 NAT 2020 2M Ans: 115 to 116

From the given circuit networks, we first find the equivalent resistance of the parallel branch:

$$R = 300\,\Omega \parallel 100\,\Omega = \frac{300 \cdot 100}{300 + 100} = 75\,\Omega$$

Replacing the parallel combination with its equivalent resistance $R = 75\,\Omega$, we obtain the simplified loop network.

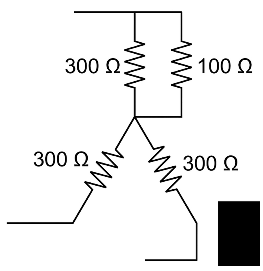


Fig. Solution Q5.20.4

Applying KVL for mesh current I_1 and mesh current I_2 considering the line voltages in their correct phase sequence:

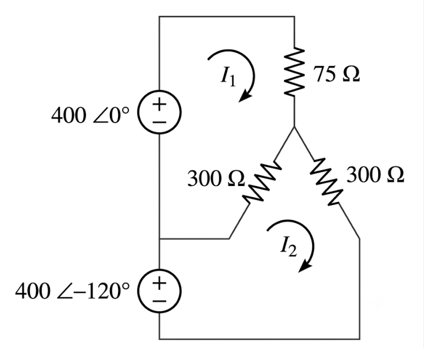


Fig. Solution Q5.20.4

For Loop 1:

$$(75 + 300)I_1 - 300I_2 = 400\angle 0^\circ$$

$$375I_1 - 300I_2 = 400\angle 0^\circ \quad \text{— (1)}$$

For Loop 2:

$$-300I_1 + (300 + 300)I_2 = 400\angle - 120^\circ$$

$$-300I_1 + 600I_2 = 400\angle - 120^\circ \quad \text{— (2)}$$

Solving equations (1) and (2) simultaneously for the mesh current I_1:

$$I_1 = 1.5396\angle - 30^\circ \text{ A}$$

The voltage V across the parallel combination (and hence across the 100 Ω resistor) is given by:

$$V = 75 \cdot I_1 = 75 \cdot 1.5396\angle - 30^\circ = 115.47\angle - 30^\circ \text{ V}$$

Thus, the magnitude of the voltage across the 100 Ω resistor is 115.47 V.

$$\boxed{115 \textit{ to } 116}$$

Q5.19.1 MCQ 2019 1M Ans: B ● ○ ○

We know that the quality factor (Q) of a series RLC circuit is given by:

$$Q = \frac{1}{R}\sqrt{\frac{L}{C}}$$

If all the components are doubled, i.e., the values become $2R$, $2L$, and $2C$, then the new quality factor Q' would be:

$$Q' = \frac{1}{2R}\sqrt{\frac{2L}{2C}}$$

$$Q' = \frac{1}{2R}\sqrt{\frac{L}{C}} = \frac{Q}{2}$$

Thus, the quality factor reduces by a factor of 2.

$$\boxed{(b) \textit{reduce by a factor of } 2}$$

Q5.19.2 | NAT | 2019 | 2M | Ans: 20000 to 20000 | ● ○ ○

Current in phase with voltage signifies resonance. This means calculating the equivalent input admittance and setting the imaginary part to zero.

The input admittance Y_{in} for a parallel RLC circuit is given by:

$$Y_{in} = \frac{1}{R} + \frac{1}{j\omega_o L} + j\omega_o C$$

where ω_o is the resonant frequency.

Simplifying the equation by rationalizing the imaginary terms:

$$Y_{in} = \frac{1}{R} + j\left(\omega_o C - \frac{1}{\omega_o L}\right)$$

At resonance, the imaginary part of the admittance must be zero:

$$\mathrm{Im}(Y_{in}) = 0$$

$$\omega_o C - \frac{1}{\omega_o L} = 0$$

$$\omega_o = \frac{1}{\sqrt{LC}}$$

Substituting the given component values ($L = 5 \times 10^{-3}$ H and $C = 500 \times 10^{-9}$ F):

$$\omega_o = \frac{1}{\sqrt{5 \times 10^{-3}\,\mathrm{H} \times 500 \times 10^{-9}\,\mathrm{F}}}$$

$$\omega_o = 20000\,\mathrm{rad/s}$$

$$\boxed{20000}$$

Q5.18.1 | MCQ | 2018 | 1M | Ans: A | ● ● ○

Consider the given series RC circuit connected to a sinusoidal voltage source $V = 1$ V (taken as a reference with a magnitude of 1).

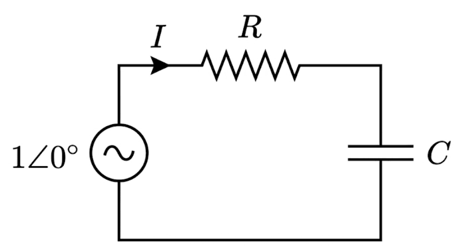


Fig. Solution Q5.18.1

The total input impedance Z of the circuit is:

$$Z = R + \frac{1}{j\omega C}$$

The current I flowing through the circuit is given by:

$$I = \frac{1}{R + \frac{1}{j\omega C}} = \frac{j\omega C}{j\omega RC + 1}$$

To find the real and imaginary parts, we rationalize the expression by multiplying the numerator and denominator by the complex conjugate of the denominator $(1 - j\omega RC)$:

$$I = \frac{j\omega C(1 - j\omega RC)}{1 + \omega^2 R^2 C^2}$$

$$I = \frac{\omega^2 RC^2}{1 + \omega^2 R^2 C^2} + j\frac{\omega C}{1 + \omega^2 R^2 C^2}$$

Let the real part be x and the imaginary part be y:

$$x = \frac{\omega^2 RC^2}{1 + \omega^2 R^2 C^2}, \quad y = \frac{\omega C}{1 + \omega^2 R^2 C^2}$$

We analyze the behavior of the current locus by checking specific boundary frequencies:

- **At $\omega = 0$:**

$$x = 0, \quad y = 0$$

- **At $\omega \to \infty$:** Approximating for $\omega^2 R^2 C^2 \gg 1$:

$$x = \frac{\omega^2 RC^2}{\omega^2 R^2 C^2} = \frac{1}{R}$$

$$y = \frac{\omega C}{\omega^2 R^2 C^2} = \frac{1}{\omega R^2 C} = 0$$

- **At $\omega = \frac{1}{RC}$:**

$$x = \frac{\left(\frac{1}{RC}\right)^2 RC^2}{1 + \left(\frac{1}{RC}\right)^2 R^2 C^2} = \frac{\frac{1}{R}}{1+1} = \frac{1}{2R}$$

$$y = \frac{\left(\frac{1}{RC}\right) C}{1 + \left(\frac{1}{RC}\right)^2 R^2 C^2} = \frac{\frac{1}{R}}{1+1} = \frac{1}{2R}$$

Since both $x \geq 0$ and $y \geq 0$ for all frequencies $\omega \in [0, \infty)$, the entire locus lies within the first quadrant.
Thus, option A is a suitable match.

Key Points

- The locus plot of the admittance or current of a series RC circuit forms a semicircle located in the upper-half complex plane (first quadrant) when plotted with respect to varying frequency ω.
- At zero frequency ($\omega = 0$), the capacitor acts as an open circuit, making the current zero. At infinite frequency ($\omega \to \infty$), the capacitor acts as a short circuit, and the current is limited only by the resistor R, reaching $\frac{1}{R}$.

$\boxed{a}$

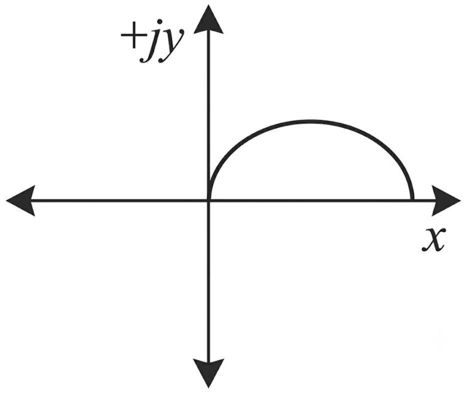


Fig. Solution Q5.18.1

Q5.18.2 NAT 2018 1M Ans: 1009 to 1011

Let us calculate the equivalent admittance between terminals P and q, taking the capacitance as C and the operating angular frequency as ω. The total input admittance Y_{pq} is given by:

$$Y_{pq} = \frac{1}{j100 + 10} + j\omega C$$

To separate the real and imaginary components, we multiply the numerator and denominator of the first term by its complex conjugate $(10 - j100)$:

$$Y_{pq} = \frac{10 - j100}{10^2 + 100^2} + j\omega C$$

$$Y_{pq} = \frac{10}{10^2 + 100^2} + j\left(\omega C - \frac{100}{10^2 + 100^2}\right)$$

At resonance, the equivalent admittance Y_{pq} is purely real, which implies that its imaginary part must equal zero:

$$\text{Im}(Y_{pq}) = 0$$

Therefore, at resonance, the admittance simplifies to:

$$Y_{pq} = \frac{10}{10^2 + 100^2}$$

The equivalent impedance Z_{pq} across the terminals at resonance is the reciprocal of the admittance:

$$Z_{pq} = \frac{1}{Y_{pq}} = \frac{10^2 + 100^2}{10}$$

$$Z_{pq} = \frac{100 + 10000}{10} = \frac{10100}{10} = 1010\,\Omega$$

$$\boxed{1009 \text{ to } 1011}$$

Q5.18.3 NAT 2018 2M Ans: 66 to 68 ● ● ○

From the given circuit diagram, an AC source of 230 V, 50 Hz supplies two parallel branches. Let the supply voltage phasor be the reference phasor:

$$V = 230\angle 0^\circ \text{ V}$$

The impedance of the first branch containing the inductor and resistor in series is:

$$Z_1 = 162.6 + j162.6\,\Omega$$

The current I_1 through the first branch is:

$$I_1 = \frac{V}{Z_1} = \frac{230\angle 0^\circ}{162.6 + j162.6} \approx 1\angle -45^\circ \text{ A}$$

The impedance of the second branch containing the capacitor is:

$$Z_2 = -j230\,\Omega = 230\angle -90^\circ\,\Omega$$

The current I_2 through the capacitive branch is:

$$I_2 = \frac{V}{Z_2} = \frac{230\angle 0^\circ}{230\angle -90^\circ} = 1\angle 90^\circ \text{ A} = j1 \text{ A}$$

Using Kirchhoff's Current Law (KCL), the total line current I is the phasor sum of branch currents:

$$I = I_1 + I_2 = 1\angle -45^\circ + j1$$

$$I = 0.707 - j0.707 + j1 = 0.707 + j0.293 \text{ A}$$

Converting the total current into polar form:

$$I \approx 0.765\angle 22.51^\circ \text{ A}$$

The phase angle θ between the supply voltage and the total line current is 22.51°. The reactive power Q taken by the circuit is given by:

$$Q = VI\sin\theta$$

$$Q = 230 \cdot 0.765 \cdot \sin(22.51^\circ) \approx 67.36 \text{ VAR}$$

$$\boxed{66 \text{ to } 68}$$

Q5.17.1 NAT 2017 1M Ans: 50 to 50 ● ○ ○

Taking the current I through the series circuit as reference, the phasor relation is as follows.

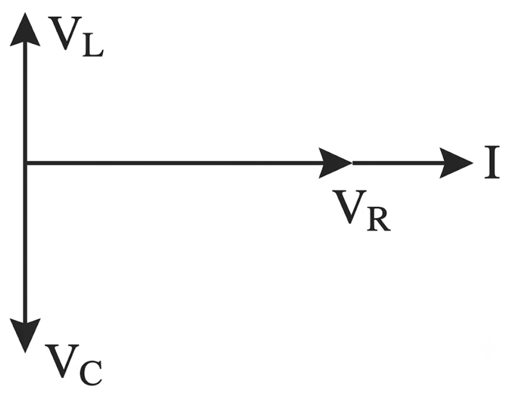


Fig. Solution Q5.17.1

V_L and V_C can be directly added or subtracted, and this resultant voltage is at a right angle to V_R.
The total source voltage phasor $\vec{V}_s$ is given by:

$$\vec{V}_s = \vec{V}_R + \vec{V}_L + \vec{V}_C$$

Taking the magnitude:

$$|V_s| = \sqrt{V_R^2 + (V_L - V_C)^2}$$

Given that $V_s = 50\,\text{V}$ and $V_R = 50\,\text{V}$:

$$50 = \sqrt{50^2 + (V_L - V_C)^2}$$

$$50^2 = 50^2 + (V_L - V_C)^2$$

$$(V_L - V_C)^2 = 0$$

$$V_L - V_C = 0 \implies V_L = V_C = 50V$$

The circuit is in resonance.

Key Points

- In a series RLC circuit, the voltage across the inductor (V_L) leads the current by 90°, while the voltage across the capacitor (V_C) lags the current by 90°.
- When the total source voltage equals the voltage across the resistor ($V_s = V_R$), the net reactive voltage is zero, which is the defining condition for series resonance.

$\boxed{50}$

Q5.17.2 | NAT | 2017 | 2M | Ans: 1.75 to 1.96 | ● ● ○

Both the loads have $200\angle 0°$ V across them.

The total complex apparent power S is given by:

$$S = VI^*$$

$$S = V(I_1 + I_2)^*$$

$$S = VI_1^* + VI_2^*$$

Substituting the given values ($V = 200\angle 0°$ V, $I_1 = 5\angle 0°$ A, and $I_2 = 5\angle 30°$ A):

$$S = 200\angle 0° \times (5\angle 0°)^* + 200\angle 0° \times (5\angle 30°)^+$$

$$S = 200 \times 5 + 200 \times 5\angle -30°$$

$$S = 1000 + 1000\angle -30°$$

Converting to polar form:

$$S = 1931.85\angle -15°\ \text{VA}$$

The total real power P is the real part of the complex apparent power S:

$$P = \text{Re}(S)$$

$$P = 1931.85\cos(-15°)$$

$$P = 1866\,\text{W} = 1.867\,\text{kW}$$

Key Points

- Total apparent complex power for parallel connected loads can be computed by summing up individual complex powers: $S = S_1 + S_2 + \cdots + S_n$.

$\boxed{1.75 \textit{ to } 1.96}$

Q5.17.3 NAT | 2017 | 2M | Ans: 6 to 7 | ● ○ ○

Given parameters from the series RLC resonant circuit: Inductance, $L = 10 \cdot 10^{-3}$ H Capacitance, $C = 4 \cdot 10^{-6}$ F Quality factor, $Q = 30$ Source voltage, $V = 15$ V

The resonant frequency ω_0 of the circuit is calculated as:

$$\omega_0 = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{10 \cdot 10^{-3} \cdot 4 \cdot 10^{-6}}} = 5000\,\text{rad/s}$$

The quality factor Q for a series resonant circuit is defined as:

$$Q = \frac{\omega_0 L}{R}$$

Substituting the known values to solve for the circuit resistance R:

$$30 = \frac{5000 \cdot (10 \cdot 10^{-3})}{R}$$

$$R = \frac{50}{30} = \frac{5}{3}\,\Omega$$

At the resonant frequency ω_0, the total net reactance is zero, making the circuit purely resistive. Thus, the current at resonance I_0 is:

$$I_0 = \frac{V}{R} = \frac{15}{5/3} = 9\,\text{A}$$

At the half-power frequency (upper cutoff frequency ω_2), the current drops to $\frac{1}{\sqrt{2}}$ times the maximum current at resonance (as power is proportional to square of current) :

$$I = \frac{I_0}{\sqrt{2}} = \frac{9}{\sqrt{2}} \approx 6.36\,\text{A}$$

Key Points

- At resonance, circuit impedance reaches its absolute minimum value ($Z = R$), driving the network current to its maximum amplitude.
- At the half-power bandwidth cutoff frequencies (ω_1 and ω_2), the current magnitude decreases to exactly 70.7% ($\frac{1}{\sqrt{2}}$) of its peak resonant value.

$\boxed{6 \textit{ to } 7}$

Q5.16.1 NAT 2016 1M Ans: 5 to 5 ● ○ ○

Applying Kirchhoff's Current Law (KCL) at the node:

$$i_1 = i_2 + i_3$$

$$i_3 = i_1 - i_2$$

Given the expressions for the currents ($i_1 = 3\cos\omega t$ and $i_2 = 4\sin\omega t$):

$$i_3 = 3\cos\omega t - 4\sin\omega t$$

To combine these trigonometric terms into a single cosine wave, we rewrite the expression as:

$$i_3 = \sqrt{3^2+4^2}\left(\frac{3}{\sqrt{3^2+4^2}}\cos\omega t - \frac{4}{\sqrt{3^2+4^2}}\sin\omega t\right)$$

$$i_3 = 5\left(\frac{3}{5}\cos\omega t - \frac{4}{5}\sin\omega t\right)$$

Let $\theta = \tan^{-1}\left(\frac{4}{3}\right)$, which gives:

$$\cos\theta = \frac{3}{5}, \quad \sin\theta = \frac{4}{5}$$

Substituting these into the equation for i_3:

$$i_3 = 5(\cos\theta\cos\omega t - \sin\theta\sin\omega t)$$

Using the trigonometric identity $\cos(A+B) = \cos A\cos B - \sin A\sin B$:

$$i_3 = 5\cos(\omega t + \theta)$$

Comparing this result with the standard sinusoidal expression for current $i_3 = I_3\cos(\omega t + \theta)$, the amplitude is found to be:

$$I_3 = 5\,\mathrm{A}$$

$$\boxed{5}$$

Q5.16.2 NAT 2016 1M Ans: 1 to 1 ● ○ ○

An air-cored coil can be modeled as a series RL circuit. The quality factor (Q) of a series coil is given by the relation:

$$Q = \frac{\omega_o L}{R} = \frac{2\pi f L}{R}$$

If the inductance (L) and resistance (R) remain constant, then the quality factor is directly proportional to the operating frequency (f):

$$Q \propto f$$

Given that at a frequency $f_1 = 100\,\mathrm{kHz}$, the quality factor is $Q_1 = 5$. We need to find the new quality factor Q_2 at a frequency $f_2 = 20\,\mathrm{kHz}$:

$$\frac{Q_2}{Q_1} = \frac{f_2}{f_1}$$

$$Q_2 = Q_1 \times \frac{f_2}{f_1}$$

Substituting the given values into the proportionality expression:

$$Q_2 = 5 \times \frac{20\,\mathrm{kHz}}{100\,\mathrm{kHz}} = 5 \times \frac{20000}{100000} = 1$$

$$\boxed{1}$$

Q5.16.3 NAT 2016 1M Ans: 0.280 to 0.283 ● ● ○

Given the circuit network containing two parallel LC tank filters connected in series, the left one named LC_1 and right one named LC_2 the input voltage consists of two harmonic components:

$$v(t) = v_1(t) + v_2(t) = 1\sin(2\pi \cdot 10000t)\text{ V} + 1\sin(2\pi \cdot 30000t)\text{ V}$$

The frequencies of the input components are:

$$f_1 = 10000\,\text{Hz}$$

$$f_2 = 30000\,\text{Hz} = 3f_1$$

Let us examine the second tank network (LC$_2$): Inductance, $L_2 = 100\,\mu\text{H} = 100 \cdot 10^{-6}\,\text{H}$ Capacitance, $C_2 = 2.53\,\mu\text{F} = 2.53 \cdot 10^{-6}\,\text{F}$

The resonant frequency f_{02} of the parallel network LC$_2$ is given by:

$$f_{02} = \frac{1}{2\pi\sqrt{L_2C_2}}$$

$$f_{02} = \frac{1}{2\pi\sqrt{100 \cdot 10^{-6} \cdot 2.53 \cdot 10^{-6}}} \approx 10000\,\text{Hz}$$

Since $f_1 = f_{02} = 10000\,\text{Hz}$, the network LC$_2$ experiences parallel resonance at frequency f_1. At parallel resonance, the ideal tank filter acts as an open circuit (infinite impedance), blocking any current loop at f_1 ($I = 0$ for component V_1).

To completely eliminate or block the second component V_2 at frequency $f_2 = 30000\,\text{Hz}$ from driving current through the load resistor R, the first tank network (LC$_1$) must be designed to present parallel resonance at f_2:

$$f_2 = 30000\,\text{Hz} = \frac{1}{2\pi\sqrt{L_1C_1}}$$

Given $L_1 = 100\,\mu\text{H} = 100 \cdot 10^{-6}\,\text{H}$, we solve for C_1:

$$C_1 = \frac{1}{(2\pi)^2 \cdot f_2^2 \cdot L_1}$$

$$C_1 = \frac{1}{(2\pi)^2 \cdot (30000)^2 \cdot 100 \cdot 10^{-6}} \approx 0.281 \cdot 10^{-6}\,\text{F} = 0.28\,\mu\text{F}$$

$$\boxed{0.280 \text{ to } 0.283}$$

Q5.16.4 NAT 2016 2M Ans: 99.5 to 100.5 ● ○ ○

The equivalent admittance Y_{eq} seen by the voltage source is:

$$Y_{eq} = \frac{1}{R + j\omega L} + j\omega C$$

Multiplying the numerator and denominator of the first term by the complex conjugate $(R - j\omega L)$:

$$Y_{eq} = \frac{R - j\omega L}{R^2 + \omega^2 L^2} + j\omega C$$

$$Y_{eq} = \frac{R}{R^2 + \omega^2 L^2} + j\left(\omega C - \frac{\omega L}{R^2 + \omega^2 L^2}\right)$$

Since the source current I_s and source voltage V_s are in phase, the circuit operates at resonance, which requires the imaginary part of the equivalent admittance to be zero:

$$\text{Im}(Y_{eq}) = 0$$

Therefore, at resonance, the equivalent admittance simplifies to its real part:

$$Y_{eq} = \frac{R}{R^2 + \omega^2 L^2}$$

Given the values from the circuit parameters ($R = 10\,\Omega$, $\omega L = 100\,\Omega$, and $V_s = 101$ V), we compute the source current magnitude I_s:

$$I_s = V_s Y_{eq} = 101 \times \frac{10}{10^2 + 100^2}$$

$$I_s = 101 \times \frac{10}{10100} = 0.1\text{ A} = 100\text{ mA}$$

$$\boxed{100}$$

Q5.15.1 NAT 2015 2M Ans: 98 to 102 ● ○ ○

At resonance in a series RLC circuit, the inductive reactance and capacitive reactance cancel each other out ($X_L = X_C$), making the circuit purely resistive. As a result, the entire source voltage appears completely across the resistor R:

$$V_R = 1\text{ V}$$

The magnitude of the voltage across the capacitor ($|V_C|$) at resonance is enhanced by the quality factor (Q) of the circuit and is given by:

$$|V_C| = QV_R$$

The quality factor (Q) for a series RLC circuit is calculated using the formula:

$$Q = \frac{1}{R}\sqrt{\frac{L}{C}}$$

Given the component values ($R = 10\,\Omega$, $L = 0.1$ H, and $C = 0.1 \times 10^{-6}$ F):

$$Q = \frac{1}{10}\sqrt{\frac{0.1}{0.1 \times 10^{-6}}}$$

$$Q = \frac{1}{10} \times \sqrt{10^6} = \frac{1}{10} \times 10^3 = 100$$

Now, we substitute the value of Q back into the expression for the capacitor voltage magnitude:

$$|V_C| = 100 \times 1\text{ V} = 100\text{ V}$$

$$\boxed{98 \textit{ to } 102}$$

Q5.15.2 NAT 2015 2M Ans: 95 to 96 ● ○ ○

The output frequency (f) of an LC tank oscillator is given by the relation:

$$f = \frac{1}{2\pi\sqrt{LC}}\text{ Hz}$$

Given the initial frequency of the oscillator:

$$\frac{1}{2\pi\sqrt{LC}} = 100 \times 10^3\text{ Hz} = 100\text{ kHz}$$

Now, if the capacitance C is increased by 10%, the new capacitance value C' becomes:

$$C' = C + 0.1C = 1.1C$$

The expression for the new operating frequency f' is:

$$f' = \frac{1}{2\pi\sqrt{L \times 1.1C}}$$

Factoring out the constant multiplier from the square root:

$$f' = \frac{1}{\sqrt{1.1}} \times \left(\frac{1}{2\pi\sqrt{LC}}\right)$$

Substituting the initial frequency value (100×10^3 Hz):

$$f' = \frac{100 \times 10^3}{\sqrt{1.1}}$$

$$f' \approx 95.34\,\text{kHz}$$

$$\boxed{95 \text{ to } 96}$$

Q5.14.1 NAT 2014 1M Ans: 32 to 32

The average real power delivered to a load network by an AC current source is independent of the load's reactive parameters and is determined exclusively by:

- The real part of the overall load impedance, $\text{Re}(Z_L)$.
- The RMS value of the current flowing into the load network, I_{rms}.

The general expression for computing average real power is given by:

$$P = (I_{rms})^2 \times \text{Re}(Z_L)$$

Given the parameters specified in the engineering problem statement:

$$\text{Re}(Z_L) = 4\,\Omega$$

$$I_{rms} = \frac{4}{\sqrt{2}}\,\text{A}$$

Substituting these operational variables into the power formula:

$$P = \left(\frac{4}{\sqrt{2}}\right)^2 \times 4$$

$$P = \frac{16}{2} \times 4 = 8 \times 4 = 32\,\text{W}$$

Key Points

- Only the resistive component ($\text{Re}(Z_L)$) of an electrical load dissipates real power as thermal energy, while the reactive elements store and return energy periodically without dissipation.
- When working with sinusoidal current waveforms, ensuring that the current magnitude is properly transformed into its effective value ($I_{rms} = \frac{I_{max}}{\sqrt{2}}$) is essential for accurate active power calculations.

32

Q5.14.2 MCQ | 2014 | 1M | Ans: A | ● ● ○

To determine the phase relationship between the two sinusoidal signals, we must convert both expressions into the same standard trigonometric form (either both sine or both cosine) with positive amplitudes.

Given time-domain expressions:

$$v_1(t) = V_m \sin(10t - 130^\circ)$$

$$v_2(t) = V_m \cos(10t + 10^\circ)$$

Using the trigonometric identity $\cos(\theta) = \sin(\theta + 90^\circ)$, we can convert $v_2(t)$ into a sine function:

$$v_2(t) = V_m \sin(10t + 10^\circ + 90^\circ)$$

$$v_2(t) = V_m \sin(10t + 100^\circ)$$

Now, compare the phase angles of both signals in their sine representations: Phase of $v_1(t)$, $\phi_1 = -130^\circ$ Phase of $v_2(t)$, $\phi_2 = 100^\circ$

The phase difference $\Delta\phi$ to see how much $v_1(t)$ leads or lags $v_2(t)$ is:

$$\Delta\phi = \phi_1 - \phi_2 = -130^\circ - 100^\circ = -230^\circ$$

Since -230° is identical to a positive rotation of $+130^\circ$ (by adding 360°):

$$\Delta\phi = -230^\circ + 360^\circ = 130^\circ$$

A positive phase difference means $v_1(t)$ leads $v_2(t)$ by 130°.

Alternatively, expressing the lag relationship:

$$\phi_1 - \phi_2 = -230^\circ \implies v_1(t) \text{ lags } v_2(t) \text{ by } 230^\circ$$

Which is equivalent to:

$$v_1(t) \text{ leads } v_2(t) \text{ by } 130^\circ$$

Key Points

- When comparing phase differences between two waveforms, always convert them into identical functions (sin or cos) with positive coefficients.
- Use the identity $\cos(\omega t + \theta) = \sin(\omega t + \theta + 90^\circ)$.
- Phase angles are periodic over 360°; hence, an angle ϕ is equivalent to $\phi \pm 360^\circ$.

Mistakes to be avoided

- Do not compare the phases directly as -130° and $+10^\circ$ without standardizing the functions; this would lead to an incorrect phase difference of -140°.
- Ensure proper tracking of signs when adding or subtracting angles across the quadrant conversions.

$$\boxed{(a) v_1(t) \; leads \; v_2(t) \; by \; 130^\circ}$$

Q5.14.3 NAT | 2014 | 2M | Ans: 186 to 188 | ● ● ○

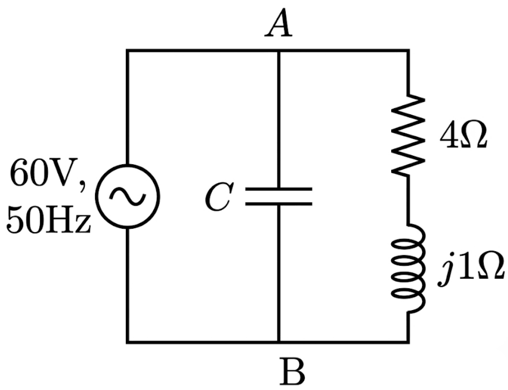


Fig. Solution Q5.14.3

Given the AC circuit operating at a frequency of 50 Hz, we need to find the capacitance C such that the circuit operates at unity power factor.

A unity power factor means the circuit is at resonance, which implies that the imaginary part of the equivalent admittance (Y_{eq}) or equivalent impedance (Z_{eq}) seen by the source is zero.

The equivalent admittance Y_{eq} seen across terminals A and B is the sum of the admittances of the capacitive branch and the parallel $R - L$ branch:

$$Y_{eq} = j\omega C + \frac{1}{4 + j1}$$

To rationalize the second term, multiply the numerator and denominator by the complex conjugate of the denominator $(4 - j1)$:

$$Y_{eq} = j\omega C + \frac{4 - j1}{4^2 + 1^2}$$

$$Y_{eq} = j\omega C + \frac{4 - j1}{17}$$

$$Y_{eq} = \frac{4}{17} + j\left(\omega C - \frac{1}{17}\right)$$

For unity power factor, the imaginary part of Y_{eq} must be zero:

$$\text{Im}(Y_{eq}) = 0$$

$$\omega C - \frac{1}{17} = 0$$

$$\omega C = \frac{1}{17}$$

Since $\omega = 2\pi f$ where $f = 50\,\text{Hz}$:

$$C = \frac{1}{17 \times 2\pi \times 50}$$

$$C \approx 1.87 \times 10^{-4}\,\text{F} = 187\,\mu\text{F}$$

$$\boxed{187}$$

Q5.13.1 MCQ 2013 1M Ans: C ● ○ ○

According to maximum power transfer theorem, the purely resistive load should have a value equal to the magnitude of the internal impedance of the source in order to have maximum power transferred to it.

$$R_L = |Z_{in}| \implies R_L = \sqrt{4^2 + 3^2} = 5\Omega$$

$$\boxed{(c)5}$$

Q5.13.2 MCQ 2013 2M Ans: C ● ○ ○

For Coil 1, the quality factor is given by:

$$q_1 = \frac{\omega L_1}{R_1}$$

Rearranging for the self-inductance of Coil 1:

$$L_1 = \frac{q_1 R_1}{\omega}$$

Similarly, for Coil 2, the quality factor is:

$$q_2 = \frac{\omega L_2}{R_2}$$

Rearranging for the self-inductance of Coil 2:

$$L_2 = \frac{q_2 R_2}{\omega}$$

When the two coils are connected in series, the equivalent parameters are:

- Equivalent inductance: $L_{eq} = L_1 + L_2$
- Equivalent resistance: $R_{eq} = R_1 + R_2$

The overall quality factor Q of the series combination is defined as:

$$Q = \frac{\omega(L_1 + L_2)}{R_1 + R_2}$$

Substituting the expressions for L_1 and L_2 into the equation:

$$Q = \frac{\omega\left[\frac{q_1 R_1}{\omega} + \frac{q_2 R_2}{\omega}\right]}{R_1 + R_2}$$

Simplifying the expression by cancelling ω:

$$\boxed{Q = \frac{q_1 R_1 + q_2 R_2}{R_1 + R_2}}$$

$$\boxed{(c)(q_1 R_1 + q_2 R_2)/(R_1 + R_2)}$$

Q5.13.3 MCQ 2013 2M Ans: C ● ● ○

To determine the open-circuit Thevenin voltage V_{th} across the load terminals where R_l is connected, we analyze the network by looking into from the load terminals (opening R_L) and thus output current is zero ($i_2 = 0\,\text{A}$).

Due to this thedependant voltage source $j40I_2$ contributes 0V and appears shorted, and the source voltage appears across the resistor ($3\,\Omega$) and the inductor ($4j\,\Omega$). The voltage drop V_l across the reactive inductive component is given by :

$$V_l = v_s \times \frac{4j}{3 + 4j}$$

$$V_l = 80\angle 90^\circ \text{ V}$$

Because the output loop terminal is completely open ($i_2 = 0$ A), there is zero potential drop across the series $j6 + 5\,\Omega$. Therefore,

$$V_{th} = 10V_l$$

Substituting the evaluated phasor value of V_l into the dependent balance relation:

$$V_{th} = 10 \times (80\angle 90^\circ) = 800\angle 90^\circ \text{ V}$$

$$\boxed{(c)800\angle 90^\circ}$$

Q5.12.1 MCQ 2012 1M Ans: B

It is given that $i(t) = 5cos(100\pi t + 100)$ A.

$$\therefore I_{rms} = \frac{5}{\sqrt{2}} \text{ A}$$

Average power delivered $P = I_{rms}^2 R$, where R = 4Ω Substituting, we get $P = 50W$

$$\boxed{(b)50W}$$

Q5.12.2 MCQ 2012 1M Ans: C

Let a ground node reference be established at the common junction point of the voltage and current sources, setting its potential to 0 V.

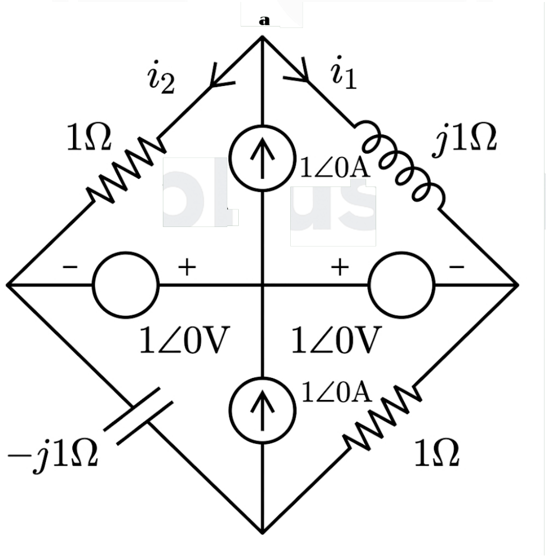


Fig. Solution Q5.12.2

Let the unknown node potential across the $1\angle 0^\circ$ A source be V.
Applying KVL along the meshes in the network :

$$1 = -V + ji_1$$

$$1 = -V + i_2$$

$$\implies ji_1 - i_2 = 0$$

Applying KCL at node ’a’ we get:

$$1 = i_1 + i_2$$

Combining the results we get :

$$ji_1 + i_1 = 1$$

$$\therefore\ i_1 = \frac{1}{1+j}\ \text{A}$$

$$\boxed{(c)\frac{1}{1+j}A}$$

Q5.12.3 | MCQ | 2012 | 2M | Ans: A | ● ● ○

First, we need to convert circuit B to its Thevenin equivalent form.

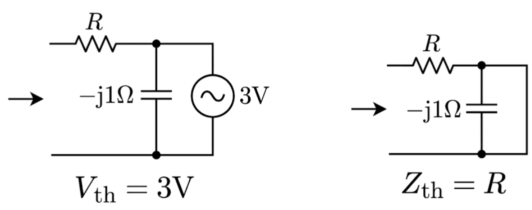


Fig. Solution Q5.12.3

The simplified circuit now looks like :

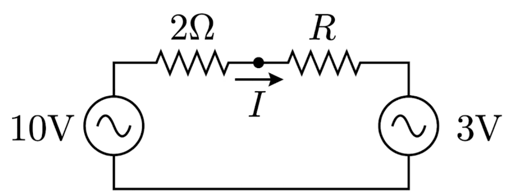


Fig. Solution Q5.12.3

The circuit current I as a function of load resistance R is expressed as:

$$I = \frac{10-3}{2+R} = \frac{7}{2+R}$$

The power transferred from circuit A to circuit B is given by the relation:

$$P = I^2R + 3I$$

Substituting the expression for current I:

$$P = I(IR + 3)$$

$$P = \left(\frac{7}{2+R}\right)\left[\frac{7R}{2+R} + 3\right]$$

$$P = \left(\frac{7}{2+R}\right)\left[\frac{7R + 3(2+R)}{2+R}\right]$$

$$P = \left(\frac{7}{2+R}\right)\left[\frac{7R + 6 + 3R}{2+R}\right]$$

$$P = \frac{7(10R+6)}{(2+R)^2} = \frac{70R+42}{(2+R)^2}$$

To find the condition for maximum power transfer, we set the first derivative of power with respect to R to zero ($\frac{dP}{dR} = 0$):

$$\frac{dP}{dR} = \frac{(2+R)^2 \cdot (70) - (70R+42) \cdot 2(2+R)}{(2+R)^4} = 0$$

Factoring out $(2+R)$ from the numerator (where $2+R \neq 0$):

$$(2+R)\left[70(2+R) - 2(70R+42)\right] = 0$$

$$140 + 70R - 140R - 84 = 0$$

$$56 - 70R = 0$$

$$R = \frac{56}{70} = 0.8\,\Omega$$

Since a negative solution ($R = -2\,\Omega$) is physically non-realizable for a passive resistor, it is ignored.

Key Points

- **Maximum Power Transfer Theorem (Modified):** When a circuit contains independent or dependent sources within the load network, the traditional condition $R_L = R_{th}$ changes. Maximization must be derived from fundamental calculus principles using $\frac{dP}{dR_L} = 0$.

Mistakes to be avoided

- Do not apply $R = R_{th} = 2\,\Omega$ here, because circuit B contains an active terminal characteristic ($3I$ component) rather than a purely passive linear resistor.

$$\boxed{(a) 0.8\Omega}$$

Q5.12.4 MCQ 2012 2M Ans: A ● ● ○

As per figure, V_2 is the voltage across a resistor, and thus is in the same phase as the circuit current I. However, the voltage V_3 across the R_L load (which contains an inductive component) will lead the current vector by the power factor angle ϕ (inductor has lagging power factor and current will lag voltage).

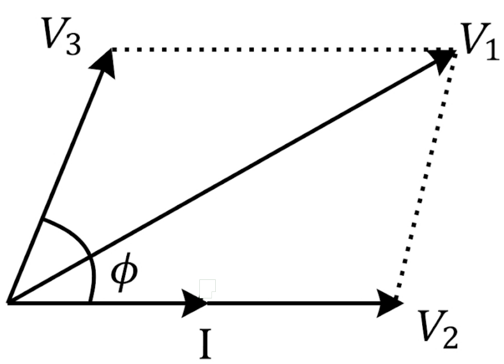


Fig. Solution Q5.12.4

From the phasor diagram, the total supply voltage V_1 is the phasor sum of V_2 and V_3. By applying the law of cosines for vector addition:

$$V_1^2 = V_2^2 + V_3^2 + 2V_2V_3\cos\phi$$

Substituting the given values from the problem ($V_1 = 220\,\text{V}$, $V_2 = 122\,\text{V}$, and $V_3 = 136\,\text{V}$):

$$(220)^2 = (122)^2 + (136)^2 + 2(122)(136)\cos\phi$$

Evaluating the squares:

$$48400 = 14884 + 18496 + 33184\cos\phi$$

$$48400 = 33380 + 33184\cos\phi$$

Solving for $\cos\phi$:

$$33184\cos\phi = 48400 - 33380$$

$$33184\cos\phi = 15020$$

$$\cos\phi = \frac{15020}{33184} \approx 0.45$$

Key Points

- For an inductive load (R_L load), the voltage leads the current by an angle ϕ, where $0° < \phi < 90°$.
- The total voltage in an AC series circuit must be calculated using phasor (vector) addition, not algebraic addition.

$$\boxed{(c)0.45}$$

Q5.12.5 MCQ 2012 2M Ans: B ● ○ ○

Given the load resistance:

$$R_L = 5\,\Omega$$

The power factor ($\cos\phi$) of the load impedance is defined as:

$$\cos\phi = \frac{R_L}{\sqrt{R_L^2 + X^2}} = \frac{R_L}{|Z|}$$

From the previously calculated power factor ($\cos\phi = 0.45$), the magnitude of the load impedance $|Z|$ can be determined as:

$$|Z| = \frac{5}{0.45} = 11.11\,\Omega$$

The RMS current I flowing through the load impedance is:

$$I = \frac{V_3}{|Z|} = \frac{136}{11.11}$$

The real power consumption in the load impedance is dissipated entirely across its resistive part, given by I^2R:

$$P = \left(\frac{136}{11.11}\right)^2 \times 5 = 749.08\,\text{W}$$

Rounding to the nearest whole number:

$$P \simeq 750\,\text{W}$$

Key Points

- We need to use the formula $P = I^2R$ as here we don't have enough information to calculate the voltage across the resistance or the phase angles of voltage and current. The voltage V_3 is the voltage across the entire RL load, and not just the resistance.

$$\boxed{(b)750W}$$

Q5.11.1 MCQ 2011 2M Ans: A

Let us convert the voltage and current expressions into phasor form by choosing $\sin(5t)$ as the reference. Given:

$$\omega = 5\,\text{rad/s}$$

The reference voltage source phasor is:

$$2\sin(5t) = 2\angle 0^\circ\ \text{V}$$

For the current source, we know that:

$$\sin(5t + 90^\circ) = \cos(5t)$$

Thus, the current phasor is:

$$\cos(5t) = 1\angle 90^\circ\ \text{A}$$

The capacitive reactance X_c is calculated as:

$$X_c = \frac{-j}{\omega C} = \frac{-j}{5 \times 0.1} = -2j\,\Omega$$

Applying KCL at the top node with voltage V:

$$\frac{V - 2\angle 0^\circ}{2} + \frac{V}{-2j} + 1\angle 90^\circ = 0$$

Rearranging the terms:

$$V\left(\frac{1}{2} - \frac{1}{j2}\right) = 1\angle 0^\circ - 1\angle 90^\circ$$

$$V\left(\frac{1}{2} + \frac{j}{2}\right) = 1 - j$$

$$\frac{V}{2}(1 + j) = 1 - j$$

Solving for V:

$$V = \frac{2(1-j)}{1+j} = \frac{2(1-j)^2}{(1+j)(1-j)} = \frac{2(1-2j-1)}{2} = -2j\ \text{V}$$

Converting the node voltage back to the time domain:

$$v(t) = -2\cos(5t)\ \text{V}$$

The current I through the capacitor is given by:

$$I = \frac{V}{X_c} = \frac{-2j}{-2j} = 1\angle 0^\circ\ \text{A}$$

Converting to the time domain relative to our reference:

$$i(t) = \sin(5t)\ \text{A}$$

Mistakes to be avoided

- Ensure both sine and cosine terms are converted using the same reference standard to avoid phase errors.
- Do not drop the negative sign or the j operator when calculating capacitive reactance X_c.

$$\boxed{(a) sin(5t) A}$$

Q5.11.2 MCQ 2011 2M Ans: A ● ● ○

Let us investigate the average power delivered or absorbed by the independent sources one by one.

1. Current Source The voltage across the current source from the previous node analysis is $V = -2j$ V. Expressing it in the time domain with respect to our sine reference:

$$v_I(t) = 2\cos(5t - 180^\circ)\ \text{V}$$

However, the polarity of this voltage is such that the current leaves the negative terminal. Therefore, power is absorbed by the current source. The average power absorbed/delivered by the current source is calculated using RMS values:

$$P_I = \frac{V_m I_m}{2}\cos\phi = \frac{2\times 1}{2}\cos(180^\circ) = -1\ \text{W}$$

Thus, the power delivered by the current source is -1 W.

2. Voltage Source The current I_R leaving through the positive terminal of the voltage source into the series resistor is given by:

$$I_R = \frac{2\angle 0^\circ - V}{2} = \frac{2\angle 0^\circ - (-2j)}{2} = 1 + j = \sqrt{2}\angle 45^\circ\ \text{A}$$

In the time domain, this current is expressed as:

$$i_R(t) = \sqrt{2}\sin(5t + 45^\circ)\ \text{A}$$

The power delivered by the voltage source is:

$$P_V = \frac{V_m I_m}{2}\cos\phi = \frac{2\times\sqrt{2}}{2}\cos(45^\circ) = 1\ \text{W}$$

Thus, the total power delivered by all the sources in the circuit is:

$$P_{\text{total}} = P_I + P_V = -1\ \text{W} + 1\ \text{W} = 0\ \text{W}$$

Mistakes to be avoided

- Pay close attention to the direction of current leaving the terminal of a source to properly identify whether power is being delivered (+) or absorbed (−).
- Do not forget to divide by 2 when using peak amplitudes (V_m, I_m) instead of RMS values (V_{rms}, I_{rms}) in the average power equation.

$$\boxed{(a) 0W}$$

Q5.10.1 MCQ | 2010 | 2M | Ans: D | ● ○ ○

Given circuit parameters:

$$L = 10\,\text{mH}$$

$$C = 100\,\mu\text{F}$$

The resonant frequency ω_0 of a series RLC circuit is given by:

$$\text{Resonant frequency} = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{10 \times 10^{-3} \times 100 \times 10^{-6}}} = 1000\,\text{rad/s}$$

Since the operating frequency matches the resonant frequency, the inductive reactance and capacitive reactance cancel each other out ($X_L = X_C$). Thus, the network behaves as a purely resistive circuit.

At resonance, the entire source voltage drops across the resistor R:

$$\text{Voltage across } R = \text{Source voltage} = 10\sqrt{2}\cos(1000t)\,\text{V}$$

The RMS value of the voltage across the resistor $V_{R,\text{RMS}}$ is:

$$V_{R,\text{RMS}} = \frac{10\sqrt{2}}{\sqrt{2}} = 10\,\text{V}$$

The power P dissipated by the resistor is given as 20 W:

$$P = \frac{V_{R,\text{RMS}}^2}{R} = 20\,\text{W}$$

Solving for R:

$$\frac{10^2}{R} = 20 \implies R = \frac{100}{20} = 5\,\Omega$$

$$\boxed{(d) 5\Omega}$$

Q5.10.2 MCQ | 2010 | 2M | Ans: B | ● ○ ○

The quality factor Q of a series resonant circuit measures the sharpness of the resonance peak and is defined as the ratio of the resonant inductive reactance to the circuit resistance.

Using the parameters from the circuit analysis:

$$\omega_0 = 1000\,\text{rad/s}$$

$$L = 10\,\text{mH} = 10 \times 10^{-3}\,\text{H}$$

$$R = 5\,\Omega$$

The expression for the quality factor Q is:

$$Q = \frac{\omega_0 L}{R}$$

Substituting the given values into the formula:

$$Q = \frac{1000 \times 10 \times 10^{-3}}{5} = 2$$

$$\boxed{(b) 2}$$

Q5.08.1 MCQ 2008 2M Ans: B

Let us find the Thevenin equivalent circuit from the load terminals.

1. Finding V_{th}

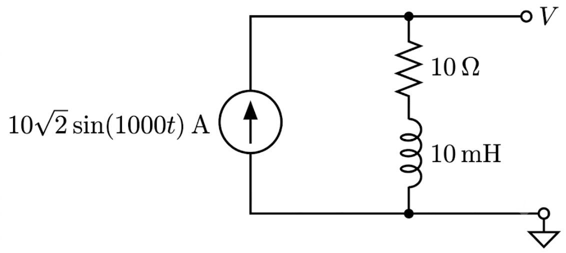


Fig. Solution Q5.08.1

From the given circuit parameters:

$$\omega = 1000\,\text{rad/s}$$

The inductive reactance X_L is:

$$X_L = \omega L = 1000 \times 10 \times 10^{-3} = 10\,\Omega$$

Let us express the independent current source in phasor form using $\sin(1000t)$ as our reference:

$$I = 10\sqrt{2}\angle 0^\circ\ \text{A}$$

The open-circuit voltage across the terminals gives the Thevenin voltage V_{th}:

$$V_{th} = I \cdot (R + jX_L)$$

$$V_{th} = 10\sqrt{2}(10 + j10) = 10\sqrt{2} \times 10\sqrt{2}\angle 45^\circ = 200\angle 45^\circ\ \text{V}$$

Converting back to the time domain:

$$v_{th}(t) = 200\sin(1000t + 45^\circ)\ \text{V}$$

2. Finding Z_{th}

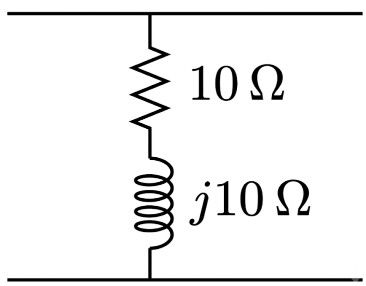


Fig. Solution Q5.08.1

To find the internal impedance, we open-circuit the independent current source. Looking back into the network from the load terminals:

$$Z_{th} = R + jX_L = 10 + j10\,\Omega$$

3. Maximum Power Transfer

For maximum power transfer to a complex load impedance Z_L, the load impedance must be the complex conjugate of the Thevenin source impedance:

$$Z_L = Z_{th}^* = 10 - j10\,\Omega$$

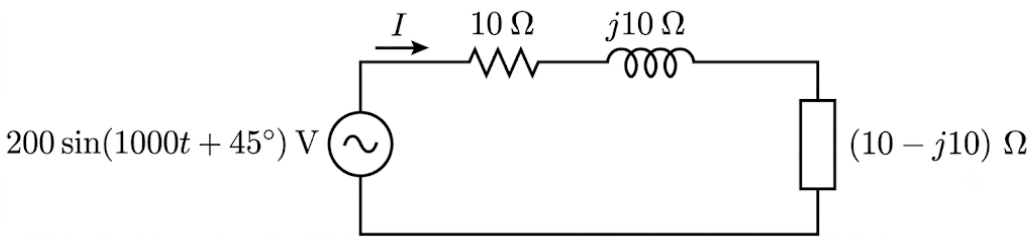


Fig. Solution Q5.08.1

Drawing the complete Thevenin equivalent circuit connected to the load, the total series loop impedance is:

$$Z_{\text{total}} = Z_{th} + Z_L = (10 + j10) + (10 - j10) = 20\,\Omega$$

The loop current phasor I_{loop} is:

$$I_{\text{loop}} = \frac{V_{th}}{Z_{\text{total}}} = \frac{200\angle 45^\circ}{20} = 10\angle 45^\circ\text{ A}$$

In the time domain:

$$i_{\text{loop}}(t) = 10\sin(1000t + 45^\circ)\text{ A}$$

The maximum average power $P_{\max,Z_L}$ transferred to the load is consumed entirely by the resistive part of the load ($R_L = 10\,\Omega$):

$$P_{\max,Z_L} = I_{\text{rms}}^2 R_L = \left(\frac{10}{\sqrt{2}}\right)^2 \times 10 = \frac{100}{2} \times 10 = 500\text{ W}$$

Key Points

- **Thevenin's Theorem**: Any linear circuit network can be simplified to an independent equivalent voltage source V_{th} in series with an equivalent internal impedance Z_{th}.
- **Maximum Power Transfer Theorem (Complex Load)**: Maximum power is delivered to a load when $Z_L = Z_{th}^*$. Under this matched condition, reactive components cancel, and the network operates at a unity power factor.

Mistakes to be avoided

- Always convert the peak current amplitude into the appropriate RMS value before evaluating active power dissipation.

$$\boxed{(b) 500W}$$

Q5.08.2 MCQ 2008 2M Ans: B

We can analyze the circuit at two different frequency components by using the principle of superposition.

Case 1: At $\omega = 1000$ rad/s To find the response due to the first source, short-circuit the $100\sqrt{2}\sin(3000t)$ V source. The inductive reactance X_{L1} at this frequency is:

$$X_{L1} = \omega_1 L = 1000 \times 1 \times 10^{-3} = 1\,\Omega$$

Converting the source voltage to phasor form using $\sin(1000t)$ as reference:

$$10\sqrt{2}\sin(1000t) \implies 10\sqrt{2}\angle 0^\circ\text{ V}$$

The resulting current phasor I_{1k} is:

$$I_{1k} = \frac{10\sqrt{2}}{1 + j1} = 10\angle -45^\circ\text{ A}$$

The average power P_{1k} dissipated in the $1\,\Omega$ resistor by this component is:

$$P_{1k} = (I_{1k,\text{rms}})^2 \times R = \left(\frac{10}{\sqrt{2}}\right)^2 \times 1 = \frac{100}{2} \times 1 = 50\,\text{W}$$

Case 2: At $\omega = 3000\,\text{rad/s}$ To find the response due to the second source, short-circuit the $10\sqrt{2}\sin(1000t)$ V source. The inductive reactance X_{L3} at this frequency is:

$$X_{L3} = \omega_3 L = 3000 \times 1 \times 10^{-3} = 3\,\Omega$$

Converting the source voltage to phasor form using $\sin(3000t)$ as reference:

$$100\sqrt{2}\sin(3000t) \implies 100\sqrt{2}\angle 0^\circ\ \text{V}$$

The resulting current phasor I_{3k} is:

$$I_{3k} = \frac{100\sqrt{2}}{1 + j3} = \frac{100\sqrt{2}}{\sqrt{1^2 + 3^2}}\angle - \tan^{-1}(3) = \frac{100\sqrt{2}}{\sqrt{10}}\angle - 71.57^\circ = 20\sqrt{5}\angle - 71.57^\circ\ \text{A}$$

The average power P_{3k} dissipated in the $1\,\Omega$ resistor by this component is:

$$P_{3k} = (I_{3k,\text{rms}})^2 \times R = \left(\frac{20\sqrt{5}}{\sqrt{2}}\right)^2 \times 1 = \frac{2000}{2} \times 1 = 1000\,\text{W}$$

Total Power Dissipated Since the two sources operate at completely different frequencies, their currents are orthogonal over a complete period. Thus, the total average power is simply the sum of the average powers contributed by each frequency component:

$$P = P_{1k} + P_{3k} = 50\,\text{W} + 1000\,\text{W} = 1050\,\text{W}$$

Mistakes to be avoided

- Never add phasor currents or voltages that belong to different frequencies directly in the frequency domain.
- Do not try to compute total power using $P = (I_{1k} + I_{3k})^2 R$; cross-frequency power terms integrate out to zero over time, making power superposition valid only for distinct frequencies.

$$\boxed{(b)1050W}$$

Q5.08.3 MCQ 2008 2M Ans: D ● ○ ○

Given that the circuit is operating under the condition of resonance:

$$V_R = V_s = 10\,\text{V}$$

The loop current I under resonance can be calculated using the resistance R of the circuit:

$$I = \frac{V_s}{R} = \frac{10}{10} = 1\,\text{A}$$

The capacitor voltage can be calculated as follows:

$$V_C = I \cdot (-jX_C) = 1 \times (-j100) = -j100\,\text{V}$$

Expressing this in rectangular complex form:

$$V_C = (0 - j100)\,\text{V}$$

$$\boxed{(d)(0 - j100)V}$$

Q5.08.4 MCQ 2008 2M Ans: A ● ○ ○

Here we can see that, at steady state ($t \to \infty$), the source voltage would die down to zero:

$$\lim_{t\to\infty} v(t) = \lim_{t\to\infty} 5\sqrt{2}e^{-5t}\cos(1000t) = 0$$

Thus, the voltage source would get shorted.

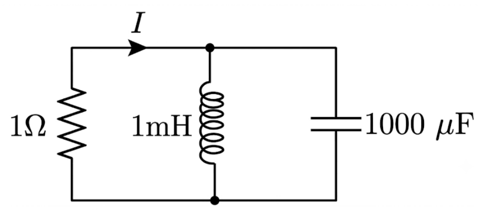


Fig. Solution Q5.08.4

The circuit becomes a source-free parallel RLC circuit.

Now, even if we consider that any of the energy-storing elements has some energy accumulated in them (appearing as voltage across the capacitor and current through the inductor), it would always have a parallel path containing a resistor through which it can discharge itself.

Thus, at steady state, the current I would be zero:

$$\boxed{0}$$

Key Points

- An exponentially decaying excitation source reduces to zero as time approaches infinity ($t \to \infty$), causing an ideal independent voltage source to behave as a short circuit.
- In a source-free parallel RLC network containing a dissipative resistive element (R), all stored magnetic and electric energies are eventually dissipated as heat, resulting in zero steady-state response currents and voltages.

Q5.08.5 MCQ 2008 1M Ans: A ● ○ ○

The given current expression is:

$$i(t) = -6\sqrt{2}\sin(100\pi t) + 6\sqrt{2}\cos(300\pi t + \frac{\pi}{4})\text{ A}$$

This current consists of three distinct components:

1. A sinusoidal AC component of frequency 50 Hz with a peak value of $I_{m1} = -6\sqrt{2}$ A
2. A cosinusoidal AC component of frequency 150 Hz with a peak value of $I_{m2} = 6\sqrt{2}$ A

Since the two AC components are at different frequencies (100π rad/s and 300π rad/s), they are orthogonal. The total RMS value of the current is given by:

$$I_{\text{rms}} = \sqrt{\left(\frac{I_{m1}}{\sqrt{2}}\right)^2 + \left(\frac{I_{m2}}{\sqrt{2}}\right)^2}$$

Substituting the given values into the formula:

$$I_{\text{rms}} = \sqrt{\left(\frac{-6\sqrt{2}}{\sqrt{2}}\right)^2 + \left(\frac{6\sqrt{2}}{\sqrt{2}}\right)^2} = 6\sqrt{2}A$$

Therefore, the true RMS ammeter reading will be $6\sqrt{2}A$.

$$\boxed{(a)6\sqrt{2}A}$$

Q5.07.1 MCQ 2007 2M Ans: A ● ● ○

By applying the principle of superposition, we can separate the calculation into the responses produced by the DC component and the AC component.

Case 1: Considering only the DC component (5 V) At DC, the angular frequency is:

$$\omega = 0\,\text{rad/s}$$

The capacitive reactance X_c behaves as:

$$X_c = \frac{1}{\omega C} \implies X_c \to \infty$$

Since an infinite reactance acts as an open circuit, no DC current can flow through the loop to reach the output resistor. Therefore, the DC component of the output voltage is:

$$V_{o1} = 0\,\text{V}$$

Case 2: Considering only the AC component ($3\cos(\omega t)$) The s-domain transfer function $\frac{V_o}{V_i}(s)$ for a high-pass RC circuit layout is:

$$\frac{V_o}{V_i}(s) = \frac{R}{R + \frac{1}{sC}} = \frac{sRC}{sRC + 1}$$

Substituting $s = j\omega$ to evaluate the frequency response:

$$\frac{V_o}{V_i}(j\omega) = \frac{j\omega RC}{j\omega RC + 1}$$

Given that $\omega RC = 2$:

$$\frac{V_o}{V_i}(j\omega) = \frac{j2}{j2 + 1}$$

Taking the magnitude of the transfer function to determine the peak output voltage:

$$|V_o(j\omega)| = |V_i(j\omega)| \cdot \left|\frac{j2}{1 + j2}\right| = 3 \times \frac{2}{\sqrt{1^2 + 2^2}} = \frac{6}{\sqrt{5}}\,\text{V}$$

Using superposition to combine the individual steady-state time-domain expressions:

$$v_o(t) = V_{o1} + |V_o(j\omega)|\cos(\omega t - \phi) = 0 + \frac{6}{\sqrt{5}}\cos(\omega t - \phi)\,\text{V}$$

Comparing this with the generic form $P + Q\cos(\omega t - \phi)$, we match the constants:

$$P = 0$$

$$Q = \frac{6}{\sqrt{5}}$$

$$\boxed{(a) P = 0 \text{ and } Q = 6/\sqrt{5}}$$

Q5.07.2 MCQ 2007 2M Ans: A ● ● ○

By using the principle of superposition, we can find the total current flowing through the resistor by analyzing the AC and DC sources separately.

Case 1: Considering only the AC current source (Shorting the DC voltage source)

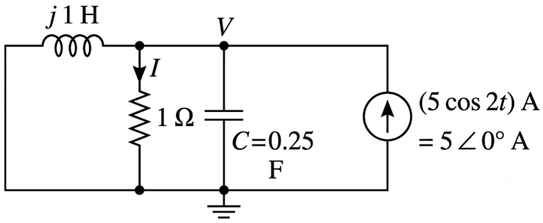


Fig. Solution Q5.07.2

From the circuit configuration at an angular frequency of $\omega = 2$ rad/s: The inductive reactance X_L is:

$$X_L = j\omega L = j(2)(1) = j2\,\Omega$$

The capacitive reactance X_C is:

$$X_C = \frac{-j}{\omega C} = \frac{-j}{2 \times 0.25} = -j2\,\Omega$$

Converting the independent AC current source into phasor notation:

$$5\cos(2t)\text{ A} \implies 5\angle 0^\circ\text{ A}$$

Applying Kirchhoff's Current Law (KCL) at the top node with phasor voltage V:

$$\frac{V}{j2} + \frac{V}{1} + \frac{V}{-j2} = 5\angle 0^\circ$$

$$V\left(\frac{1}{j2} + 1 - \frac{1}{j2}\right) = 5\angle 0^\circ$$

$$V(-j0.5 + 1 + j0.5) = 5\angle 0^\circ$$

$$V = 5\angle 0^\circ\text{ V}$$

Converting the node voltage back to the time domain:

$$v(t) = 5\cos(2t)\text{ V}$$

The component of the current through the 1 Ω resistor due to the AC source is:

$$i_1(t) = \frac{v(t)}{1} = 5\cos(2t)\text{ A}$$

Case 2: Considering only the DC voltage source (Shorting the AC current source)

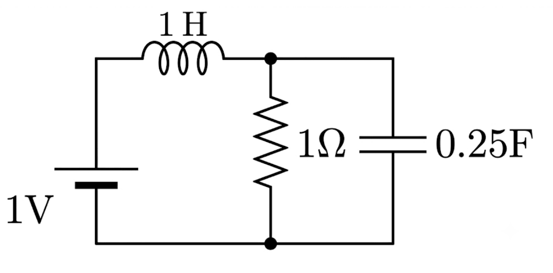


Fig. Solution Q5.07.2

Under steady-state DC conditions ($\omega = 0$ rad/s), reactive components change their characteristics:

- The capacitor acts as an open circuit (OC).

- The inductor acts as a short circuit (SC).

With the AC current source open-circuited, the entire DC loop contains only the 1 V voltage source connected directly across the parallel combination of the branch containing the shorted inductor and the 1 Ω resistor. Thus, the entire DC current flows through the resistor:

$$I_2 = \frac{1\,\text{V}}{1\,\Omega} = 1\,\text{A}$$

Summing the individual contributions using the superposition principle gives the total time-domain current $i(t)$ through the resistor:

$$i(t) = I_2 + i_1(t) = 1 + 5\cos(2t)\ \text{A}$$

$$\boxed{(a) 1 + 5\cos(2t)}$$

Q5.06.1 MCQ 2006 1M Ans: C

The two components of the total voltage waveform are a DC component $V_o(t)$ and an AC square wave component $V_1(t)$:

$$V_o(t) = 2\,\text{V}$$

$$V_1(t) = \text{Square wave with p-p value } 4\,\text{V}$$

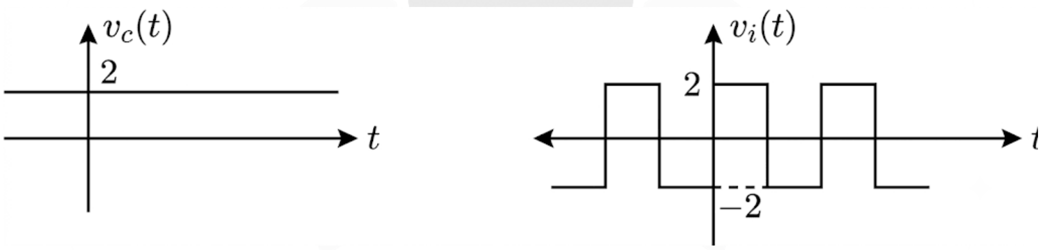


Fig. Solution Q5.06.1

The RMS value of the AC square wave component $V_1(t)$ over a period T is given by:

$$V_{1,\text{RMS}} = \sqrt{\frac{1}{T}\left(\int_0^{\frac{T}{2}} 2^2\,dt + \int_{\frac{T}{2}}^{T} (-2)^2\,dt\right)}$$

$$V_{1,\text{RMS}} = \sqrt{\frac{1}{T}\int_0^{T} 4\,dt} = 2\,\text{V}$$

Since the DC component and the AC symmetric square wave component are orthogonal over the period, the total RMS value V_{RMS} is calculated as:

$$V_{\text{RMS}} = \sqrt{V_o^2 + (V_{1,\text{RMS}})^2}$$

$$V_{\text{RMS}} = \sqrt{2^2 + 2^2} = \sqrt{4+4} = \sqrt{8}\,\text{V}$$

$$\boxed{(c)\sqrt{8}}$$

Q5.05.1 MCQ 2005 2M Ans: B

The quality factor Q of a series resonant circuit is defined as the ratio of the inductive reactance to the resistance:

$$Q = \frac{\omega_0 L}{R} = \frac{X_L}{R}$$

Since it is a series circuit, the current flowing through the components R, L, and C is identical (say I). Therefore, we can multiply the numerator and the denominator by I:

$$Q = \frac{I \cdot X_L}{I \cdot R}$$

At resonance, the inductive reactance equals the capacitive reactance ($X_L = X_C$), and the voltage across the resistor equals the total source voltage ($V_R = V$). Thus, the expression can also be written in terms of the capacitive voltage V_C or output voltage V_o:

$$Q = \frac{I \cdot X_C}{I \cdot R} = \frac{V_C}{V_R} = \frac{V_o}{V}$$

$$\boxed{(b)\frac{V_0}{V}}$$

Q5.04.1 MCQ 2004 2M Ans: A

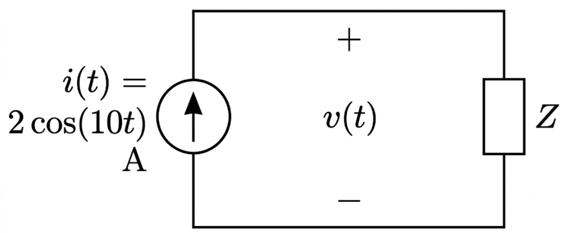


Fig. Solution Q5.04.1

Given the voltage and current expressions:

$$v(t) = 6\sqrt{2}\cos(10t + 45°)\text{ V}$$

$$i(t) = 2\cos(10t)\text{ A}$$

Taking $2\cos(10t)$ as the reference, the voltage and current in phasor form are:

$$I = 2\angle 0°\text{ A}$$

$$V = 6\sqrt{2}\angle 45°\text{ V}$$

From Ohm's law in frequency domain:

$$V = I \cdot Z$$

$$|V| = |I| \cdot |Z|$$

$$\angle V = \angle I + \angle Z$$

Since the current lags the voltage ($\angle I = 0°$ and $\angle V = 45°$), the circuit is inductive.
Calculating the magnitude of the impedance:

$$|Z| = \frac{|V|}{|I|} = \frac{6\sqrt{2}}{2} = 3\sqrt{2}\,\Omega$$

Calculating the angle of the impedance:

$$\angle Z = \angle V - \angle I = 45^\circ - 0^\circ = 45^\circ$$

Thus, the total impedance Z is:

$$Z = 3\sqrt{2}\angle 45^\circ\ \Omega = 3\sqrt{2}\left(\frac{1}{\sqrt{2}} + j\frac{1}{\sqrt{2}}\right)\ \Omega = 3 + j3\ \Omega$$

For the given option A, with $\omega = 10\,\text{rad/s}$:

$$R = 3\,\Omega$$

$$X_L = \omega L = 10 \cdot 0.3 = 3\,\Omega$$

$$Z = R + jX_L = 3 + j3\,\Omega$$

This matches the required impedance condition.

Mistakes to be avoided

- It is critical to check both angle and magnitude conditions. On only checking the angle condition, multiple options may match, but the magnitude condition establishes the correct unique answer.

$$\boxed{(a) 3\,\Omega \textit{ resistor in series with 0.3 H inductor}}$$

Q5.99.1 MCQ | 1999 | 1M | Ans: A

The total root-mean-square voltage V_{RMS} of a composite waveform containing a DC offset and a sinusoidal AC component is given by:

$$V_{\text{RMS}} = \sqrt{V_o^2 + \frac{V_{1m}^2}{2}}$$

Where V_{1m} is the peak value of the AC component. Substituting the given values:

$$V_{\text{RMS}} = \sqrt{2^2 + \frac{1^2}{2}} = \frac{3\sqrt{2}}{2}\ \text{V}$$

The average voltage V_{avg} is equal to the DC offset voltage V_{DC}:

$$V_{\text{avg}} = V_{\text{DC}} = 2\,\text{V}$$

This is because the average value of a pure sinusoidal waveform over a complete cycle is zero:

$$\text{Average value of } \cos\left(\omega t + \frac{\pi}{6}\right) = 0\,\text{V}$$

Therefore, the ratio of the RMS voltage to the average voltage is:

$$\frac{V_{\text{RMS}}}{V_{\text{average}}} = \frac{\frac{3\sqrt{2}}{2}}{2} = \frac{3\sqrt{2}}{4} = \frac{3}{2\sqrt{2}}$$

$$\boxed{(a)\frac{3}{2\sqrt{2}}}$$

Q5.98.1 MCQ 1998 2M Ans: C ● ● ○

Given that the current through branch AB is $I_{AB} = 1$ A.

The voltage across branch AB can be calculated as:

$$V_{AB} = 1 \cdot (30 - j40) = (30 - j40)\text{ V}$$

Since the branch with impedance $(30 + j40)\,\Omega$ is connected in parallel with branch AB, the same voltage V_{AB} appears across it. Therefore, the current I_1 through that impedance is:

$$I_1 = \frac{30 - j40}{30 + j40} = 1\angle - 106.26^\circ\text{ A}$$

The total supply current I_s flowing through the $5\,\Omega$ resistor is the phasor sum of these two parallel branch currents:

$$I_s = I_{AB} + I_1$$

$$I_s = 1 + 1\angle - 106.26^\circ = 1.2\angle - 53.13^\circ\text{ A}$$

The total input voltage V_{in} can be determined by applying KVL:

$$V_{in} = I_s \cdot 5 + V_{AB}$$

$$V_{in} = 1.2\angle - 53.13^\circ \cdot 5 + (30 - j40)$$

$$V_{in} = 56\angle - 53.13^\circ\text{ V}$$

Taking the magnitude of the input voltage:

$$|V_{in}| = 56\text{ V}$$

$$\boxed{(c)56}$$

DEBUG INFO

Chapter V

AC Circuit Analysis

Total Questions : 57

MCQ : 29

MSQ : 0

NAT : 28

SOLUTIONS

VI

Two Port Parameters

Topics

1. Z-Parameters
2. Y-Parameters, Reciprocity and Symmetry
3. H, G and ABCD Parameters
4. Series and Parallel Connection of Two Port Networks

Q6.26.1 MCQ | 2026 | 2M | Ans: B | ● ● ○

To find the maximum power transfer in a linear network, the load resistance must be equal to the Thevenin equivalent resistance., i.e., $R_L = R_{th}$. Given the z-parameters:

$$z_{11} = 10\,\Omega, \quad z_{22} = 10\,\Omega, \quad z_{12} = z_{21} = 5\,\Omega$$

The source voltage $V_s = 5$ V is connected in series with a source resistance $R_s = 5\,\Omega$.
For the two-port network,

$$V_1 = z_{11}I_1 + z_{12}I_2$$
$$V_2 = z_{21}I_1 + z_{22}I_2$$

To determine the Thevenin resistance seen at Port 2(Load Terminals), deactivate the voltage source. Hence, the source is replaced by its internal resistance R_s, giving

$$V_1 = -R_sI_1.$$

Substituting into the first z-parameter equation,

$$-R_sI_1 = z_{11}I_1 + z_{12}I_2$$
$$(z_{11} + R_s)I_1 = -z_{12}I_2$$
$$I_1 = -\frac{z_{12}}{z_{11} + R_s}I_2.$$

Substituting this expression for I_1 into the second equation,

$$V_2 = z_{21}I_1 + z_{22}I_2$$
$$V_2 = z_{21}\left(-\frac{z_{12}}{z_{11} + R_s}I_2\right) + z_{22}I_2$$
$$V_2 = \left(z_{22} - \frac{z_{12}z_{21}}{z_{11} + R_s}\right)I_2.$$

Hence, the Thevenin resistance seen from Port 2 is

$$R_{\text{th}} = \frac{V_2}{I_2} = z_{22} - \frac{z_{12}z_{21}}{z_{11} + R_s}.$$

For maximum power transfer,

$$R_L = R_{\text{th}}.$$

Substituting the given values,

$$R_L = 10 - \frac{5 \times 5}{10 + 5} = 10 - \frac{25}{15} = \frac{25}{3}\,\Omega \approx 8.33\,\Omega.$$

$$\boxed{B)\ 8.33}$$

Q6.17.1 MCQ | 2017 | 1M | Ans: A | ● ● ○

To find the $ABCD$ parameters of a cascaded connection of two 2-port networks, the overall transmission matrix is the product of the individual transmission matrices in the order of connection:

$$\begin{bmatrix} A & B \\ C & D \end{bmatrix}_{combined} = \begin{bmatrix} A & B \\ C & D \end{bmatrix}_{N_1} \times \begin{bmatrix} A & B \\ C & D \end{bmatrix}_{N_2}$$

Given the matrices:

$$\begin{bmatrix} A & B \\ C & D \end{bmatrix}_{N_1} = \begin{bmatrix} 1 & 5 \\ 0 & 1 \end{bmatrix}, \quad \begin{bmatrix} A & B \\ C & D \end{bmatrix}_{N_2} = \begin{bmatrix} 1 & 0 \\ 0.2 & 1 \end{bmatrix}$$

Perform matrix multiplication:

$$\begin{bmatrix} A & B \\ C & D \end{bmatrix}_{combined} = \begin{bmatrix} 1 & 5 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0.2 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} (1 \times 1 + 5 \times 0.2) & (1 \times 0 + 5 \times 1) \\ (0 \times 1 + 1 \times 0.2) & (0 \times 0 + 1 \times 1) \end{bmatrix}$$

$$= \begin{bmatrix} (1+1) & (0+5) \\ (0+0.2) & (0+1) \end{bmatrix} = \begin{bmatrix} 2 & 5 \\ 0.2 & 1 \end{bmatrix}$$

$$\boxed{\text{A)} \begin{bmatrix} 2 & 5 \\ 0.2 & 1 \end{bmatrix}}$$

Q6.13.1 MCQ 2013 2M Ans: A ● ● ○

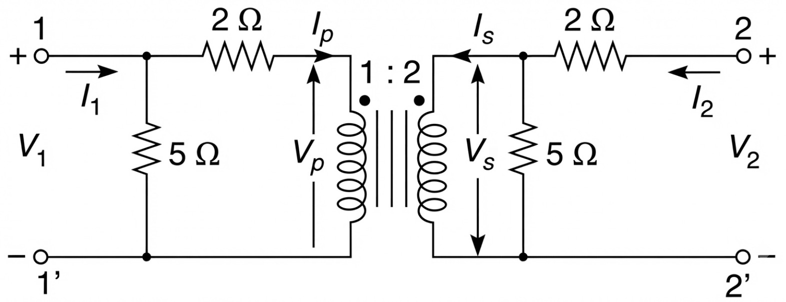


Fig. Solution Q6.13.1

The transmission parameter A is defined as:

$$A = \left.\frac{V_1}{V_2}\right|_{I_2=0}$$

For an ideal transformer with a turns ratio of 1 : 2:

$$I_p = 2I_s \quad \text{and} \quad V_s = 2V_p \quad \ldots \text{(i)}$$

At $I_2 = 0$:

$$V_2 = V_s = 5I_s = 5\left(\frac{I_p}{2}\right) = 2.5I_p \quad \ldots \text{(ii)}$$

From the primary side circuit:

$$V_1 = V_p + 2I_p$$

Substituting $V_p = \frac{V_s}{2} = \frac{V_2}{2}$:

$$V_1 = \frac{V_2}{2} + 2I_p$$

Using $V_2 = 2.5I_p$ from equation (ii), we get $I_p = \frac{V_2}{2.5} = 0.4V_2$:

$$V_1 = \frac{V_2}{2} + 2(0.4V_2) = 0.5V_2 + 0.8V_2 = 1.3V_2$$

Thus:

$$A = \frac{V_1}{V_2} = 1.3$$

$$\boxed{\text{A) } 1.3}$$

Q6.12.1 MCQ 2012 2M Ans: C

To determine the transmission parameters (ABCD parameters), we use the following equations for conditions (i) and (ii):

$$V_1 = AV_2 - BI_2$$

For condition (i), where $V_1 = 10\,\text{V}$, $V_2 = 3\,\text{V}$, and $I_2 = -3\,\text{A}$:

$$10 = 3A + 3B \quad \ldots(1)$$

For condition (ii), where $V_2 = (2.5)(2) = 5\,\text{V}$ and $I_2 = -2\,\text{A}$:

$$10 = 5A + 2B \quad \ldots(2)$$

Solving the simultaneous equations (1) and (2) yields the values for parameters A and B:

$$A = \frac{10}{9} \quad \ldots(3)$$

$$B = \frac{20}{9} \quad \ldots(4)$$

Given the circuit conditions $V_1 = 10\,\text{V}$ and $V_2 = -7I_2$, we use the transmission parameter equation:

$$V_1 = AV_2 - BI_2$$

Substituting $V_2 = -7I_2$ and $A = \frac{10}{9}$ and $B = \frac{20}{9}$ into the equation:

$$10 = -7I_2A - BI_2$$

$$10 = -7I_2\left(\frac{10}{9}\right) - \left(\frac{20}{9}\right)I_2$$

$$10 = -\frac{70}{9}I_2 - \frac{20}{9}I_2$$

$$10 = -\frac{90}{9}I_2$$

$$10 = -10I_2$$

$$I_2 = -1\,\text{A}$$

The negative sign signifies that current is drawn from the port.

$$\boxed{C)\ 1\,\text{A}}$$

Q6.12.2 MCQ 2012 2M Ans: C

To find the open-circuit voltage $(V_2)_{OC}$, we use the given conditions $V_1 = 8\,\text{V}$ and $I_2 = 0\,\text{A}$, knowing $A = \frac{10}{9}$:

$$V_1 = A(V_2)_{OC} - B(0)$$

$$8 = A(V_2)_{OC}$$

$$(V_2)_{OC} = \frac{8}{A} = \frac{8}{10/9} = 7.2\,\text{V}$$

We see that 7.2V is not in options.
A passive two-port network assumes that if external inputs are zero, the outputs must be zero proportionally. However, when a network is specified as active or contains internal independent bias sources, it maintains non-zero internal energy

even when external port excitations change. To account for this internal independent contribution mathematically, we must include an offset voltage translation constant V_0 in the transmission equation:

$$V_1 = AV_2 + BI_L + V_0$$

Hence we resolve for A and B with V_0, using this equation:

$$AV_2 + BI_L + V_0 = V_1$$

$$3A + 3B + V_0 = 10$$

$$5A + 2B + V_0 = 10$$

$$\frac{7}{3}A + \frac{7}{3}B + V_0 = 6$$

Solving Simultaneously we get,

$$A = 2, B = 4, V_0 = -8V$$

Even if we again solve part 1, we get

$$2(7I_2) + 4I_2 - 8 = 10, \implies I_2 = 1A$$

Hence, solving part 2, we get

$$V_1 = 8V, I_B = I_L = 0A \therefore V_1 = AV_2 + BI_L + V_0$$

$$8 = 2V_2 + 4(0) - 8$$

$$8 = 2V_2 - 8$$

$$16 = 2V_2 \implies V_2 = 8\,\mathrm{V}$$

$$\boxed{\text{C) 8V}}$$

Q6.08.1 MCQ 2008 2M Ans: D

By applying KCL at point A,

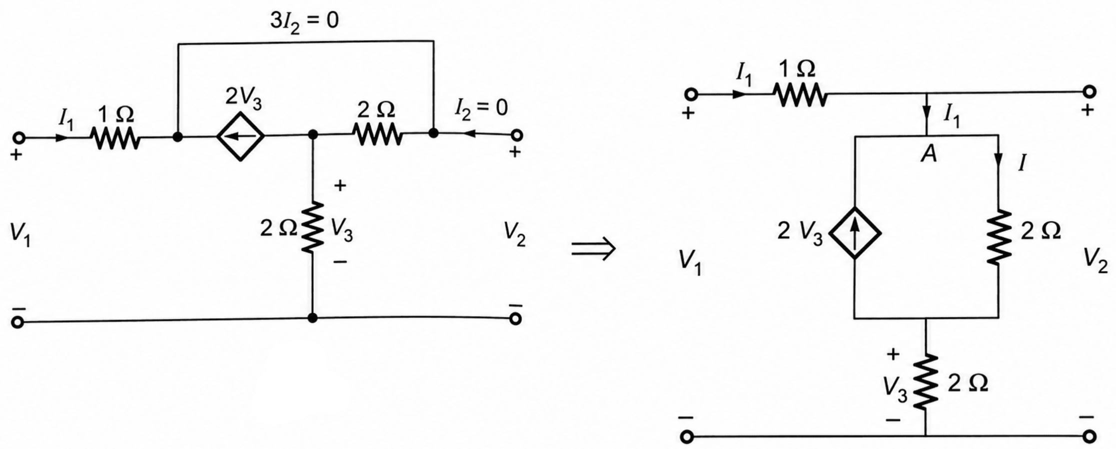


Fig. Solution Q6.08.1

$$I_1 + 2V_3 = I$$

and by applying KVL in the outer loop,

$$V_1 = I_1 + 2(I_1 + 2V_3) + 2I_1$$

$$V_1 = 5I_1 + 4V_3 \quad \ldots(i)$$

Given that $V_3 = 2I_1$, we substitute this into equation (i):

$$V_1 = 5I_1 + 8I_1$$

$$\frac{V_1}{I_1} = 13\,\Omega$$

$$\boxed{D)\ 13\,\Omega}$$

Q6.07.1 MCQ 2007 1M Ans: A

By applying the voltage division rule across the resistor R_2, the voltage V is given by:

$$V = \frac{R_2}{R_1 + R_2} \cdot V_i$$

The dependent voltage source provides an output voltage V_o that is proportional to V, defined by the gain A_v:

$$V_o = A_v V$$

$$V_o = A_v \cdot \left(\frac{R_2}{R_1 + R_2}\right) \cdot V_i$$

DC voltage gain=

$$\frac{V_o}{V_i} = \frac{R_2}{R_1 + R_2} A_v$$

$$\boxed{A)\ \frac{R_2}{R_1 + R_2} A_v}$$

Q6.07.2 MCQ 2007 2M Ans: A

For the coupled circuit shown in the figure, we apply KVL to both circuits:

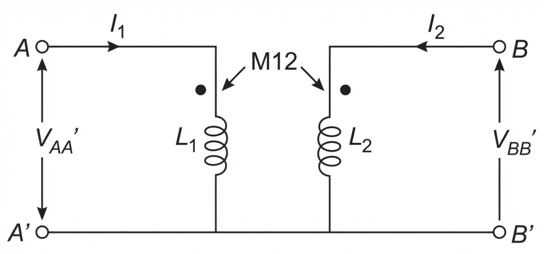


Fig. Solution Q6.07.2

$$V_{AA'} = j\omega L_1 I_1 + j\omega M_{12} I_2 \quad \ldots(i)$$

$$V_{BB'} = j\omega M_{12} I_1 + j\omega L_2 I_2 \quad \ldots(ii)$$

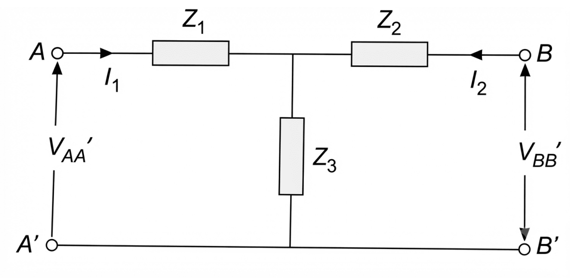


Fig. Solution Q6.07.2

For the equivalent T-network, the KVL equations are:

$$V_{AA'} = (Z_1 + Z_3)I_1 + Z_3 I_2 \quad \ldots(iii)$$

$$V_{BB'} = Z_3 I_1 + (Z_2 + Z_3)I_2 \quad \ldots(iv)$$

By comparing equation (i) with (iii) and equation (ii) with (iv), we get:

$$Z_1 + Z_3 = j\omega L_1$$

$$Z_2 + Z_3 = j\omega L_2$$

$$Z_3 = j\omega M_{12}$$

Solving these simultaneous equations, we obtain the equivalent impedances:

$$\boxed{A)\ Z_1 = j\omega(L_1 - M_{12}), \quad Z_2 = j\omega(L_2 - M_{12}), \quad Z_3 = j\omega M_{12}}$$

Q6.01.1 MCQ 2001 1M Ans: B ● ● ●

Convert the y-parameters to $ABCD$ parameters, multiply the matrices, and convert back to y-parameters.
Given y-parameters for each identical network:

$$[y] = \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} \text{S}$$

The conversion from y-parameters to $ABCD$ parameters is:

$$A = -\frac{y_{22}}{y_{21}}, \quad B = -\frac{1}{y_{21}}, \quad C = -\frac{\Delta y}{y_{21}}, \quad D = -\frac{y_{11}}{y_{21}}$$

Where $\Delta y = y_{11}y_{22} - y_{12}y_{21} = (1)(1) - (-1)(-1) = 0$.

$$A = -\frac{1}{-1} = 1, \quad B = -\frac{1}{-1} = 1, \quad C = -\frac{0}{-1} = 0, \quad D = -\frac{1}{-1} = 1$$

So, $[ABCD] = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$. For two identical networks in cascade, multiply the matrices:

$$[ABCD]_{total} = [ABCD] \times [ABCD] = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$$

Convert back to y-parameters using:

$$y_{11} = \frac{D}{B}, \quad y_{12} = -\frac{\Delta ABCD}{B}, \quad y_{21} = -\frac{1}{B}, \quad y_{22} = \frac{A}{B}$$

Where $\Delta ABCD = AD - BC = (1)(1) - (2)(0) = 1$.

$$y_{11} = \frac{1}{2} = 0.5\,\text{S}, \quad y_{12} = -\frac{1}{2} = -0.5\,\text{S}, \quad y_{21} = -\frac{1}{2} = -0.5\,\text{S}, \quad y_{22} = \frac{1}{2} = 0.5\,\text{S}$$

$$\boxed{\text{B)}\ y_{12} = -\frac{1}{2}\,\text{S}}$$

DEBUG INFO

Chapter VI

Two Port Parameters

Total Questions : 9

MCQ : 9

MSQ : 0

NAT : 0

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